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MONFERRER et al.(10) **Pub. No.: US 2025/0283055 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **BROAD SPECTRUM VIRUS-TRAPPING
NANOSHELLS****Publication Classification**(71) Applicant: **TECHNISCHE UNIVERSITÄT
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(2013.01)(21) Appl. No.: **18/859,607**

(57)

ABSTRACT(22) PCT Filed: **Apr. 28, 2023**(86) PCT No.: **PCT/EP2023/061264**

§ 371 (c)(1),

(2) Date: **Oct. 24, 2024**

The present invention relates to a DNA-based nanostructure for encapsulating viruses or viral particles, to a composition comprising one or more viruses or viral particles encapsulated by such a DNA-based nanostructure according to the present invention, and to a method for encapsulating one or more viruses or viral particles by using such a DNA-based nanostructure.

(30) **Foreign Application Priority Data**

Apr. 28, 2022 (EP) 22170578.3

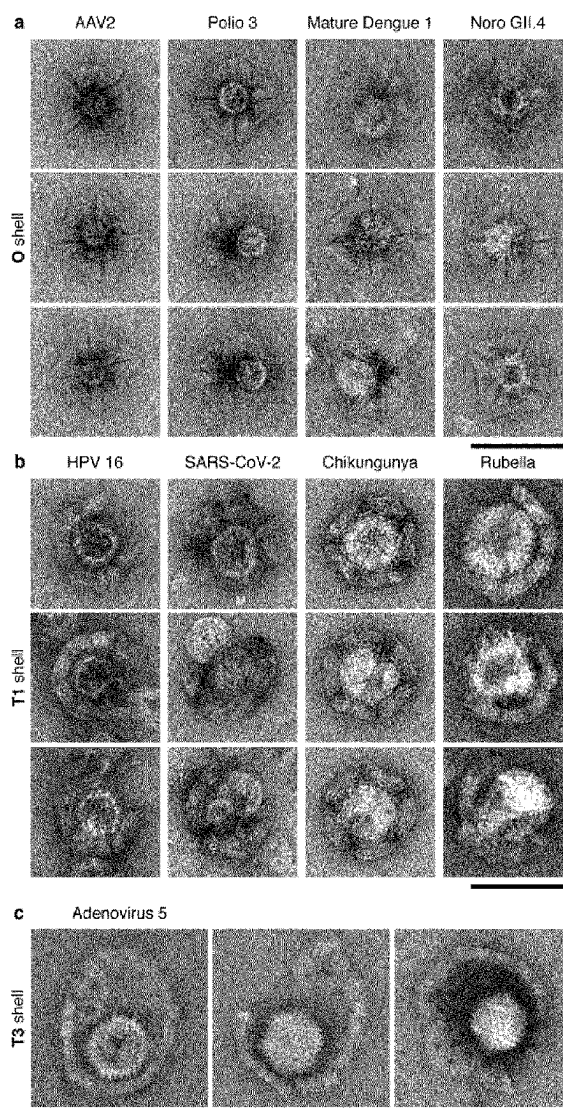
Specification includes a Sequence Listing.

FIGURE 1:

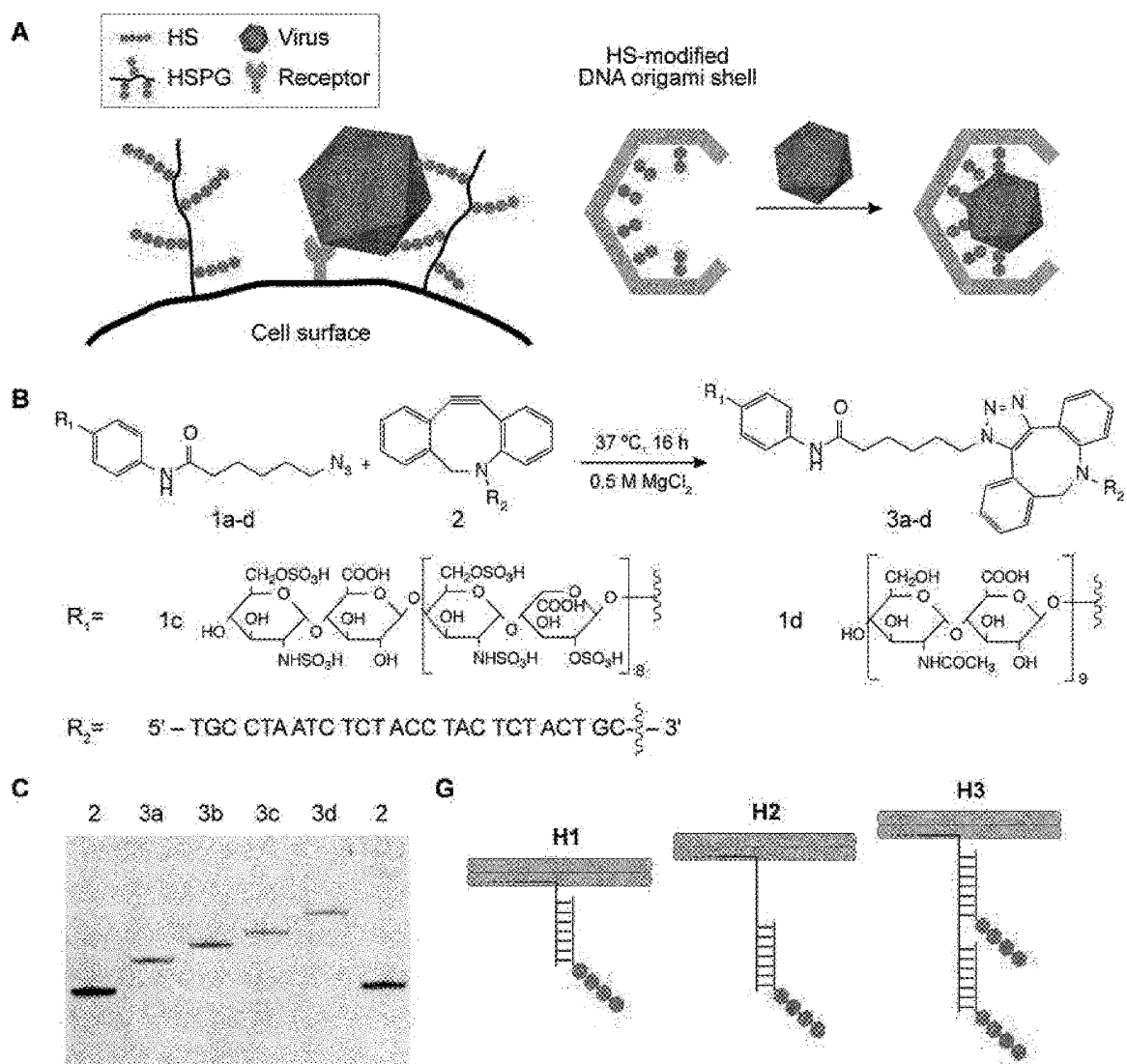


FIGURE 1 (contd.):

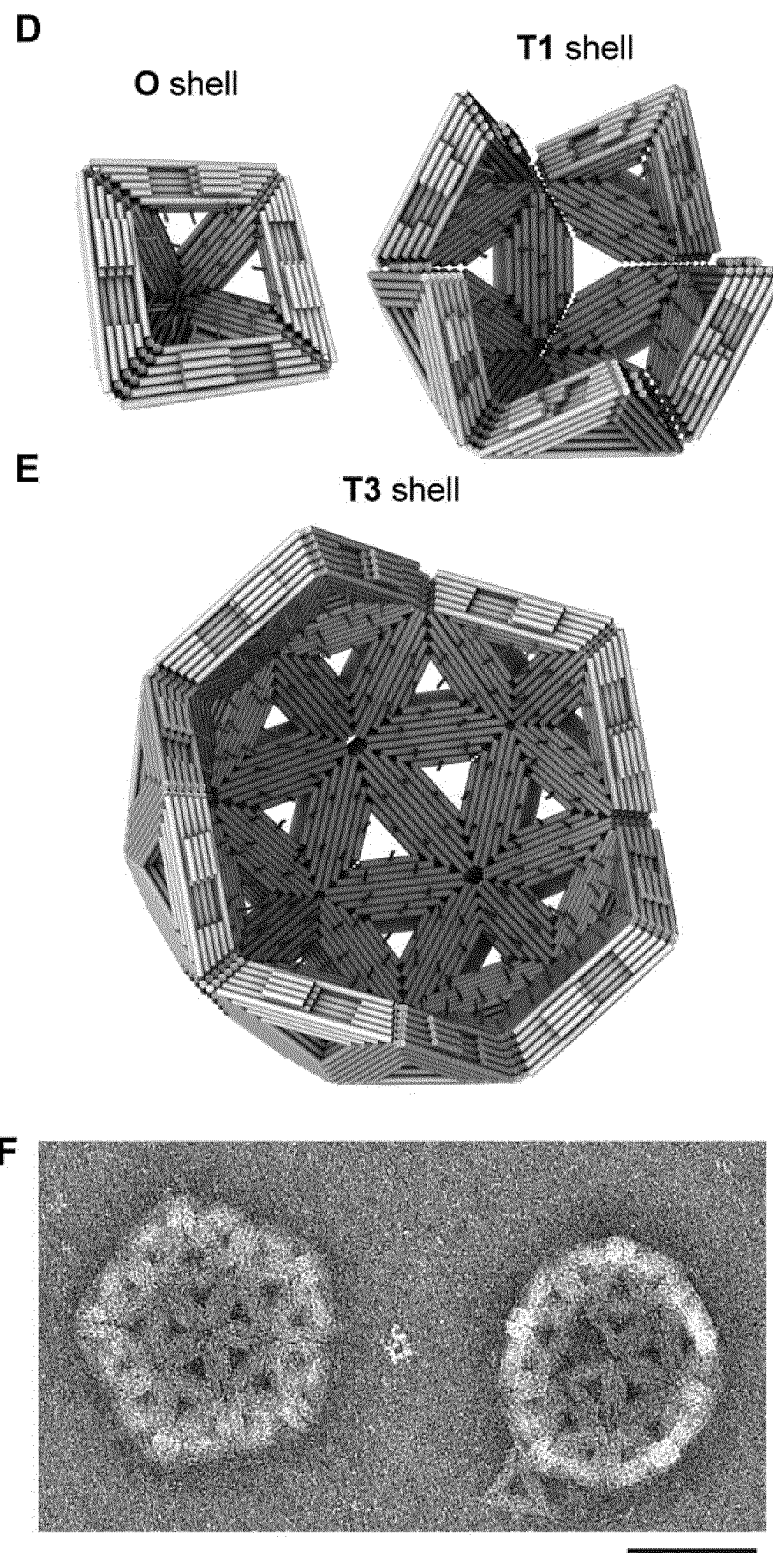


FIGURE 2:

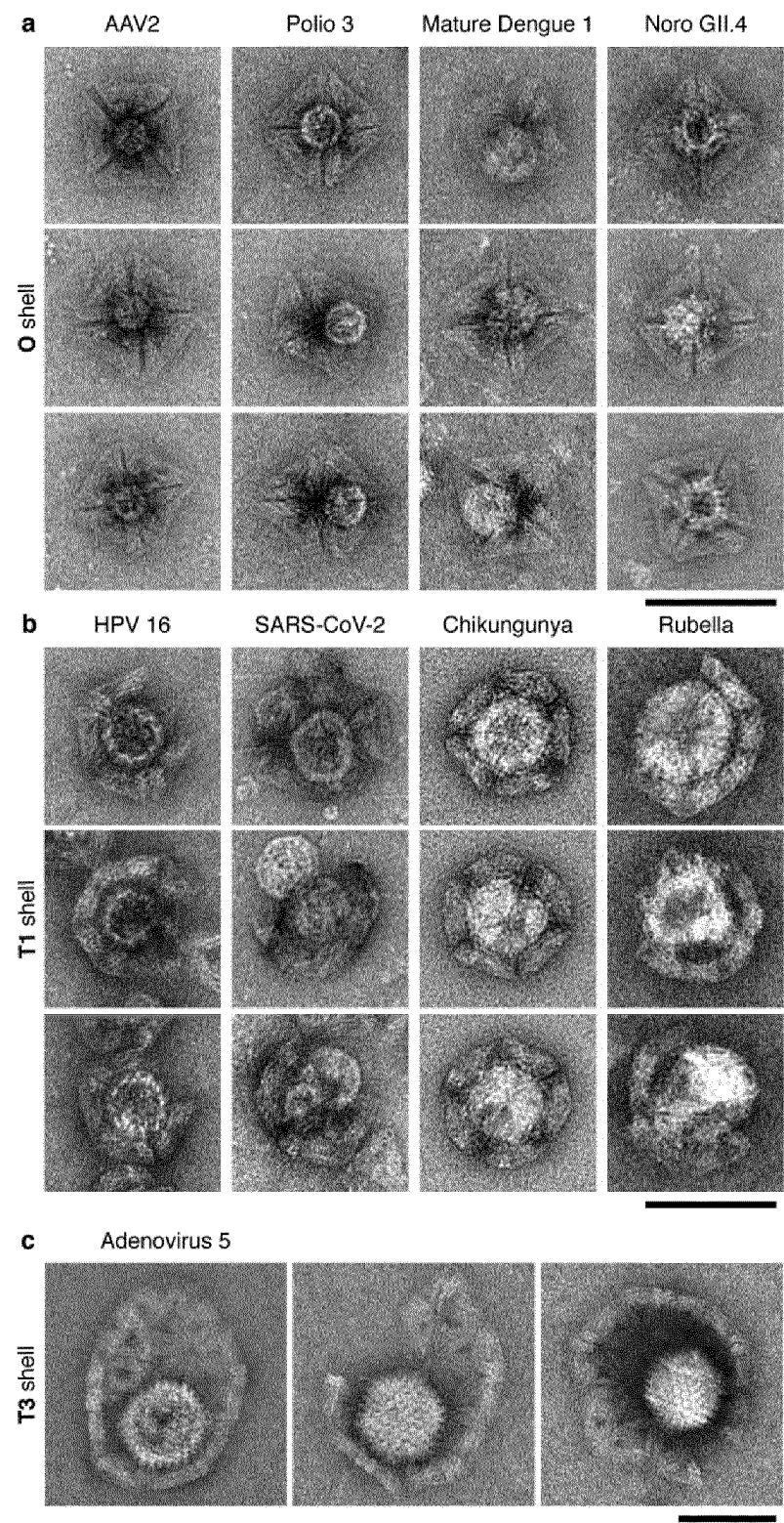


FIGURE 3:

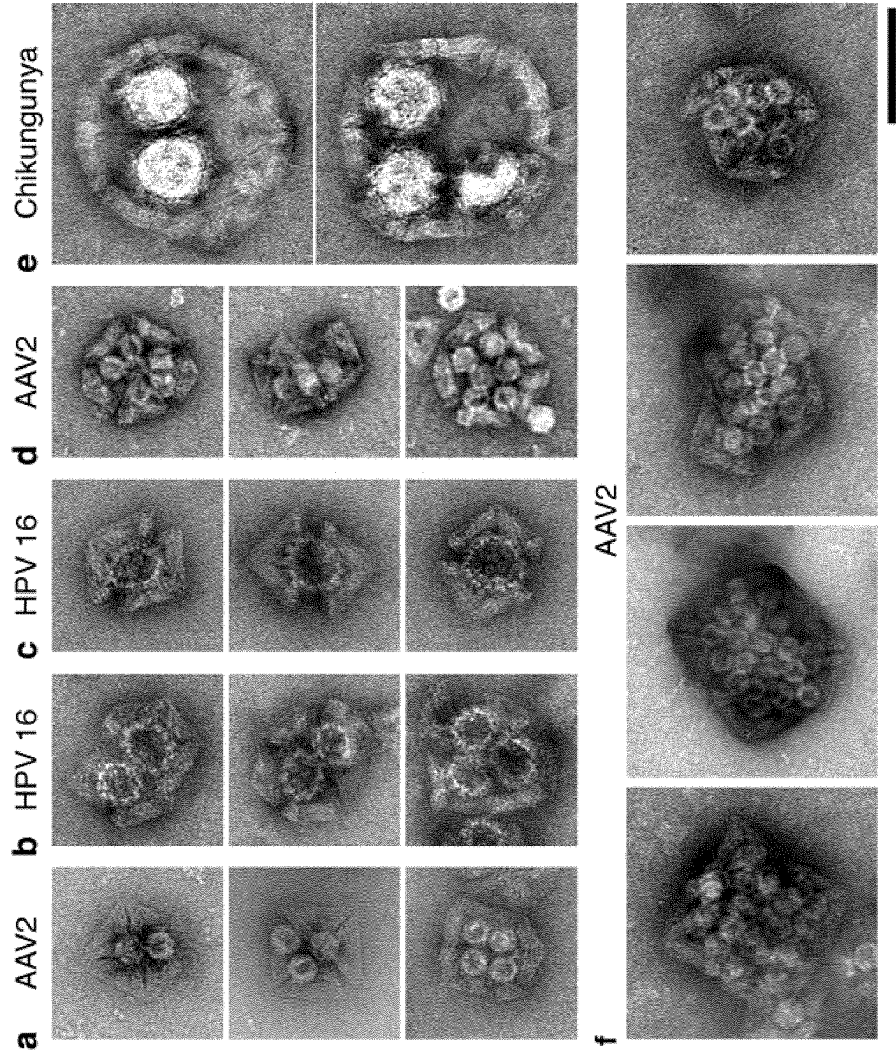


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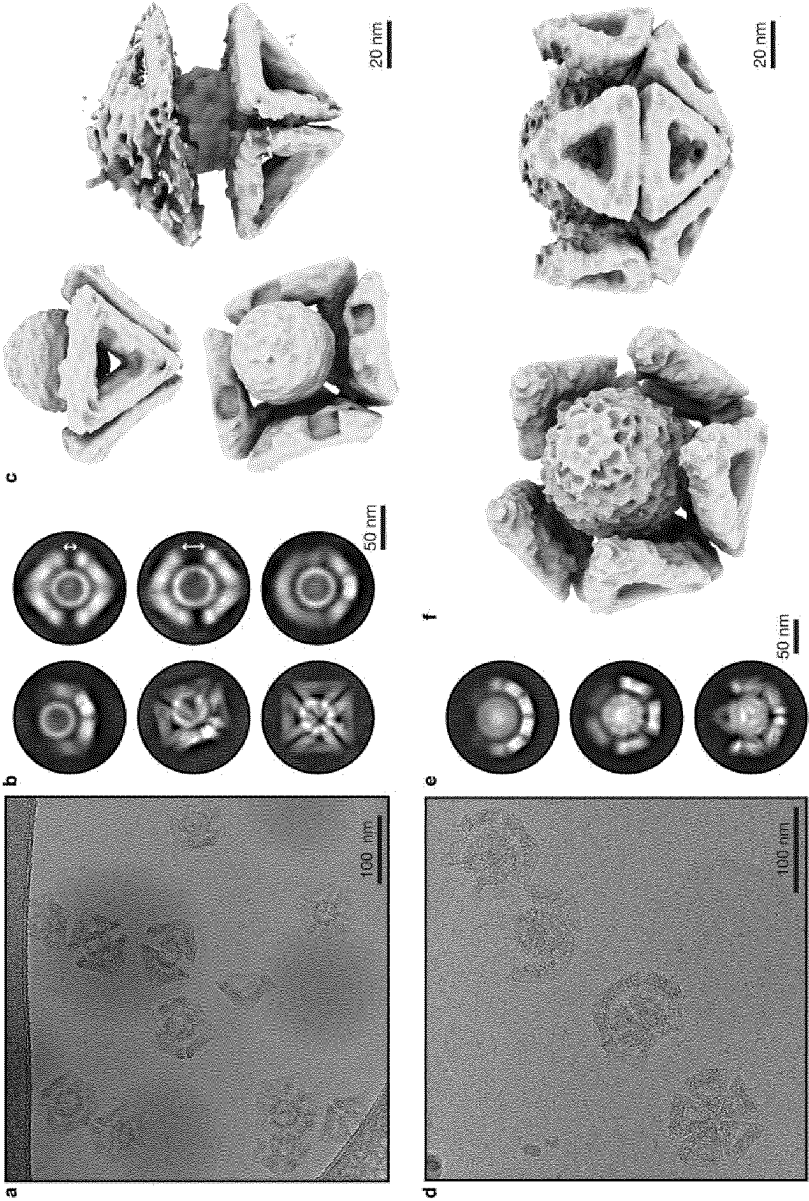


FIGURE 5:

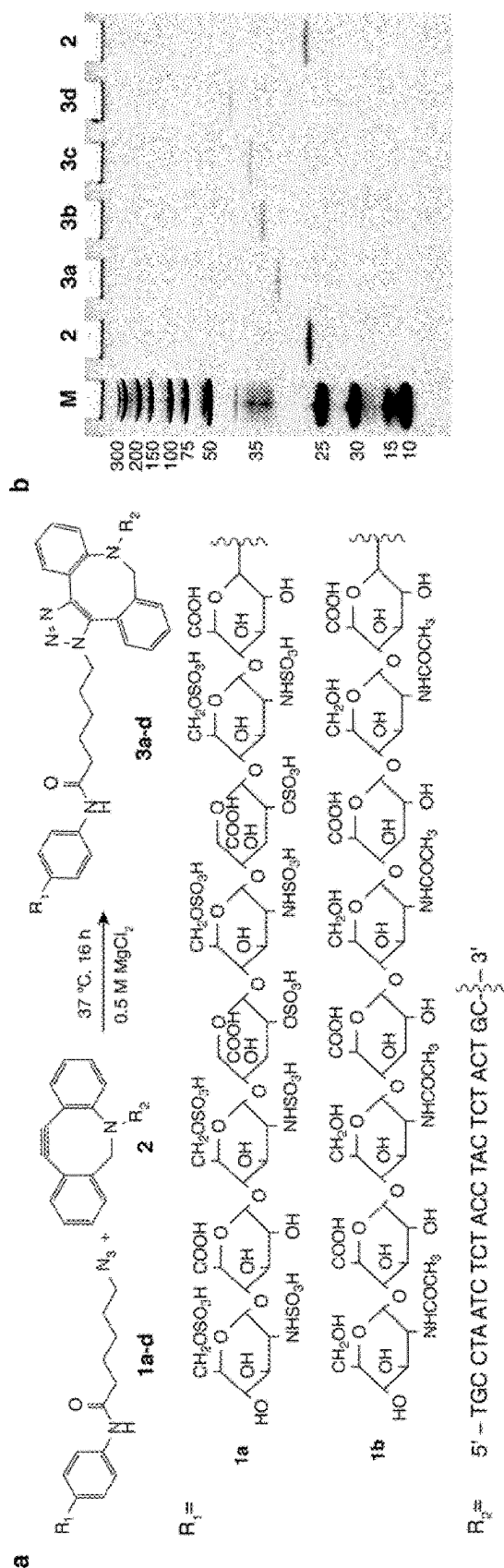


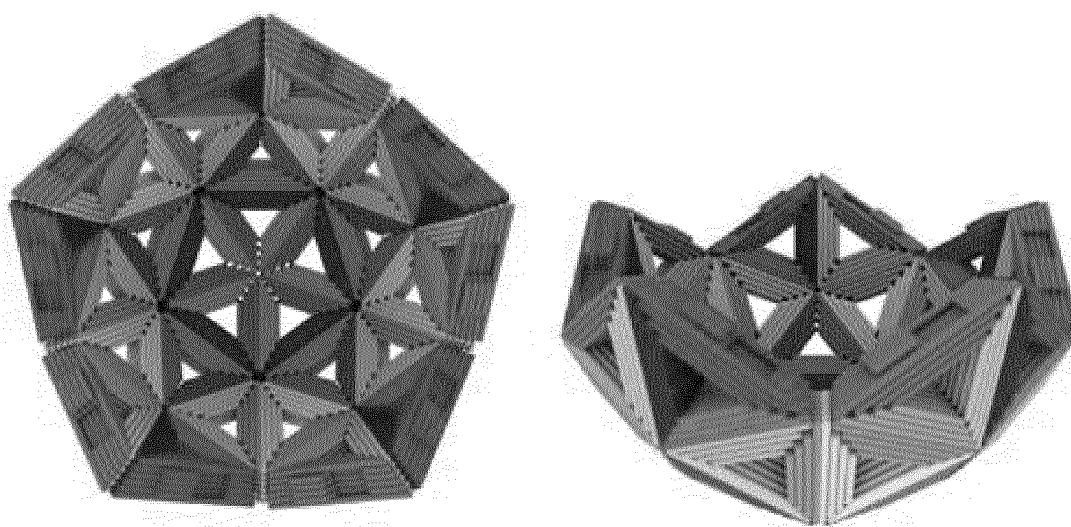
FIGURE 6:**a**

FIGURE 6 (contd.):

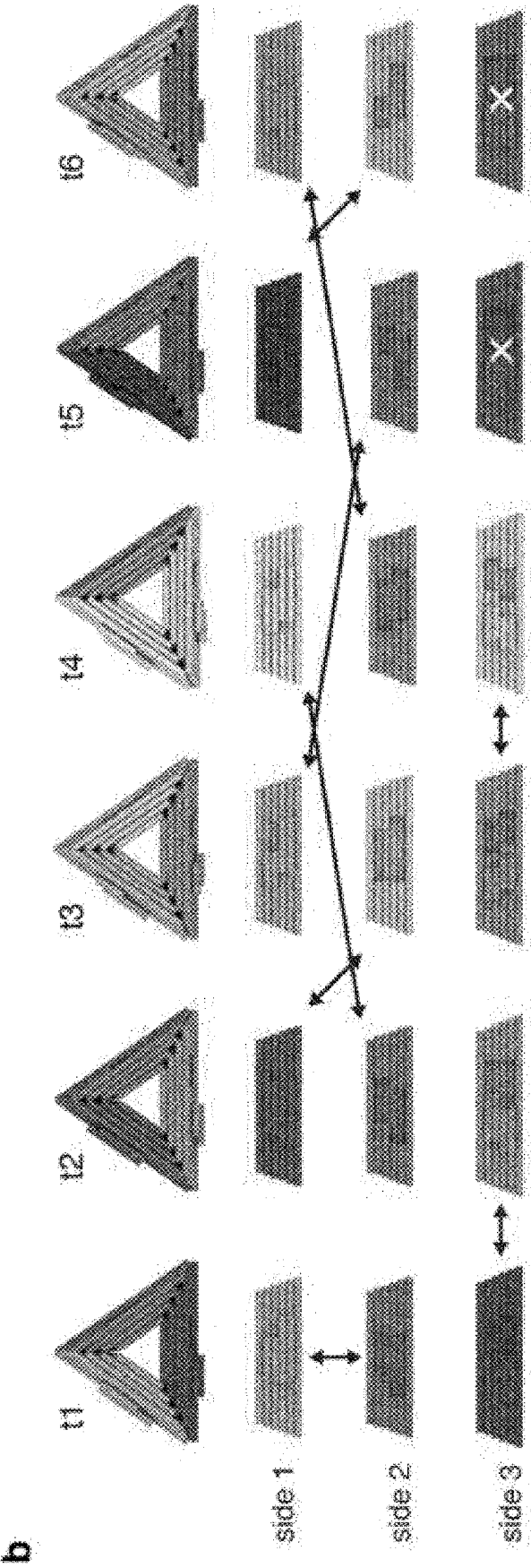


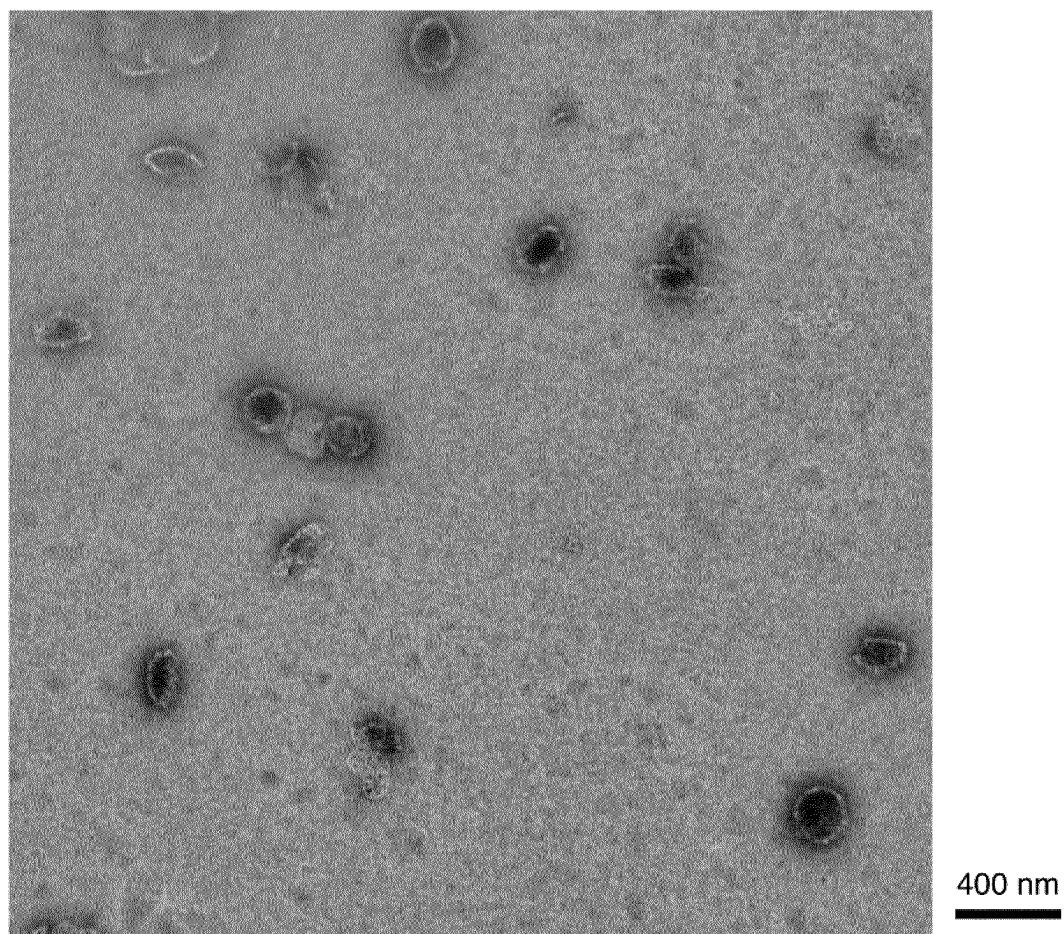
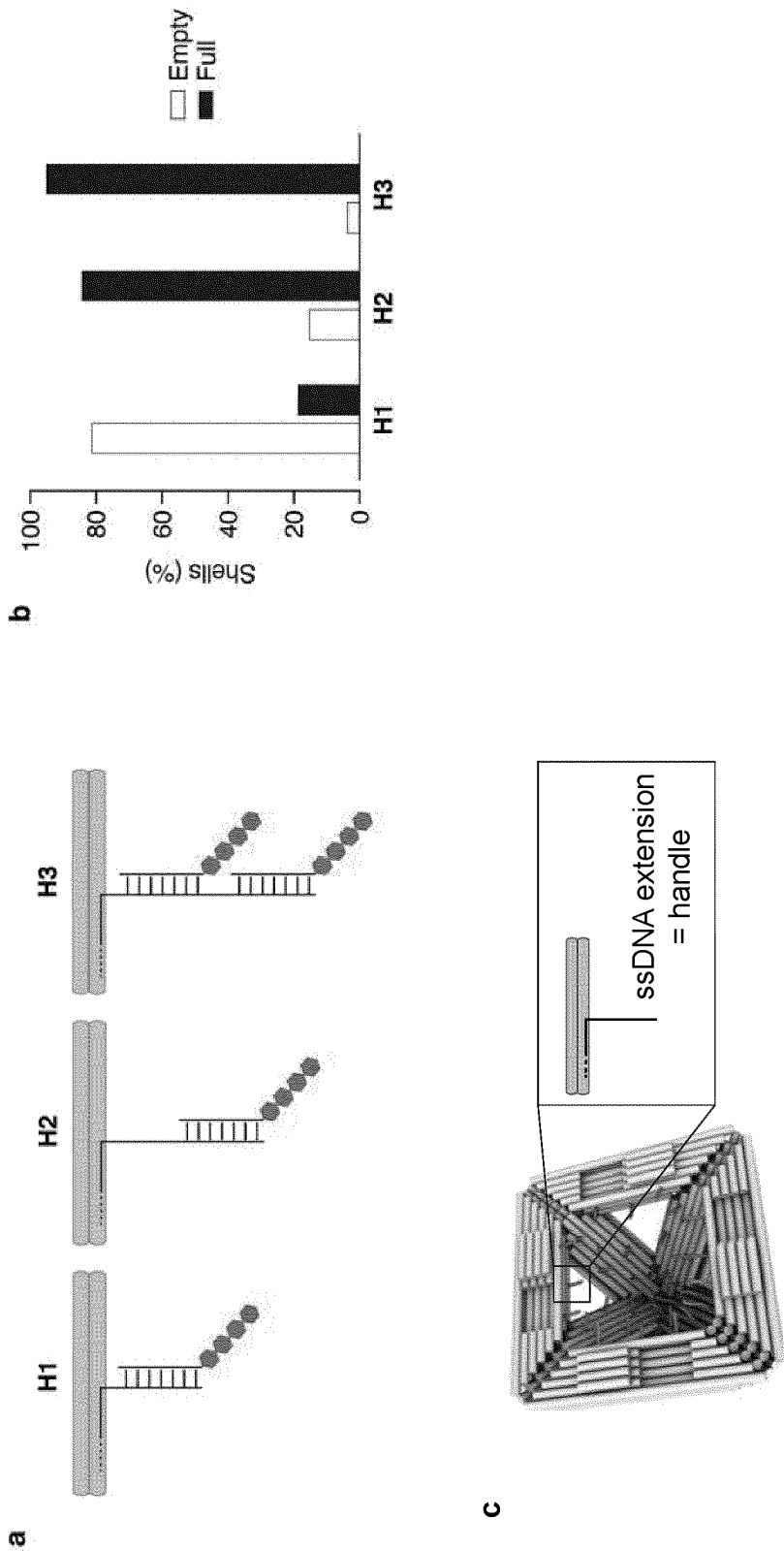
FIGURE 7:

FIGURE 8:



100 nm

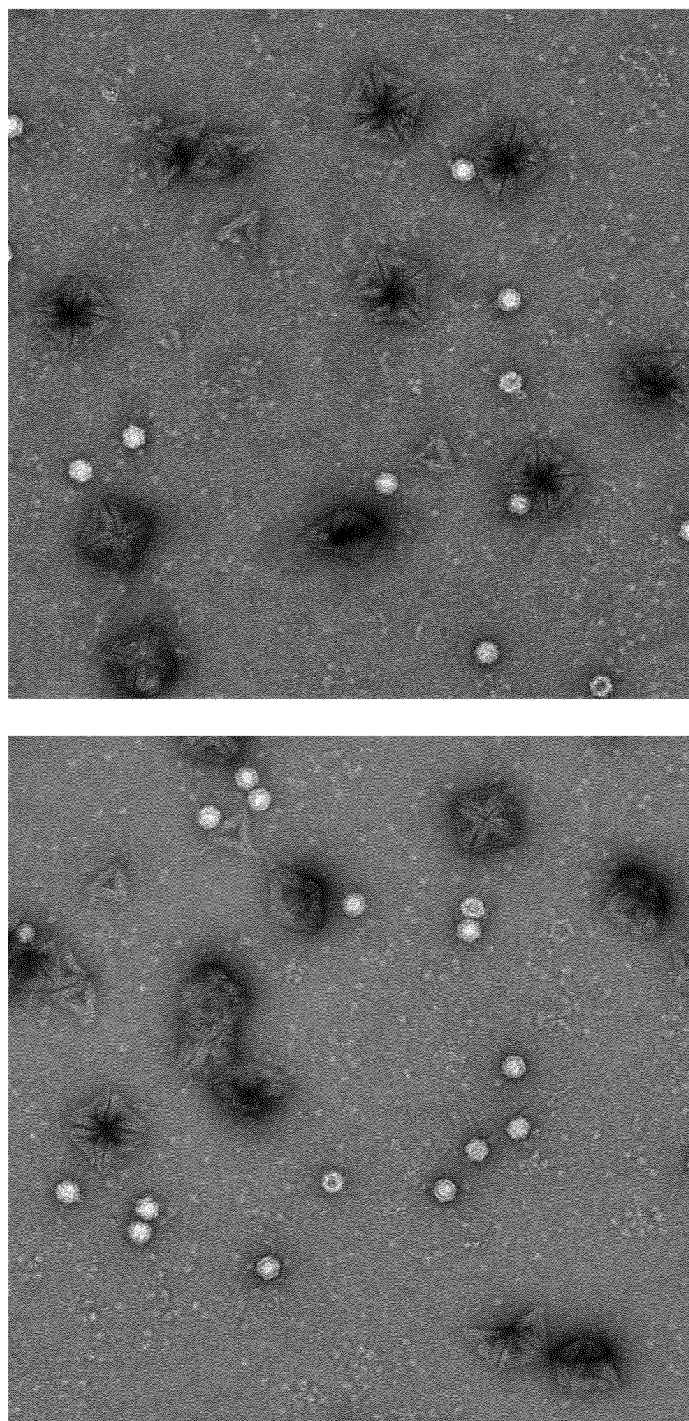


FIGURE 9:

100 nm

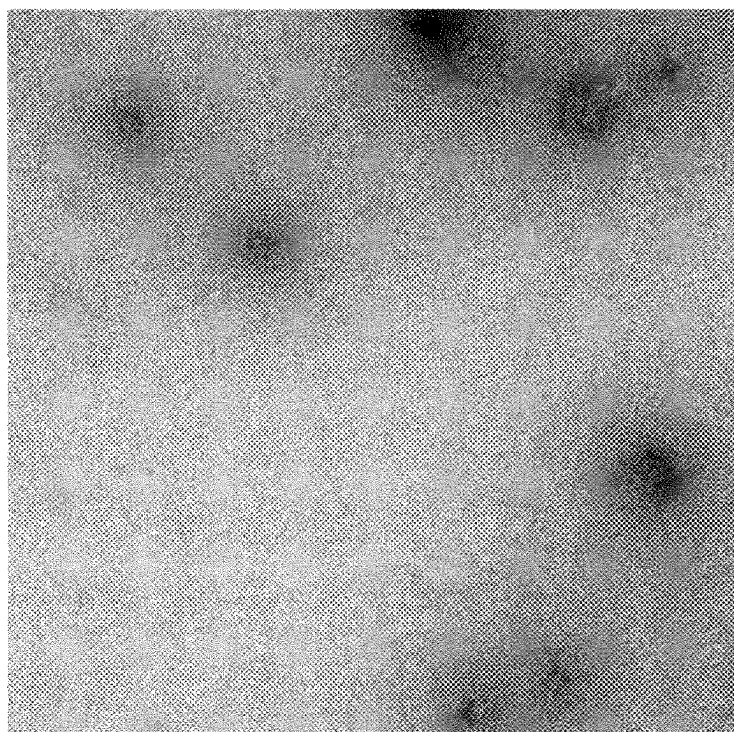
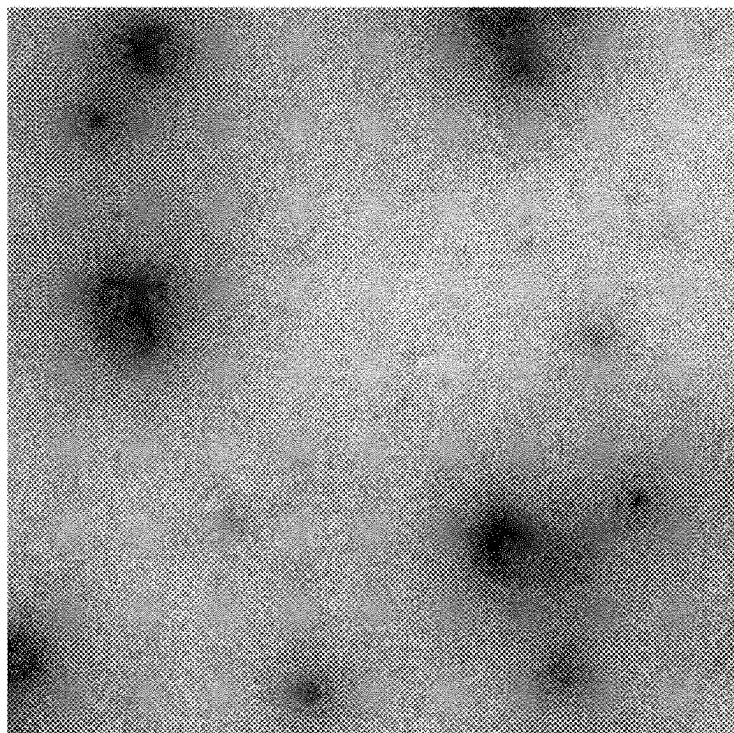


FIGURE 10:

FIGURE 11:

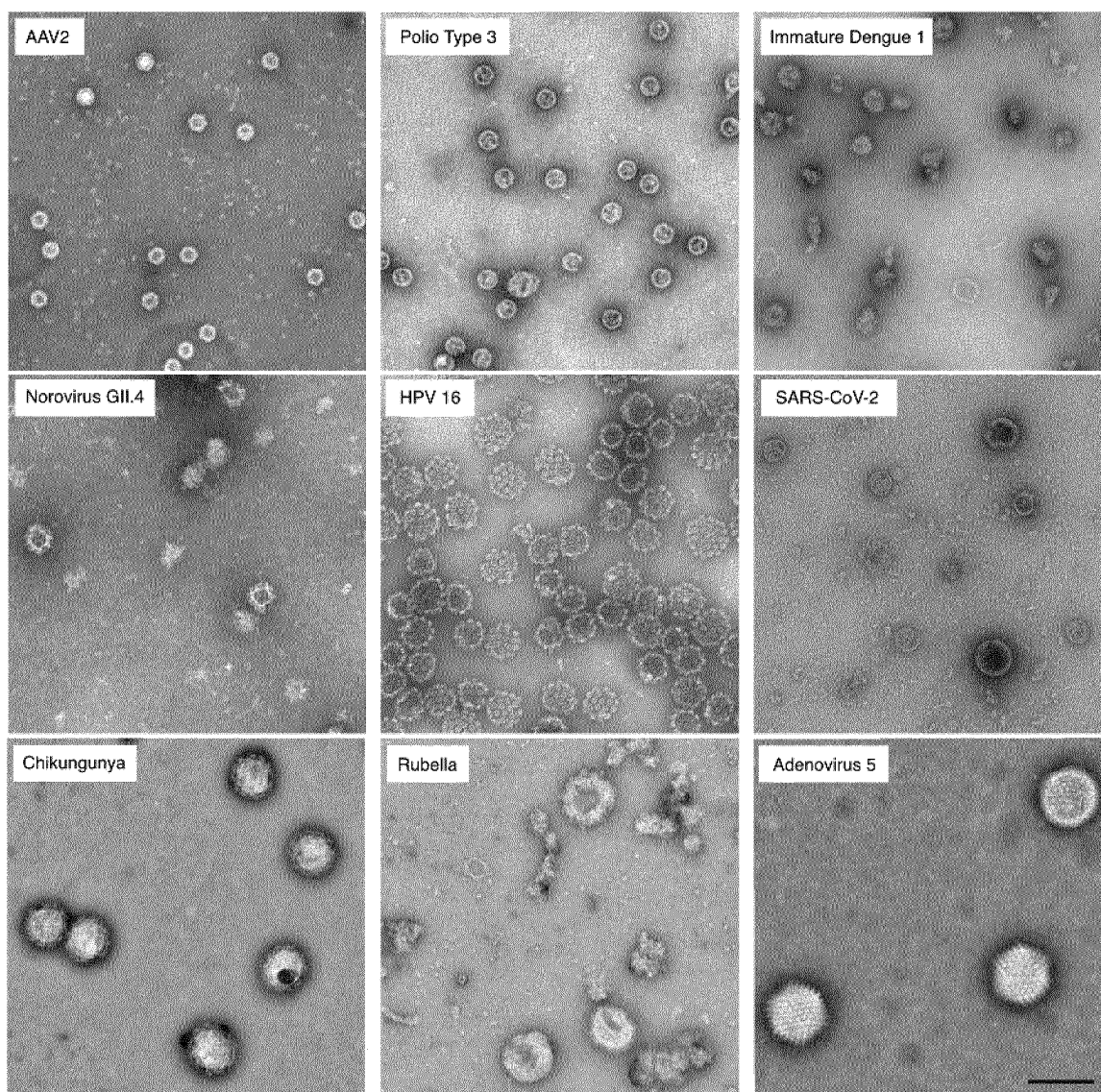


FIGURE 12:

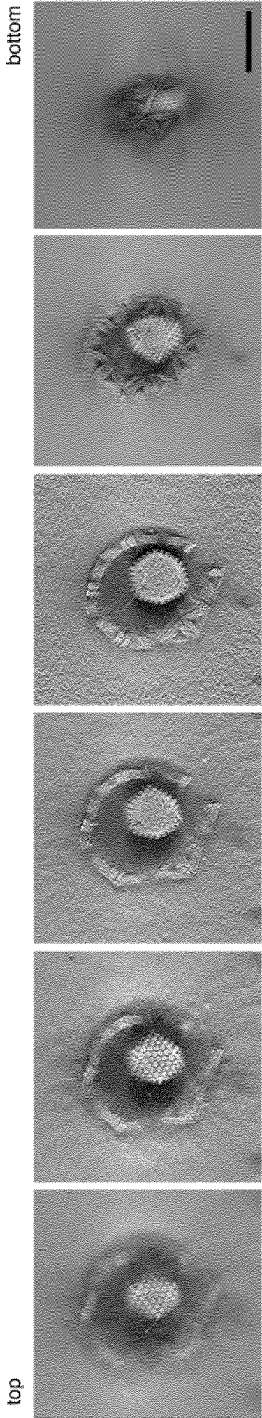


FIGURE 13:

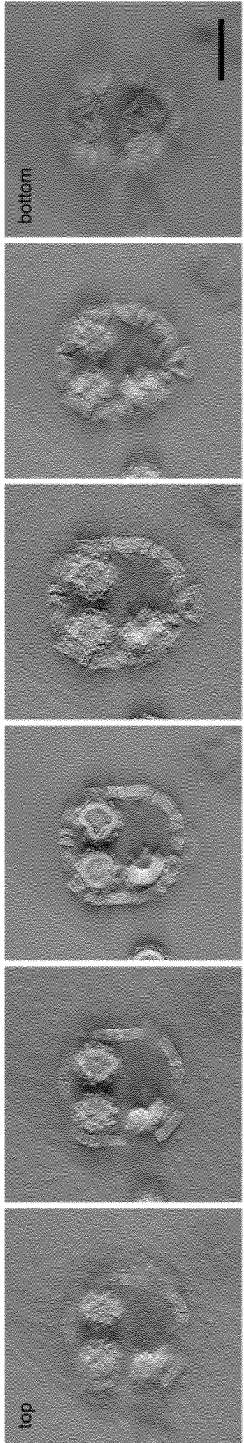
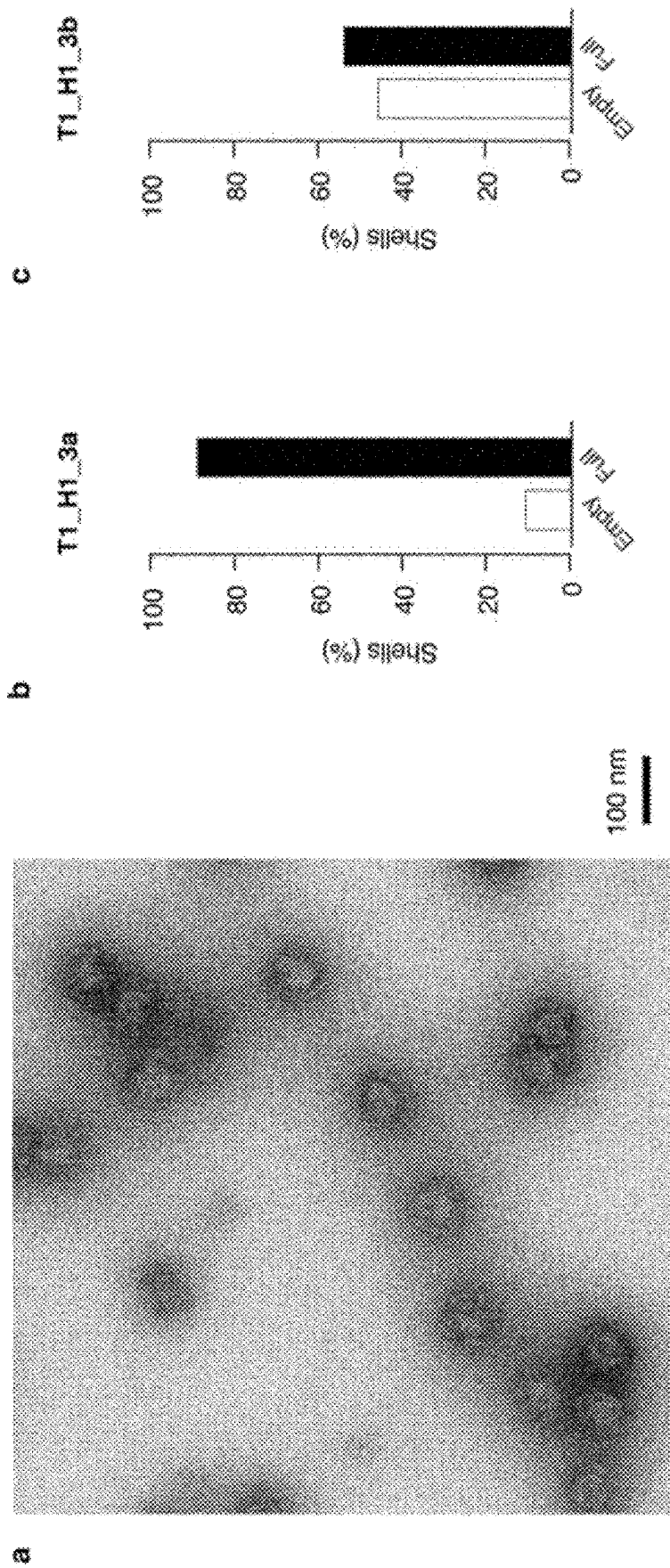


FIGURE 14:



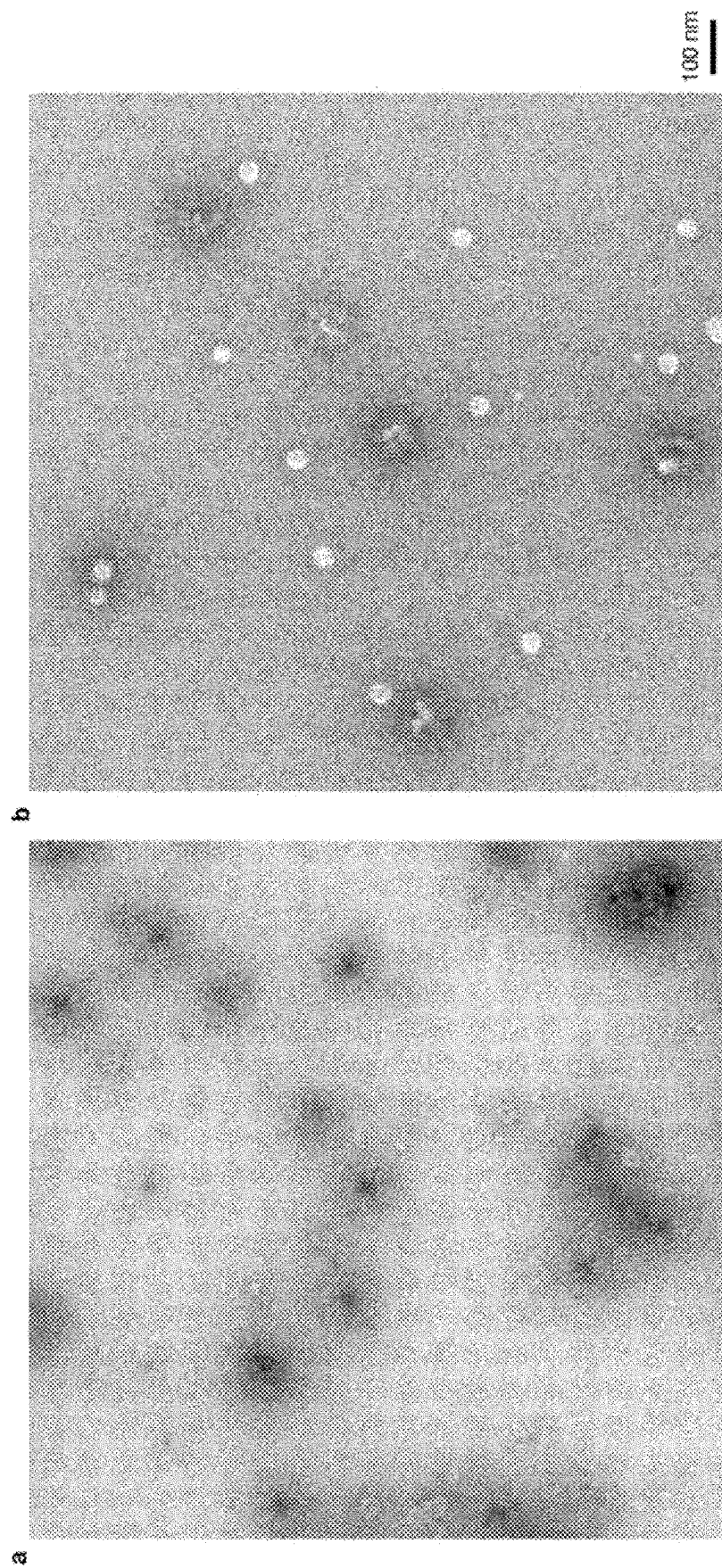


FIGURE 15:

FIGURE 16:

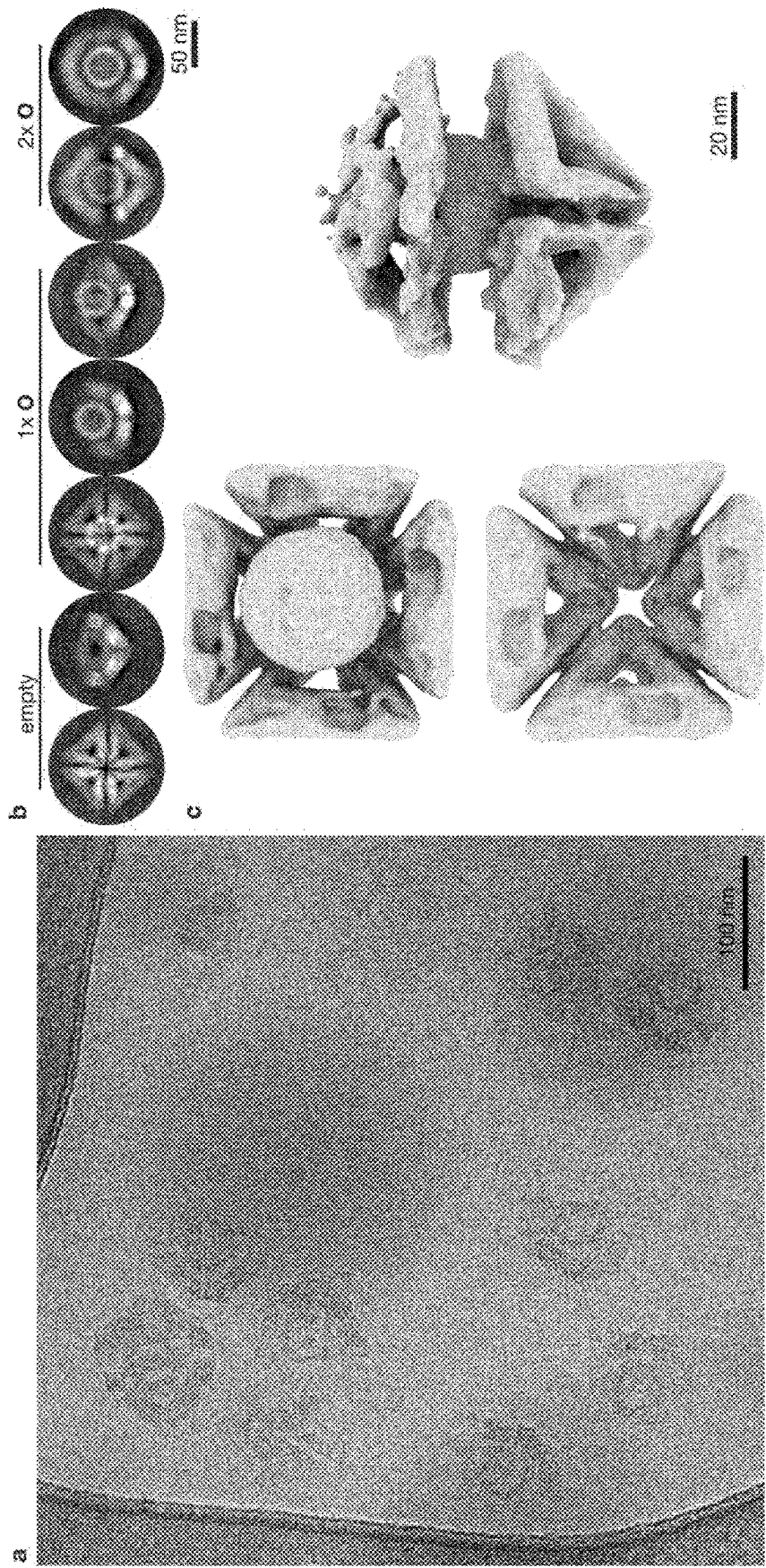


FIGURE 16 (contd.):

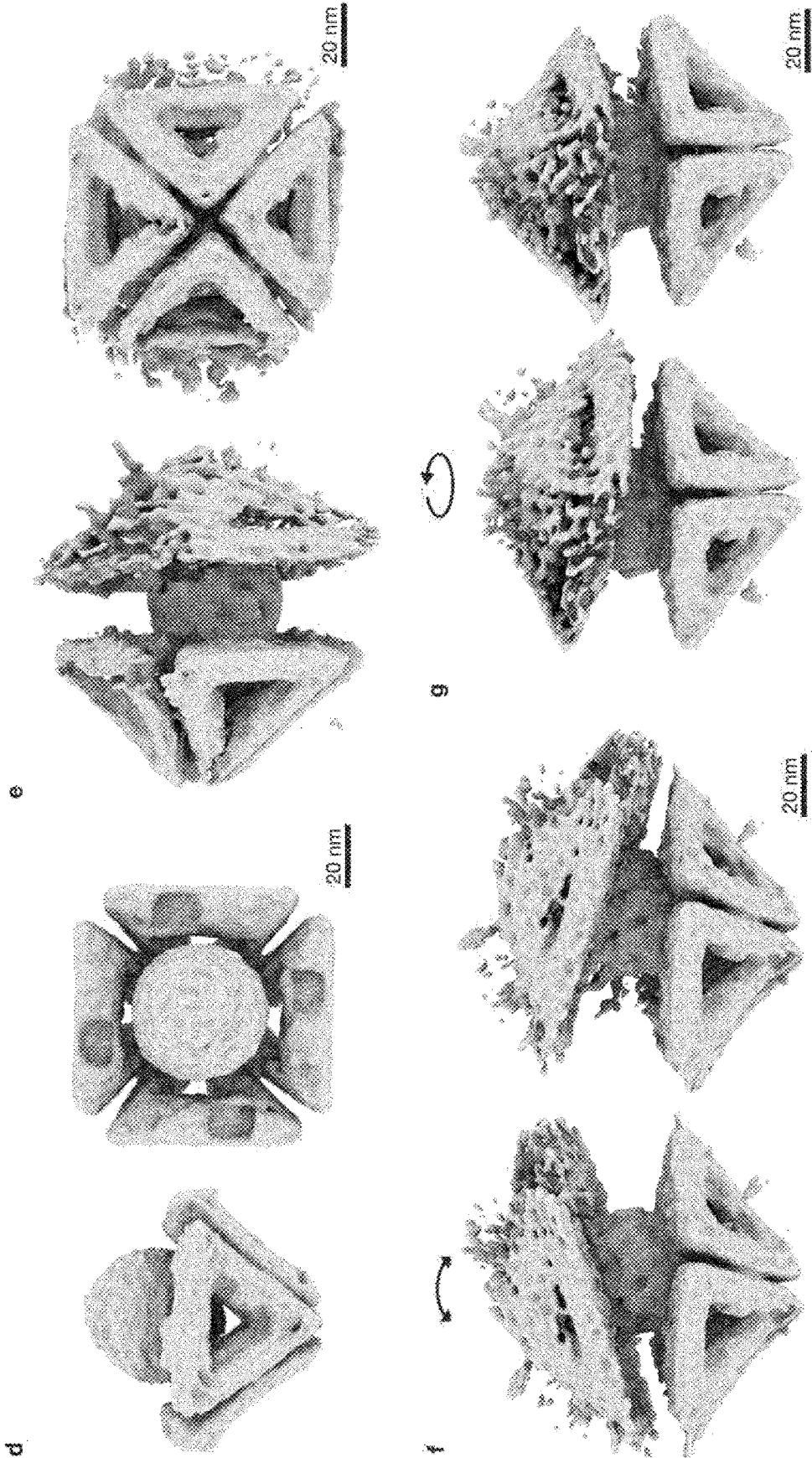


FIGURE 17:

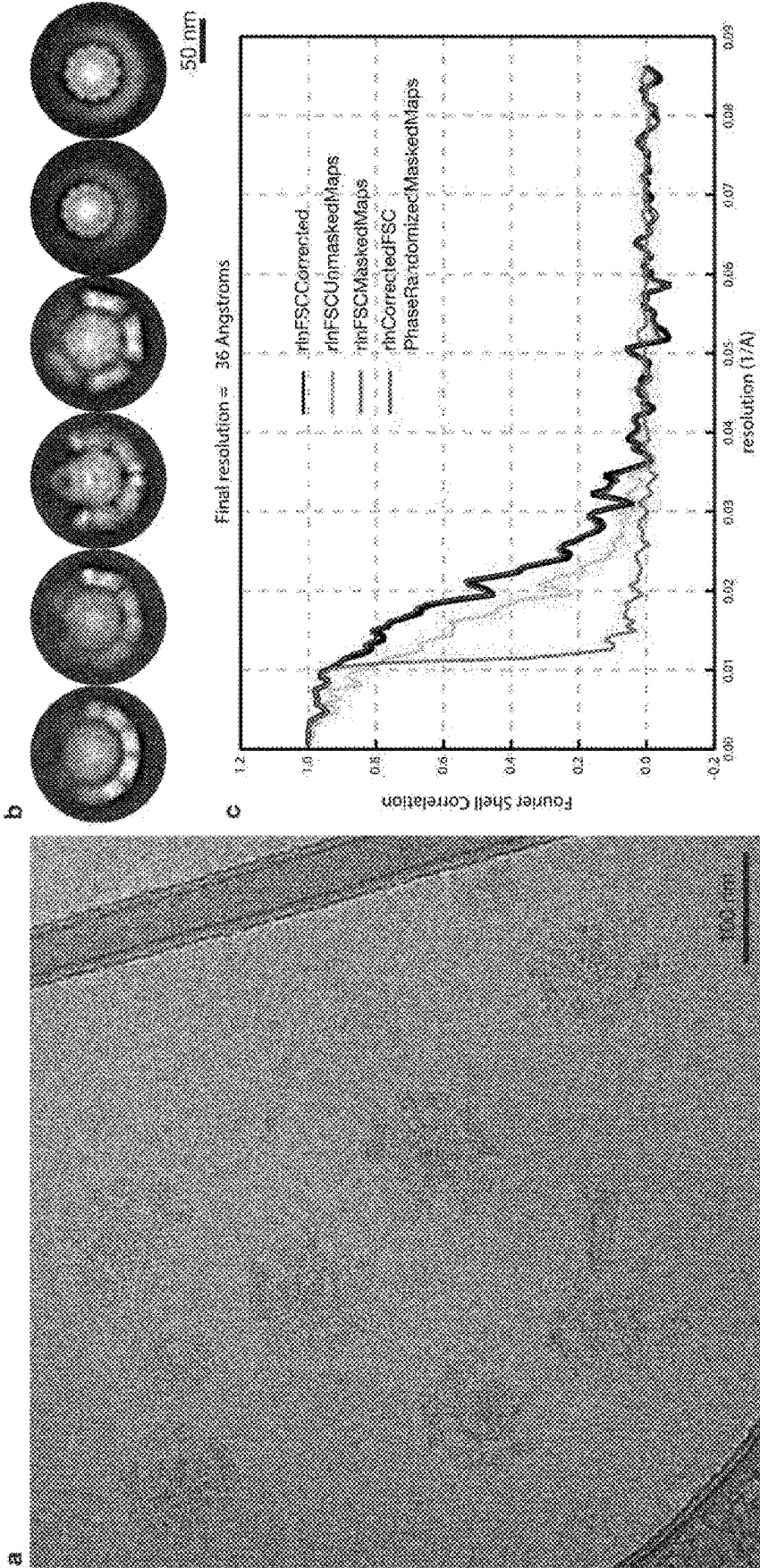


FIGURE 17 (contd.):

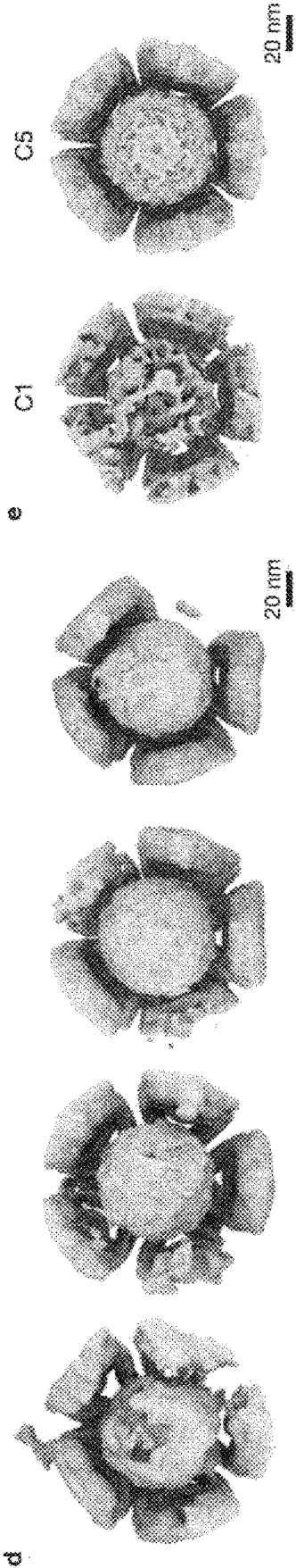


FIGURE 18:

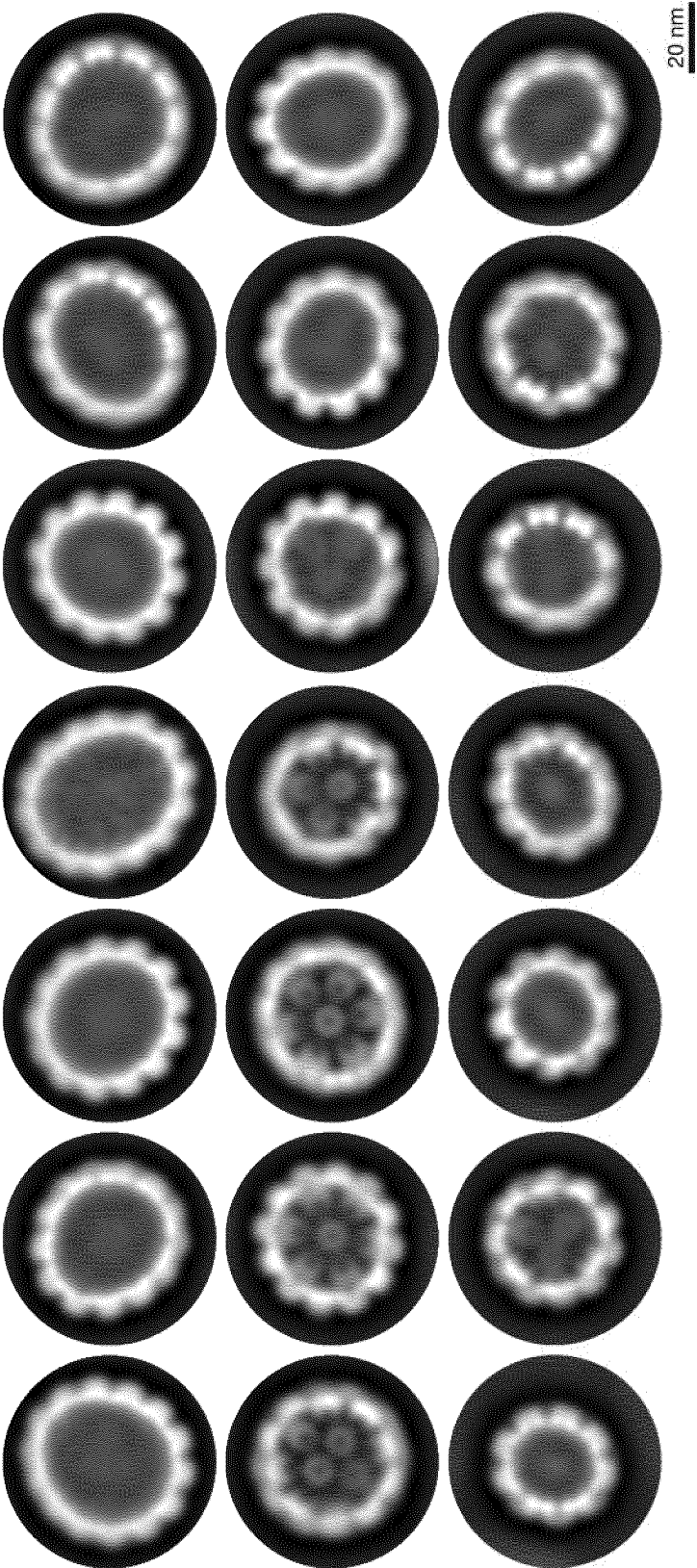


FIGURE 19:

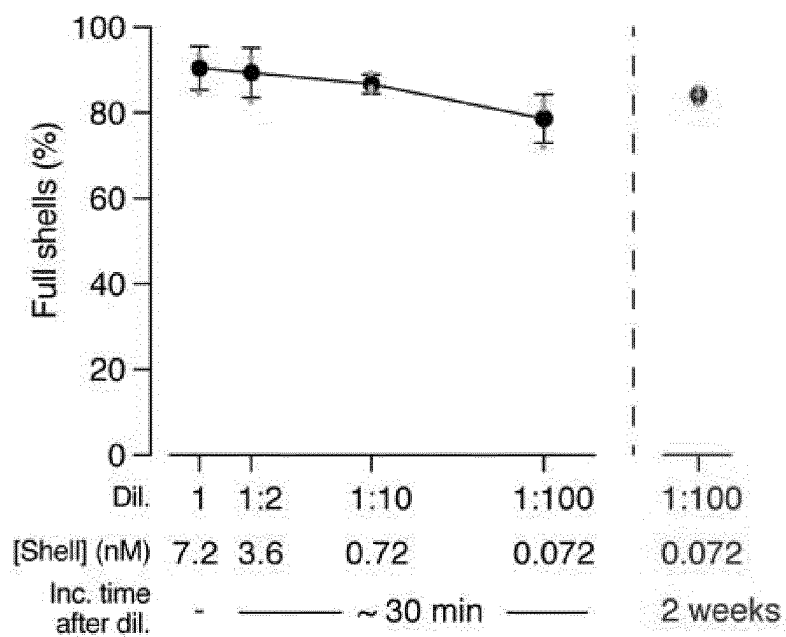


FIGURE 20:

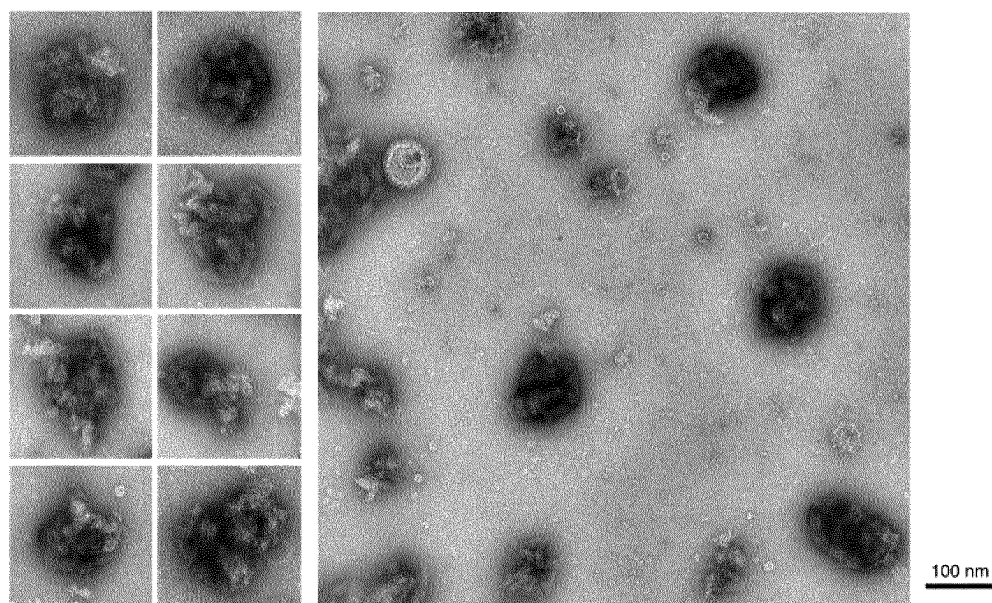


FIGURE 21:

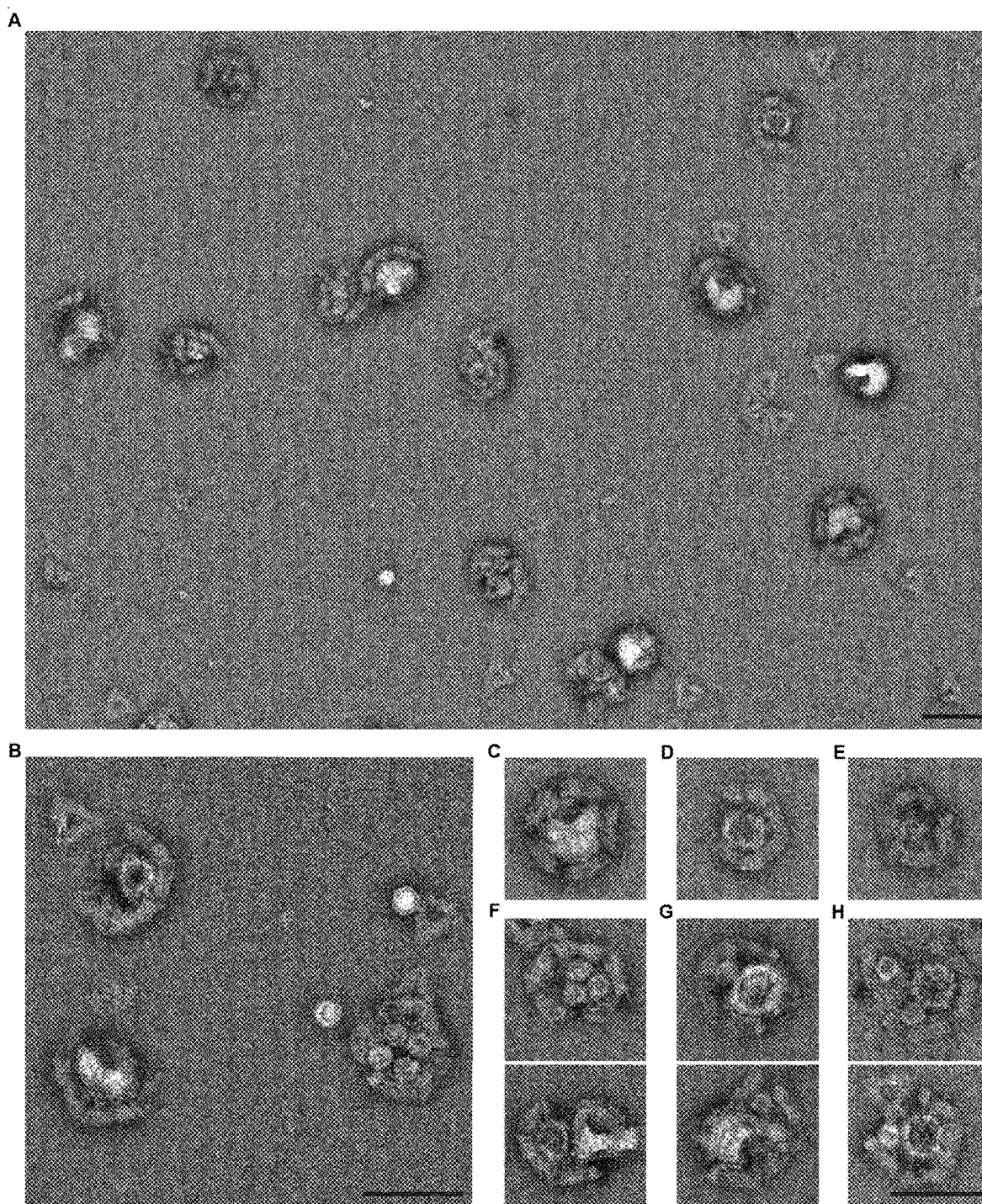
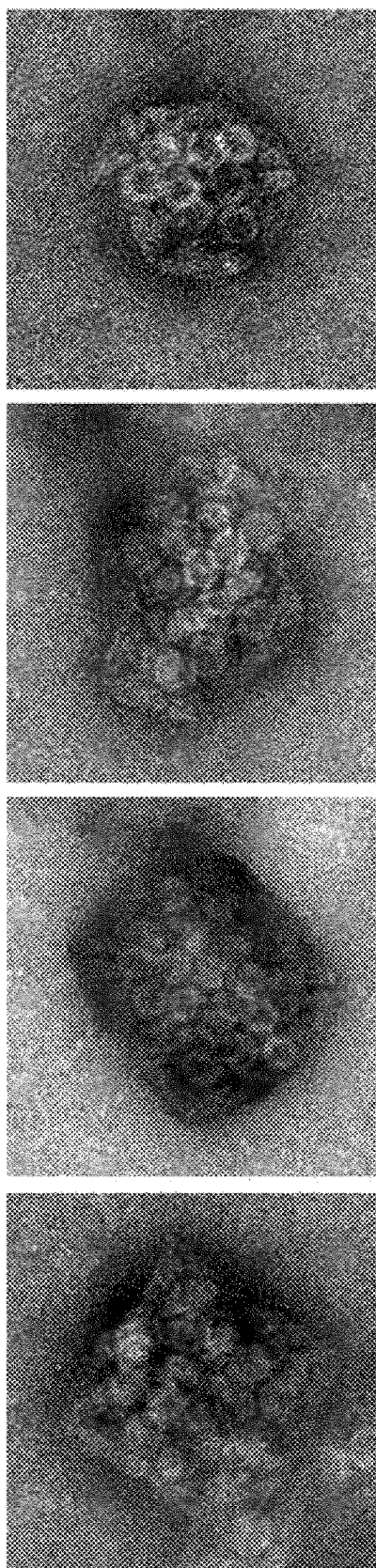


FIGURE 22:



BROAD SPECTRUM VIRUS-TRAPPING NANOSHELLS

RELATED APPLICATIONS

[0001] This application is the U.S. national phase of International Patent Application No. PCT/EP2023/061264, filed Apr. 28, 2023, which claims priority to European Patent Application No. 22170578.3, filed Apr. 28, 2022, which are hereby incorporated by reference in their entirety.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0002] This application includes, as a separate part of disclosure, a Sequence Listing in computer-readable form (Filename: SubSequenceListing.xml, 2,370,592 bytes; created Mar. 20, 2025). The contents of the Sequence Listing xml file are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0003] The present invention relates to a DNA-based nanostructure for encapsulating a broad spectrum of viruses or viral particles, to a composition comprising one or more viruses or viral particles encapsulated by such a DNA-based nanostructure according to the present invention, and to a method for encapsulating one or more viruses or viral particles by using such a DNA-based nanostructure.

BACKGROUND OF THE INVENTION

[0004] At present, there are over 200 known viral-vector borne human diseases, of which only nine are treatable with current antiviral drugs (Heida et al., *Drug Discov. Today* 26 (2021) 122-137). In the search for effective antiviral therapies, neutralizing antibodies are increasingly being considered for treating acute viral infections (see, for example, Wang et al., *Science*, 2021 Aug. 13; 373 (6556): eabh1766. doi: 10.1126/science.abh1766. Epub 2021 Jul. 1. PMID: 34210892; Taylor, P. C. et al., *Nat. Rev. Immunol.* 21 (2021) 382-393). Antiviral antibodies often derive their virus-neutralizing function by blocking the interactions that viruses undergo with specific receptors on the surface of host cells that are required for receptor-mediated cell invasion. However, antibodies are prone to losing their function due to mutational drift, take time to develop, and will only be effective for one virus or virus serotype at a time. Furthermore, antibodies or other proteinaceous virus binders may cause adverse immunogenic effects in organisms and create substantial additional fabrication hurdles and costs.

[0005] Recently, a new concept for neutralizing viruses by encapsulation in macromolecular shells fabricated with DNA origami has been presented (WO 2021/165528). The shells mechanically prevent interactions between trapped viruses and host cells. For the proof-of-concept experiments, the inside of the shell was coated with antibodies to sequester virus particles in the shells. Heparin is mentioned as a potential alternative binding moiety, but no data are shown. One key advantage of the shells is that the virus-binding moieties used in their interior themselves do not need to have a neutralizing function, since this task is performed by the shell material. Nonetheless, as stated above, the use of antibodies in the virus trapping shells presents several challenges that may limit the usefulness of the virus-trapping concept. WO 2021/165528 used up to 90 antibodies per virus-trapping shells, which were attached via interaction with single-stranded oligonucleotide handles with 16-mer or 26-mer overhangs for hybridization. WO 2021/165528 dem-

onstrates that virus particles can successfully be encapsulated in shells equipped with antibodies, but does not quantify the rate of encapsulations being achieved.

[0006] The concept laid out in WO 2021/165528 was described in a scientific publication as well (Sigl et al., *Nat. Mater.* 20 (2021) 1281-1289). Sigl et al. show the successful encapsulation of virus particles using antibody-equipped virus particles. Heparin is not mentioned as an alternative binding moiety, and no quantification of the rate of encapsulations is provided.

[0007] Knappe et al. (*ACS Nano* 2021; 14316-14322) describe DNA origami particles that are functionalized via click chemistry, so that different types of functional moieties, including antibodies and carbohydrates, can be coupled to the DNA origami particles. Heparin is not specifically mentioned in Knappe et al., and functionalization is performed at the outside of the DNA origami particles, since the DNA origami particles are closed shells. Thus, no information can be derived from Knappe et al. about the options for encapsulating virus particles in the interior of DNA origami shells and about any particularities of the optimal design of the attachment sites.

[0008] Another attractive avenue is the packaging of viral payload for the purposes of delivering viruses or other vectors for genetic information or other cargo to target cells or target tissue, as discussed e.g., in Antigen-Triggered Logic-Gating of DNA Nanodevices, Engelen et al., *J. Am. Chem. Soc.* 2021, 143, 51, 21630-21636, Dec. 20, 2021. Also here the use of antibodies as moieties to attach viruses or virus-like particles or other assemblies within DNA-based shells may limit the scope of such applications, due to the challenges listed above arising from the use of antibodies.

[0009] Thus, while different strategies for the treatment of viral infections have been developed or suggested up to date, there is still a need for the development of a concept of a generic antiviral drug platform for targeting a variety of viral pathogens. In particular, a concept would be highly desirable that does not rely on prior detailed knowledge about genetics and properties of the target virus. Additionally, there is an unmet need for the development of a system that permits the encapsulation of virus particles with high efficiency.

SUMMARY OF THE INVENTION

[0010] It is an object of the present invention to provide constructs that enable the encapsulation of one or more viruses or viral particles. The solution to that problem, i. e. the use of macromolecular building blocks, such as DNA-based nanostructures, has not yet been taught or suggested by the prior art.

[0011] Therefore, in one aspect, the present invention relates to a DNA-based nanostructure, wherein said DNA-based nanostructure is a shell comprising a cavity enclosed by said DNA-based nanostructure, wherein said DNA-based nanostructure is formed by self-assembling DNA-based building blocks, wherein each of said self-assembling DNA-based building blocks is formed by a single-stranded DNA template strand and a set of oligonucleotides complementary to said single-stranded DNA template, wherein each of said oligonucleotides is either complementary to one contiguous DNA sequence stretch or to at least two non-contiguous DNA sequence stretches on said single-stranded DNA template, wherein each of said self-assembling DNA-based building blocks is a triangular and/or a rectangular prismoid, particularly a triangular prismoid, and wherein a subset of one or more of said oligonucleotides in one or more of said self-assembling DNA-based building blocks is/are each linked to a construct comprising at least one sulfonated or

sulfated polysaccharide group pointing to the interior of said cavity, particularly a construct comprising one or two sulfonated or sulfated polysaccharide groups, wherein each said construct comprises (i) a handle comprising at least one binding site for said sulfonated or sulfated polysaccharide group, and (ii) said sulfonated or sulfated polysaccharide group(s) bound to said handle; wherein said handle has a length corresponding to at least the length of a single-stranded oligonucleotide comprising 30 nucleotides.

[0012] In a second aspect, the present invention relates to a composition comprising a DNA-based nanostructure according to the present invention encapsulating one or more viruses or viral particles.

[0013] In a third aspect, the present invention relates to a method for encapsulating one or more viruses or viral particles, comprising the steps of: providing a DNA-based nanostructure according to the present invention, and contacting said DNA-based nanostructure with a medium comprising, or suspected to comprise, said viruses or viral particles.

[0014] The disclosure contemplates all combinations of any one or more of the foregoing aspects and/or embodiments, as well as combinations with any one or more of the embodiments set forth in the detailed description and examples.

[0015] Other features, objects, and advantages of the compositions and methods herein will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 shows the DNA origami shells and functionalization with HS derivatives. A: A heparan sulfate proteoglycan (HSPG) interacts with a virus pathogen and mediates its cellular uptake (left). DNA origami shell schematics, with HS modifications in its interior, capable of binding and sequestering a viral particle (right). B: SPAAC reaction in between azide-modified HS oligomers and a DBCO-modified DNA oligo. The DNA sequence is complementary to the handles of the DNA origami shells in (e,f). C: PAGE characterization of the HS-modified DNA oligos. Products containing HS with sulfate and sulfonate groups (3a and 3c) migrate at a faster rate through the gel than the analog negative controls (3b and 3d) due to their increased anionic character. D: Cylindrical models of O and T1 shells made of 4 and 10 triangle subunits respectively, containing single stranded protruding oligos (termed handles, shown in red) decorating their interior. Each triangle subunit contains 9 handle positions. E: New T3 shell design consisting of 30 triangle subunits and featuring an inner cavity of 150 nm. Each triangle subunit also contains 9 handle positions. F: Negative stain TEM micrograph of T3 shells. Scale bar is 100 nm. G: Schematic representation of three different handle designs. H1 contains one HS modification per handle placed as close to the origami surface as possible. H2 also contains one HS modification per handle but has a polyT extension of 20 bases, allowing the handle to reach further than H1. H3 mimics a branched polymer containing two HS modifications per handle unit, therefore doubling the local HS density.

[0017] FIG. 2 shows viruses and VLPs trapped within HS-modified O, T1 and T3 shells. Negative stain TEM captions of a: AAV2, polio 3, mature dengue 1 and norovirus GII.4 successfully trapped in O shells; b: HPV 16, SARS-CoV-2, chikungunya and rubella engulfed by T1 shells; c: adenovirus 5 captured with T3 shells. Scale bars are 100 nm.

[0018] FIG. 3 shows multiple viruses and VLPs trapped in HS-modified O, T1 and T3 shells. Negative stain TEM

images of a: up to four AAV2 in one O shell, b: up to three HPV 16 in one T1 shell, c: one HPV 16 coordinated by two O shells for complete occlusion of the virus particle; d: up to six AAV2 per T1 shell; e: up to three chikungunya VLPs per T3 shell; f: cooperative effect of multiple O shells capturing numerous AAV2 particles. Scale bar is 100 nm.

[0019] FIG. 4 shows cryo-EM analysis of virus-like particles trapped in DNA origami shells. a: Cryo-EM micrograph of O shells binding to HPV 16 VLPs; b: 2D class average images of one or two O shells binding to one HPV 16 particle, demonstrating different orientations of the complexes. The white arrows indicate the gap difference in between the two O shells, confirming the capture of differently sized VLP particles; c: 3D reconstructions of HPV 16 bound to one and two O shells; d: Cryo-EM micrograph of T1 shells binding to chikungunya VLPs; e: 2D class average images of T1 shells binding to chikungunya particles showing different orientations of the complex; f: Two different views of the 3D reconstruction of a T1 shell engulfing a chikungunya virus particle.

[0020] FIG. 5 shows the SPAAC reaction for 8-mer HS derivatives (1a and 1b). a: Click chemistry reaction in between azide-modified HS polymers and a DBCO-modified DNA oligo. The DNA sequence is complementary to the handles of the DNA origami shells. b: PAGE characterization of all the HS-modified DNA oligos.

[0021] FIG. 6 shows T3 shell design. a: Top and front view of a T3 cylindrical model. b: Cylindrical models of the triangles t1-t6 involved in the T3 shell assembly. Arrows indicate complementary side interactions.

[0022] FIG. 7 shows TEM field of view of T3 shells. The T3 DNA origami shells presented an inner diameter of ~150 nm. Due to their flexibility, they appear deformed on the grid. Scale bar is 400 nm.

[0023] FIG. 8 shows TEM quantification of O shells for AAV2 trapping with the different handle designs. a: Schematic representation of the three different handle designs H1, H2 and H3. HS represented as red hexagons. H1 contains one HS modification per handle, placed closely to the origami surface. H2 also contains one HS modification per handle but has a polyT extension of 20 bases, allowing the handle to reach further than H1. H3 mimics a branched polymer containing two HS modifications per handle unit, therefore doubling the local HS density. b: Blind TEM quantification of full vs. empty O shells of each handle design when functionalized with 3c HS derivative and AAV2 excess. H1 presented ~20% of full shells, H2~84%, and H3~96%. c: Schematic representation of an O half-shell and its ssDNA handles in the inner cavity.

[0024] FIG. 9 shows TEM of negative control for AAV2 trapping in O shells. Two fields of view of the same sample showing that no binding was observed when the 3d negative control HS modification was hybridized to H3 handle design. Scale bar is 100 nm.

[0025] FIG. 10 shows TEM of AAV2 trapping with O shell excess. Two fields of view of the same sample showing that all AAV2 particles were encapsulated when the origami shell was used in excess. Scale bar is 100 nm.

[0026] FIG. 11 shows TEM of free viruses and VLPs. TEM data showed that AAV2, poliovirus, HPV 16, chikungunya and adenovirus 5 are the purest samples of our library. Dengue, norovirus, SARS-CoV-2 and rubella visibly contained a high amount of protein debris and presented a variable range of particle sizes. Scale bar is 100 nm.

[0027] FIG. 12 shows TEM tomography of adenovirus 5 in a T3 shell. The slices of the tomogram calculated from an

EM tilt series proved the full encapsulation of an adenovirus in the selected shell particle. Scale bar is 100 nm.

[0028] FIG. 13 shows TEM tomography of chikungunya VLPs in a T3 shell. The slices of the tomogram calculated from an EM tilt series proved the full encapsulation of three chikungunya VLPs in the selected shell particle. The last image slice showed a disruption of the triangles' connectivity. It was not clear if this discontinuity was due to the shell's rearrangement to encapsulate multiple VLPs or a consequence deforming on the grid during the sample preparation. Scale bar is 100 nm.

[0029] FIG. 14 shows TEM quantification of T1 shells trapping chikungunya VLPs with the 3a and 3b HS derivatives. a: negative stain TEM micrograph of T1 shells functionalized with the 3b negative control HS derivative. Due to the size and shape complementarity, weak electrostatic interactions in between the DNA and the chikungunya VLP were sufficient to keep the virus particles encapsulated when the negative control handles were used. Scale bar is 100 nm. b: TEM quantification of full vs. empty T1 shells functionalized with the 3a HS derivative on a H1 handle design. ~90% of shells were full c: TEM quantification of full vs. empty T1 shells functionalized with the 3b negative control HS derivative on a H1 handle design. ~54% of shells were full.

[0030] FIG. 15 shows TEM of immature and mature dengue 1 VLPs trapping with O shells. a: The immature configuration of the dengue VLPs showed no binding to the HS-modified origami shells. b: Mature dengue VLPs were recognized and encapsulated by the O shells. VLPs were used in excess. Scale bar is 100 nm.

[0031] FIG. 16 shows Cryo-EM imaging of O shells trapping HPV 16 particles (EMD-13884). a: Exemplary micrograph of O shells trapping HPV 16 vitrified on lacey carbon grids with ultrathin carbon support. b: 2D class averages of empty shells (left), HPV 16 trapped by one O shell (middle), and two O shells trapping an HPV 16 (right). c: 3D classes of selected particles showing similar particles as in b. d: 3D reconstruction of HPV 16 particles trapped in one O shell. e: Multibody refinement of HPV 16 particles encapsulated by two O shells. f+g: Multi-component analysis of two O shells trapping an HPV 16.

[0032] FIG. 17 shows Cryo-EM imaging of T1 shells trapping chikungunya VLPs (EMD-13883). a: Exemplary micrograph of T1 shells trapping chikungunya VLPs vitrified on lacey carbon grids with ultrathin carbon support. b: 2D class averages of extracted particles. c: FSC estimation of the reconstruction shown in e (C5). d: 3D classification of extracted particles. e: 3D reconstruction of T1 shells trapping chikungunya of selected particles from multiple rounds of 3D classification without (C1) and with symmetry (C5).

[0033] FIG. 18 shows 2D class averages of free HPV 16 VLPs extracted from cryo-EM images. The HPV 16 VLPs' diameter ranged from 35 nm to 50 nm.

[0034] FIG. 19 shows the stability of virus trapping by the shells. TEM quantification of AAV2 trapping in O shells subjected to a dilution series. The fraction of occupied shells remained the same prior and after 100-fold dilution and incubation for 14 days at RT in the diluted sample relative to the non-diluted sample. The overall shell concentration was 0.072 nM.

[0035] FIG. 20 shows TEM of rubella protein debris trapping in T1 shells. Successful encapsulation of protein debris from the rubella VLP sample.

[0036] FIG. 21 shows the results of negative stain TEM imaging of a virus cocktail trapping with heparan sulfate-modified T1 half-shells. (A,B) TEM fields of view of T1

half-shells trapping AAV2, Chikungunya and HPV16 virus particles in different ratios, with homogeneous and heterogeneous complexes. Selected trapped virus particles as (C) one Chikungunya, (D) one HPV16, (E) one AAV2, (F top) several AAV2, (F bottom) HPV16-Chik, (G) AAV2-Chik, (H) AAV2-HPV16. All scale bars: 100 nm.

[0037] FIG. 22 shows the cooperative effect of multiple O shells capturing numerous AAV2 particles. Scale bar: 50 nm.

DETAILED DESCRIPTION OF THE INVENTION

[0038] The present disclosure provides constructs that enable the encapsulation of viruses or viral particles.

[0039] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those of ordinary skill in the art to which this invention pertains.

[0040] The terms "comprising" and "including" are used herein in their open-ended and non-limiting sense unless otherwise noted. With respect to such latter embodiments, the term "comprising" thus includes the narrower term "consisting of".

[0041] The terms "a" and "an" and "the" and similar references in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. For example, the term "a cell" includes a plurality of cells, including mixtures thereof. Where the plural form is used for compounds, salts, and the like, this is taken to mean also a single compound, salt, or the like.

[0042] Therefore, in one aspect, the present invention relates to a DNA-based nanostructure, wherein said DNA-based nanostructure is a shell comprising a cavity enclosed by said DNA-based nanostructure, wherein said DNA-based nanostructure is formed by self-assembling DNA-based building blocks, wherein each of said self-assembling DNA-based building blocks is formed by a single-stranded DNA template strand and a set of oligonucleotides complementary to said single-stranded DNA template, wherein each of said oligonucleotides is either complementary to one contiguous DNA sequence stretch or to at least two non-contiguous DNA sequence stretches on said single-stranded DNA template, wherein each of said self-assembling DNA-based building blocks is a triangular and/or a rectangular prismoid, particularly a triangular prismoid, and wherein a subset of one or more of said oligonucleotides in one or more of said self-assembling DNA-based building blocks is/are each linked to a construct comprising at least one sulfonated or sulfated polysaccharide group pointing to the interior of said cavity, particularly a construct comprising one or two sulfonated or sulfated polysaccharide groups, wherein each said construct comprises (i) a handle comprising at least one binding site for said sulfonated or sulfated polysaccharide group, and (ii) said sulfonated or sulfated polysaccharide group(s) bound to said handle.

[0043] In an alternative embodiment, a subset of one or more of said oligonucleotides in one or more of said self-assembling DNA-based building blocks is linked to a construct comprising at least one sialic acid group pointing to the interior of said cavity, particularly a construct comprising one or two sialic acid groups.

[0044] In particular embodiments, said handle has a length corresponding to at least the length of a single-stranded oligonucleotide comprising 30 nucleotides.

[0045] By using such extended versions of the linking moieties, it could surprisingly be shown that the number of

DNA-based nanostructures that actually encapsulated a virus particle was drastically increased.

[0046] In the context of the present disclosure, the term “DNA-based nanostructure” refers to a nanostructure that is formed by a set of DNA-based macromolecules. DNA-based nanostructure of the type used in accordance with the present invention are described in detail in of WO 2021/165528 and in Sigl et al., loc. cit.

[0047] In the context of the present disclosure, the term “DNA” refers to deoxyribonucleic acid composed of a single-strand of monomeric units called nucleotides, wherein each nucleotide is composed of a nitrogen-containing nucleobase, a 2-deoxyribose sugar moiety, and a phosphate group, wherein the individual nucleotides are linked in the single-strand by a phosphate group linking the OH group in position 5' of a 2-deoxyribose sugar moiety to the OH group in 3' of a neighboring 2-deoxyribose sugar moiety. In particular embodiments, the nitrogen-containing nucleobases are independently selected from cytosine [C], guanine [G], adenine [A] and thymine [T]. In particular embodiments, one or more of the nucleobases are non-canonical bases, in particular a non-canonical base selected from the list of: a modified adenosine, in particular N6-carbamoylmethyladenine or N6-methyladenine; a modified guanine, in particular 7-deazaguanine or 7-methylguanine; a modified cytosine, N4-methylcytosine, 5-carboxylcytosine, 5-formylcytosine, 5-glycosylhydroxymethylcytosine, 5-hydroxycytosine, or 5-methylcytosine; a modified thymidine, in particular α -glutamyl thymidine or α -putrescinyll thymine; a uracil or a modification thereof, in particular uracil, base J, 5-dihydropentauracil; or 5-hydroxymethyldeoxyuracil; deoxyarchaeosine and 2,6-diaminopurine. A stretch of a single strand of DNA may interact with a complementary stretch of DNA by interaction of complementary nucleobases, wherein cytosine and guanine, and adenine and thymine, are complementary to each other, respectively by forming two (A/T) and three (G/C) hydrogen bonds between the nucleobases. Two single-strands of DNA may be fully complementary to each other, as in the case of genomic DNA, or may be partially complementary to each other, including situations, where one single-strand of DNA is partially complementary to two or more other single-stranded DNA strands. The interaction of two complementary single-stranded DNA sequences results in the formation of a double-stranded DNA double helix.

[0048] As is well known, DNA has evolved in nature as carrier of the genetic information encoding proteins. DNA further includes non-coding regions that include regions having regulatory functions. Thus, any DNA-based application usually critically depends on the specific DNA sequence and is almost always only enabled by naming the specific DNA sequence. In contrast, in the context of the present invention, such coding and/or regulatory functions do not play any role and may or may not be present, since the underlying DNA sequences are solely designed and selected in a way that the desired arrangement of double-helical subunits is formed.

[0049] Thus, in one embodiment any form of a long single-stranded DNA sequence, whether naturally occurring DNA (such as the DNA of a bacteriophage) or synthetically produced DNA may be selected as template, and a set of short single-stranded DNA sequences may be designed, wherein each sequence is complementary to one or more different parts of the template and thus forms one or more double-helical sections. Collectively, all such double-helical sections created by interaction of the full set of short single-stranded DNA sequences with the template, then

form the desired three-dimensional arrangement. Starting from a given single-stranded template sequence, the design of a set of complementary can be set up using known techniques, such as, for example, the methods described for the synthesis of megadalton-scale discrete objects with structurally well-defined 3D shapes (18, 24-35). In particular, iterative design with caDNAno (37) paired with elastic-network-guided molecular dynamics simulations (38) can be used.

[0050] In addition to the interaction of complementary nucleobases of different stretches of single-stranded DNA via hydrogen bonds, additional interactions between different DNA strands are possible, including the stacking interactions between the blunt ends of two double-stranded DNA helices (36), thus enabling the design and the formation of complex DNA-based nanostructures via the shape-complementarity of double-helical subunits. Thus, two three-dimensional arrangements formed in accordance with the previous paragraph, may interact with each other by stacking interactions between double-helical subunits present on the two three-dimensional arrangements, including specific interactions between two three-dimensional arrangements having complementary protrusions and recessions (or knobs and holes), as shown, for example, in FIGS. 7-13D of WO 2021/165528.

[0051] In an alternative aspect of the present invention, the invention relates to a macromolecule-based nanostructure, which is an RNA-based nanostructure.

[0052] In the context of the present disclosure, the term “RNA” refers to ribonucleic acid composed of a single-strand of monomeric units called nucleotides, wherein each nucleotide is composed of a nitrogen-containing nucleobase, a ribose sugar moiety, and a phosphate group, wherein the individual nucleotides are linked in the single-strand by a phosphate group linking the OH group in position 5' of a ribose sugar moiety to the OH group in 3' of a neighboring ribose sugar moiety. In particular embodiments, the nitrogen-containing nucleobases are independently selected from cytosine [C], guanine [G], adenine [A] and uracil [U]. In particular embodiments, one or more of the nucleobases are non-canonical bases, in particular a non-canonical base selected from the list of: pseudouridine, ribothymidine, and inosine. Unlike DNA, RNA is most often in a single-stranded form, but the formation of double-stranded forms is possible by interaction of complementary nucleobases, wherein cytosine and guanine, and adenine and uracil, are complementary to each other, respectively by forming two (A/U) and three (G/C) hydrogen bonds between the nucleobases.

[0053] In the context of the present invention, the term “cavity” relates to the space enclosed by said DNA-based nanostructure. In particular embodiments, said cavity resembles a sphere, where a spherical segment has been cut off, with the cutting plane being formed by the self-assembling DNA-based building blocks at the borders of said DNA-based nanostructure. In particular embodiments, the cutting plane is a great circle so that the DNA-based nanostructure is a half-shell. In other cases, where the three-dimensional geometric shape of said DNA-based nanostructure is derived from a spherocylinder or a polyhedron, in particular a tetrahedron, an octahedron or an icosahedron, said cavity is to be understood as the space resulting from cutting a corresponding spherocylinder or polyhedron by a plane with the cutting plane being formed by the self-assembling DNA-based building blocks at the borders of said DNA-based nanostructure.

[0054] In certain other embodiments, said DNA-based nanostructure resembles a spherical segment. Since in such embodiments, only part of a virus interacting with such DNA-based nanostructure is covered, the encapsulation of one or more viruses or viral particles in accordance with the present invention requires binding of two or more of such DNA-based nanostructures to said one or more viruses or viral particles.

[0055] In the context of the present application, the term “sulfonated or sulfated polysaccharide group” relates to a group comprising a polysaccharide comprising at least one sulfated hydroxy group or at least one sulfonated glycosylamino group.

[0056] Importantly, in addition to targeting specific receptors, many viruses also weakly interact with different biological substances, including sulfated or sulfonated polysaccharides (Cagno, V. et al., *Viruses* 11 (2019) 596; see Table 2).

[0057] In particular embodiments, each of said handles comprises two binding sites for said sulfonated or sulfated polysaccharide groups.

[0058] In particular embodiments, each of said polysaccharide comprising at least one sulfated hydroxy group or at least one sulfonated glycosylamino group is independently selected from the list of heparin, heparan sulfate, hybrid heparan sulfates, carrageenans, cellulose sulfate, and dextrin 2-sulfate, in particular a heparan sulfate or a hybrid heparan sulfate.

[0059] In particular embodiments, said polysaccharide comprising at least one sulfated hydroxy group or at least one sulfonated glycosylamino group consists of between 3 and 10 disaccharide units, in particular 4, 5, 6, 7, 8 of 9 units, particularly 4 or 9 monosaccharide units.

[0060] In particular embodiments, said disaccharide units comprise two or three O- and/or N-sulfonate groups per disaccharide unit, in particular three O- and/or N-sulfonate groups.

[0061] In particular embodiments, each of said sulfonated or sulfated polysaccharide groups is independently selected from the list of heparin, heparan sulfate, hybrid heparan sulfates, carrageenans, cellulose sulfate, and dextrin 2-sulfate.

[0062] In particular such embodiments, each of said sulfonated or sulfated polysaccharides is independently selected from a heparan sulfate and a hybrid heparan sulfate, in particular is a heparan sulfate.

[0063] In the context of the present invention, the terms “heparin” and “heparan sulfate” both relate to a family of linear sulfated, heterogeneous polysaccharides found on the cell membrane and in the extracellular matrix as part of heparan sulfate proteoglycans (HSPGs). They are composed of repeating 1→4 linked disaccharide units, in which one monosaccharide is an α -D-glucosamine residue and the other an uronic acid (or, in a salt form, an uronate). Heparin is a structurally similar polysaccharide found within mast cells as a component of serglycin proteoglycans. Heparan sulfate and heparin can be defined as follows: first, in heparin, the uronates are predominantly α -L-iduronate, whereas in heparan sulfate, the uronates are mainly, β -D-glucuronates, the C-5 epimers of α -L-iduronate. Second, in heparan sulfate, the D-glucosamine residues are predominantly N-acetylated, whereas in heparin, they are N-sulfonated. Finally, whereas at least 70-80% of heparin is composed of the disaccharide L-iduronate 2-O-sulfate a (1→4) D-glucosamine N,6-sulfate, in heparan sulfate around 40-60% of the disaccharides consist of (1→4) D-glucuronate β (1→4) D-glucosamine, that can be either

N-acetylated or N-sulfonated. Together, these structural characteristics make heparin more sulfated and, hence, more charged than heparan sulfate. It has become apparent, however, that the designations heparin or heparan sulfate are less clear-cut than this description implies, and that polysaccharides isolated from some organisms appear to be hybrid constructs. In the context of the present invention, the term “hybrid heparan sulfate” is used to refer to such hybrids having structures being a mixture of the “typical” heparin structural elements (L-iduronates; high degree of sulfonation) and the “typical” heparan sulfate structural elements (D-glucuronate; N-acetylation and 6-O-sulfonation).

[0064] Heparan sulfate proteoglycans (HSPG) (Cagno, V. et al., *Viruses* 11 (2019) 596; Zhang, Q. et al., *Cell Discov.* 6 (2020) 1-14) are commonly found on the surface of mammalian cells. The weak interactions of viruses with HSPG are conserved across virus families and thus appear generically beneficial for the virus lifecycle. For example, HSPG-virus interactions may enable an infection-enhancing diffusive search of virus particles for their specific host cell receptors on the surface of cells (FIG. 1A, left panel). The interactions of heparan sulfate (HS) with viruses have already been exploited for medical purposes, for example in virus-sequestering coatings of condoms that are based on HS-decorated dendrimers (Tyssen, D. et al., *PLOS ONE* 5, e12309 (2010); Price, C. F. et al., *PLOS ONE* 6, e24095 (2011); Zelikin, A. N. & Stellacci, F., *Adv. Healthc. Mater.* 10 (2021) 2001433). Other investigations have frequently involved the surface functionalization of nanoparticles and polymers with HS derivatives to create virus-binding complexes with antiviral activity (Cagno, V. et al., *Nat. Mater.* 17 (2018) 195-203; Al-Mahtab, M. et al., *PLOS ONE* 11 (2016) e0156667; Vaillant, A., *Antiviral Res.* 133 (2016) 32-40; Cagno, V. et al., *Antimicrob. Agents Chemother.* 64 (2020) e02001-20). Commonly, a high level of multivalency is required to increase the strength of binding between the HS-nanoparticles and viruses. The reversible nature of the binding can lead to undesirable unbinding and release of infectious viruses from the virus-sequestering coatings, or the requirement for high concentrations of the therapeutically active agent to be maintained (Zelikin, loc. cit.).

[0065] In particular embodiments, said subset of self-assembling DNA-based building blocks consists of between 1 and 100% of all self-assembling DNA-based building blocks forming said DNA-based nanostructure. In particular embodiments, said subset of self-assembling DNA-based building blocks consists of between 50 and 100% of all self-assembling DNA-based building blocks forming said DNA-based nanostructure, more particularly between 75 and 100%, and in particular 100% of all self-assembling DNA-based building blocks.

[0066] In particular embodiments, one or more of said self-assembling DNA-based building blocks in said subset comprise n single-stranded oligonucleotides as said handles, wherein each handle is independently linked to at least one of said sulfonated or sulfated polysaccharide groups, wherein n is an integer independently selected from 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12, particularly wherein n is 9.

[0067] In a particular embodiment, said handles are single-stranded oligonucleotides having a length of between 30 and 60 nucleotides, in particular between 40 and 55 nucleotides, more particularly between 45 and 50 nucleotides.

[0068] Examples of self-assembling DNA-building blocks in the form of frusta, wherein the small base of each of said frusta comprises nine of said polynucleotides are shown in the examples, for example T_{octa} self-assembling DNA-

based building blocks, T1_pentamer_triangle self-assembling DNA-based building blocks, T1_ring_triangle self-assembling DNA-based building blocks or T3_6_triangle based self-assembling DNA-based building blocks.

[0069] In particular embodiments, each member of said n oligonucleotides comprises at least one oligonucleotide stretch as binding site, and wherein each of said sulfonated or sulfated polysaccharide groups comprises an oligonucleotide having a sequence that is complementary to one of said oligonucleotide stretches comprised in said handles. In particular embodiments, each member of said n oligonucleotides comprises one or two oligonucleotide stretches as binding sites.

[0070] In particular embodiments, each member of said n oligonucleotides comprises two oligonucleotide stretches as binding sites.

[0071] By using oligonucleotides comprising two binding sites, it could surprisingly be shown that the number of DNA-based nanostructures that actually encapsulated a virus particle was further increased, even when compared to the already advantageous version with an extended handle design described above. This is particularly surprising in light of the fact that the additional second binding site is at the same position as the initial binding site in the H1 handle design with a short oligonucleotide (26-mer) and despite the fact that the addition of a second binding site and HS moiety might have been expected to decrease the encapsulation efficiency due to steric hindrance.

[0072] Examples of subsets of nine of such oligonucleotide handles, which can be linked to constructs comprising a sulfonated or sulfated polysaccharide group, can be found in the sets of Hx oligos according to SEQ ID. NOs: 177-185, 186-194, 389-397, 398-406, 607-615, 616-624, 821-829, 1018-1026, 1216-1224, 1416-1424, 1613-1621, and 1812-1820, each oligo comprising one handle, and the sets of Hx oligos according to SEQ ID. NOs: 195-203, 407-415, and 625-633, each oligo comprising two binding sites.

[0073] In a particular embodiment, said cavity has a diameter of at least 15 nm, at least 25 nm, at least 50 nm, at least 100 nm, at least 150 nm, at least 200 nm or at least 250 nm.

[0074] In particular embodiments, said cavity has a diameter of at most 1,000 nm.

[0075] In the context of the present invention, the term “diameter” refers to the diameter of the smallest circle that is encompassed by the surface of the DNA-based nanostructure. For the sake of clarity, in the case of a DNA-based nanostructure in the form of a capsule (or spherocylinder), the diameter is the diameter of the hemispherical ends and/or the diameter of the cylindrical central part.

[0076] In a particular embodiment, the DNA-based nanostructure has a molecular mass of at least 1 MDa, particularly at least 10 MDa, particularly at least 20 MDa, more particularly at least 30 MDa. In other particular embodiments, the DNA-based nanostructure has a molecular mass of at least 50 MDa, at least 80 MDa, at least 100 MDa, at least 200 MDa, or at least 500 MDa. In particular embodiments, the DNA-based nanostructure has a molecular mass of at most 1,500 MDa.

[0077] In particular embodiments, the ratio between the numerical value of the molecular mass of the DNA-based nanostructure (in MDa) and the numerical value of the volume of the cavity encased by said DNA-based nanostructure (in nm^3) is less than 10,000, particularly less than 9,000. In particular embodiments said ratio has a value of between 1,000 and 10,000, particularly between 2,000 and 9,000. For example, in the case of certain octahedral nanostructures,

where the molecular mass is about 40 MDa, and where the encased volume is about $113,000 \text{ nm}^3$, said ratio is about 2,800.

[0078] In particular embodiments, the ratio between the outer surface area of the DNA-based nanostructure covered by the macromolecules forming said DNA-based nanostructure and the outer surface area not covered by said macromolecules (excluding the area of the opening of a DNA-based nanostructure in the form of a shell) is at least 1, in particular at least 2, in particular at least 4, in particular at least 6, in particular at least 8. In other particular embodiments, the ratio is at least 10. In particular embodiments, the ratio is between 1 and 20, in particular between 2 and 18, between 4 and 16, between 6 and 14, and more particularly between 8 and 12. For example, in a case, where the DNA-based nanostructure is a shell in the form of a half sphere, only the area of the curved surface, but not that of the opening, i.e. the area of the flat face of the half sphere, is used for calculating said ratio.

[0079] In a particular embodiment, the molecular weight of each self-assembling DNA-based building block is between 4.5 and 5.5 MDa.

[0080] In a particular embodiment, each self-assembling DNA-based building block comprises between 7,500 and 8,500 base pairs.

[0081] In a particular embodiment, the DNA-based nanostructure consists of between 4 and 180 of such self-assembling DNA-based building blocks.

[0082] In particular embodiments, said single-stranded DNA template is single-stranded DNA of filamentous bacteriophage, or is derived from single-stranded DNA of filamentous bacteriophage.

[0083] In the context of the present invention, the term “filamentous bacteriophage” refers to a type of bacteriophage, or virus of bacteria, which is characterized by its filament-like shape that usually contains a genome of circular single-stranded DNA and infects Gram-negative bacteria. Filamentous phage includes Ff phage, such as M13, f1 and fd1 phage, and Pfl phage.

[0084] In particular embodiments, said single-stranded DNA template has the sequence of SEQ ID NO: 1 (M13 8064) (see Table 1). In particular embodiments, said single-stranded DNA is circular.

[0085] In the context of the present invention, a single-stranded DNA template that is “derived from single-stranded DNA of filamentous bacteriophage” refers to a DNA construct that is derived from a naturally occurring of published DNA sequence of a filamentous bacteriophage by one or more of: (i) opening of the circular structure to a linear sequence; (ii) deletion of one or more nucleotides; (iii) insertion of one or more nucleotides; (iii) substitution of one or more nucleotides; (iv) addition of one or more nucleotides; and (v) modification of one or more nucleotides. While any such variation might have detrimental, or at least rather unpredictable, effects on bacteriophage biology, its infectivity and its ability to propagate, such effects do not play any role in the context of the present invention, since, as already mentioned above, said single-stranded DNA template is only used as naked template without any requirement for having any functional property, and all structural aspects, such as the correct formation the three-dimensional shape of said self-assembling DNA-based building blocks, are implemented by the proper choice of said set of complementary oligonucleotides.

[0086] In particular embodiments, said single-stranded DNA template has at least 80%, particularly at least 90%, more particularly at least 95%, sequence identity to the

sequence of a naturally occurring or published sequence of a filamentous bacteriophage, in particular to a M13, f1 or fd1 phage, in particular to a sequence selected from SEQ ID NO: 1 (M13 8064) and M13 7249 (SEQ ID NO: 2 of WO 2021/165528).

[0087] In a particular embodiment, the DNA-based nanostructure is a closed three-dimensional geometric shape, in particular a closed three-dimensional geometric shape selected from a sphere, a spherocylinder, and a polyhedron, in particular a tetrahedron, an octahedron or an icosahedron, formed in situ from said self-assembling DNA-based building blocks in the presence of said viruses or viral particles to be encapsulated.

[0088] In a particular such embodiments, said self-assembling DNA-based building blocks forming said DNA-based nanostructure in situ are a collection of individual DNA-based building blocks, each comprising a single single-stranded DNA template.

[0089] In particular other embodiments, said self-assembling DNA-based building blocks forming said DNA-based nanostructure in situ are a collection of one or more preassembled DNA-based building blocks consisting of two or more individual DNA-based building blocks, each comprising a single single-stranded DNA template.

[0090] In certain embodiments, all of said self-assembling DNA-based building blocks are preassembled DNA-based building blocks. In an alternative embodiment, said self-assembling DNA-based building blocks are a mixture of preassembled DNA-based building blocks and of individual DNA-based building blocks, each comprising a single single-stranded DNA template.

[0091] In such embodiments, said preassembled DNA-based building blocks form a curved geometrical shape, wherein said handles or said constructs comprising at least one sulfonated or sulfated polysaccharide group, which are linked to said handles, are present on the negative curvature of said curved geometrical shape, so that said handles or constructs are pointing to the interior of the cavity formed from the self-assembly of said preassembled DNA-based building blocks.

[0092] In another particular embodiment, the DNA-based nanostructure is a shell with an opening for accessing said cavity.

[0093] In the context of the present invention, the term “shell” refers to a structure that is a part of a closed three-dimensional geometric shape, in particular a closed three-dimensional geometric shape selected from a sphere, a spherocylinder, and a polyhedron, in particular a tetrahedron or an octahedron.

[0094] In yet another particular embodiment, the DNA-based nanostructure is a combination of a first subshell and a second subshell, each with an opening to access a first and a second inner cavity, respectively, wherein said first and said second inner cavity together form said cavity.

[0095] In a particular embodiment, said first and said second subshells are connected by at least one linker.

[0096] In particular embodiments, said linker is a linker selected from a DNA linker, an RNA linker, a polypeptide linker, a protein linker and a chemical linker.

[0097] In the context of the present invention, the term “DNA linker” refers to a linker formed from DNA, wherein the sequence of said DNA linker is not complementary to the DNA of said single-stranded DNA template or to any of said set of oligonucleotides complementary to said single-stranded DNA template, wherein said DNA linker is linked at one terminus to a DNA sequence forming a self-assembling DNA-based building block of said first shell, and at the

other terminus to a DNA sequence forming a self-assembling DNA-based building block of said second shell.

[0098] In the context of the present invention, the term “polypeptide linker” refers to a linker formed from at least 2, particularly at least 5, at least 10, or at least 20 amino acid residues linked by peptide bonds, wherein said polypeptide has no tertiary or quaternary structure, and wherein said polypeptide linker is linked at one terminus to a DNA sequence forming a self-assembling DNA-based building block of said first shell, and at the other terminus to a DNA sequence forming a self-assembling DNA-based building block of said second shell.

[0099] In the context of the present invention, the term “protein linker” refers to a linker formed from at least 20, particularly at least 50, at least 100, at least 200 amino acid residues, at least 500 amino acid residues, or at least 1,000 amino acid residues, particularly less than 1,500 amino acid residues linked by peptide bonds, wherein said polypeptide has tertiary and/or quaternary structure, and wherein said protein linker is linked at one terminus to a DNA sequence forming a self-assembling DNA-based building block of said first shell, and at the other terminus to a DNA sequence forming a self-assembling DNA-based building block of said second shell. In particular embodiments, said protein linker is covalently attached to said DNA sequences. In particular other embodiments, said protein linker is non-covalently attached to said DNA sequences, in particular, wherein said protein linker is an antibody-based protein linker, in particular selected from a diabody and a full antibody, including an IgG antibody.

[0100] In the context of the present invention, the term “chemical linker” refers to a continuous chain of between 1 and 30 atoms (e.g. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 atoms; thus, in the context of the present invention, the term “between” is used so that the borders mentioned are included) in its backbone, i.e. the length of the linker is defined as the shortest connection as measured by the number of atoms or bonds between the two DNA sequences linked by said chemical linker. In the context of the present invention, a chemical linker preferably is an C_{1-20} -alkylene, C_{1-20} -heteroalkylene, C_{2-20} -alkenylene, C_{2-20} -heteroalkenylene, C_{2-20} -alkynylene, C_{2-20} -heteroalkynylene, cycloalkylene, heterocycloalkylene, arylene, heteroarylene, aralkylene, or a heteroaralkylene group, which may optionally be substituted. The linker may contain one or more structural elements such as carboxamide, ester, ether, thioether, disulfide, urea, thiourea, hydrocarbon moieties and the like. The linker may also contain combinations of two or more of these structural elements. Each one of these structural elements may be present in the linker more than once, e.g. twice, three times, four times, five times, or six times. In some embodiments the linker may comprise a disulfide bond. It is understood that the linker has to be attached either in a single step or in two or more subsequent steps to the two DNA sequences linked by said chemical linker. To that end the linker to be will carry two groups, preferably at a proximal and distal end, which can (i) form a covalent bond to a group present in one of the two DNA sequences to be linked, or (ii) which is or can be activated to form a covalent bond with one of the two DNA sequences.

[0101] In a particular embodiment, the DNA-based nanostructure is based on an icosahedral structure.

[0102] In a particular embodiment, each of said self-assembling DNA-based building blocks is a prismoid.

[0103] In the context of the present invention, the term “prismoid” refers to a polyhedron, wherein all vertices lie in two parallel planes.

[0104] In particular embodiments, said prismoid is a triangular prismoid. In other embodiments, said prismoid is a rectangular prismoid.

[0105] In particular embodiments, the DNA-based nanostructure is based on a mixture of a triangular and a rectangular prismoid.

[0106] In a particular embodiment, the present invention relates to a DNA-based nanostructure, wherein each said triangular, or said rectangular prismoid, is formed by m triangular, or rectangular, respectively, planes, wherein m is an integer independently selected from 4, 5, 6, 7 and 8, in particular independently selected from 5, 6 and 7, more particularly wherein said integer is 6.

[0107] wherein the three, or four, respectively, edges of each of said m planes are formed by n parallel stretches of DNA double helices, wherein n is an integer independently selected from 1, 2, 3, 4, 5 and 6 in particular independently selected from 2, 3, 4 and 5, more particularly independently selected from 3 and 4,

[0108] wherein each plane is connected to a plane above and/or a plane beyond said plane (i) by stacking interactions between the DNA double helices forming said planes, and (ii) partially by DNA stretches within said single-stranded DNA template and/or said oligonucleotides forming said DNA-based building block bridging at least two of said planes, and

[0109] wherein at least two of the three, or four, respectively, side trapezoids comprise a specific pattern of recesses and/or extrusions formed by missing or additional DNA double helical stretches for specific interaction with a complementary pattern on the side trapezoid of another one of said self-assembling DNA-based building blocks.

[0110] In a particular embodiment, the average length of each of the n stretches of DNA double helices in the m planes of a triangular, or rectangular, respectively, prismoid is between 80 and 200 base pairs.

[0111] In particular embodiments, said triangular prismoid is a triangular frustum. In particular embodiments, said rectangular prismoid is a rectangular frustum.

[0112] In the context of the present invention, the term “triangular frustum” refers to a three-dimensional geometric shape in the form of a triangular pyramid, and the term “rectangular frustum” refers to a three-dimensional geometric shape in the form of a rectangular pyramid, where the tip of the pyramid has been removed resulting in a plane on the top parallel to the basis of the pyramid.

[0113] In a particular embodiment, for at least part of said self-assembling DNA-based building blocks the length of at least one edge of each of said m planes is decreasing from the first to the m^{th} plane, so that a bevel angle results between planes perpendicular to said first plane and the trapezoid plane formed by said m edges (see FIG. 5 of WO 2021/165528). In particular embodiments, all three, or four, respectively, trapezoid planes exhibit a bevel angle.

[0114] In a particular embodiment, a bevel angle is between 16° and 26° , particularly between 18° and 24° , more particularly between 20° and 22° , most particularly about 20.9° .

[0115] In a particular embodiment, said DNA-based nanostructure comprises at least one set of self-assembling DNA-based building blocks, wherein all three, or four, respectively, side trapezoids comprise a specific pattern of recesses and/or extrusions formed by missing or additional DNA

double helical stretches for specific interaction with a complementary pattern on the side trapezoid of another one of said self-assembling DNA-based building blocks.

[0116] In a particular embodiment, particularly in the case of a DNA-based nanostructure closed three-dimensional geometric shape, all said self-assembling DNA-based building blocks are identical.

[0117] In a particular embodiment, said DNA-based nanostructure comprises two or more sets of self-assembling DNA-based building blocks.

[0118] In a particular embodiment, said DNA-based nanostructure is rod-shaped.

[0119] In particular embodiments, said DNA-based nanostructure comprises two or more sets of self-assembling DNA-based building blocks.

[0120] In particular such embodiment, said rod-shaped DNA-based nanostructure comprises at least a first and a second set of self-assembling DNA-based building blocks, wherein said first and set second set differ at least with respect to the bevel angles. In a particular embodiment, at least one set consists of self-assembling DNA-based building blocks exhibiting only two bevel angles. In a particular embodiment, said at least one set consists of rectangular frusta, which comprise a bevel angle on each of two opposing trapezoids.

[0121] In a particular embodiment, the side trapezoids forming the rim of said shell, or of said first and second shell, respectively, do not comprise a specific pattern of recesses and/or extrusions formed by missing or additional DNA double helical stretches for specific interaction with a complementary pattern on the side trapezoid of another one of said self-assembling DNA-based building blocks.

[0122] In a particular embodiment, said DNA-based nanostructure is a shell selected from

[0123] (i) a half octahedron T_{octa} (FIG. 4A of WO 2021/165528), which consists of a set of four copies of a triangular frustum, wherein the base-pair stacking contacts on one of the triangular edges of the triangular frustum are inactivated by either strand shortening or by adding unpaired thymidines (FIG. 4A of WO 2021/165528, see FIG. 24A,D of WO 2021/165528);

[0124] (ii) a half $T=1$ shell (FIG. 4B of WO 2021/165528), which consists of two sets of in each case five copies of two different triangular frusta, wherein the five copies of the first set form a closed pentamer, and the five copies of the second set dock onto the edges of said pentamer (FIG. 4B of WO 2021/165528, see FIG. 24B,E of WO 2021/165528); and

[0125] (iii) a “trap” $T=1$ shell with a missing pentagon vertex (FIG. 4C of WO 2021/165528), which consists of three sets of in each case five copies of three different triangular frusta, wherein the five copies of the first set form a closed pentamer, the five copies of the second set dock onto the edges of said pentamer the five copies of the second set dock onto the edges of said pentamer, and the five copies of the third set dock into the gaps between the five copies of said second set (FIG. 4C of WO 2021/165528, see FIG. 24C,F of WO 2021/165528);

[0126] (iv) a $T=3$ icosahedral half shell, which consists of a total of 30 triangular subunits partitioned as five copies of six different full-size DNA triangle designs with specific edge docking rules

[0127] In a particular embodiment, the present invention relates to a DNA-based nanostructure further comprising one or more types of DNA brick constructs, each type of such DNA brick constructs being characterized by one or

more interaction sites for specific interaction by edge-to-edge stacking contacts with one or more complementary interaction sites present on the plane of said triangular, or rectangular, respectively, prismoid on the outer surface of said DNA-based nanostructure, wherein said DNA brick constructs cover the free space between the three, or four, respectively, edges of said plane (see FIG. 33 of WO 2021/165528).

[0128] In a particular embodiment, the present invention relates to a DNA-based nanostructure further comprising one or more cross-linkages within one of said triangular, or rectangular, respectively, prismoids, and/or between two of said triangular, or rectangular, respectively, prismoids.

[0129] In the context of the present invention, the term “cross-linkage” refers to any permanent or intermittent linkage within one of said triangular, or rectangular, respectively, prismoids, and/or between two of said triangular, or rectangular, respectively, prismoids. Any such linkage may be achieved a priori by linking two of the oligonucleotides being used for forming the self-assembling DNA-based building blocks prior to the assembly, or a priori, e.g. by chemically or photochemically adding linkages between different parts of the three-dimensional nanostructure. Permanent linkages may, for example, be created by photochemically cross-linking T residues appropriately positioned in the structure under formation of covalent cyclobutane pyrimidine dimer (CPD) bonds (41), and intermittent linkages may, for example, be created by photochemically cross-linking the blunt ends of two double-helical subunits between a 3-cyanovinylcarbazole (cnvK) moiety positioned at a first blunt end and a thymine residue (T) positioned at the other blunt end (40).

[0130] In a second aspect, the present invention relates to a composition comprising a DNA-based nanostructure according to the present invention encapsulating one or more viruses or viral particles.

[0131] In particular embodiments, said composition is formed in a process of removing said viruses or viral particles from a medium containing said viruses or viral particles. In particular other embodiments, said composition is formed in a process of incorporating said one or more viruses or viral particles as cargo in said DNA-based nanostructure.

[0132] In a third aspect, the present invention relates to a method for encapsulating a one or more viruses or viral particles, comprising the steps of: providing a DNA-based nanostructure according to the present invention, and contacting said DNA-based nanostructure with a medium comprising, or suspected to comprise, said viruses or viral particles.

[0133] In particular embodiments, (i) a DNA-based half shell nanostructure based on T_{octa} self-assembling DNA-based building blocks is selected for a virus of a size up to 50*50*50 nm³; (ii) a DNA-based half shell nanostructure based on T1_{pentamer_triangle} self-assembling DNA-based building blocks is selected for a virus of a size between 15*15*15 and 100*100*100 nm³; (iii) a DNA-based half shell nanostructure based on a combination of T1_{pentamer_triangle} and T1_{ring_triangle} self-assembling DNA-based building blocks is selected for a virus of a size between 15*15*15 and 100*100*100 nm³; and/or (iv) a DNA-based half shell nanostructure based on T3_{6_triangle} self-assembling DNA-based building blocks is selected for a virus of a size of 50*50*50 nm³ or larger.

[0134] In particular embodiments, said method is for removing said one or more viruses or viral particles from said medium. In particular embodiment, said method is for encapsulating said one or more viruses or viral particles in order to transport said virus or viral particle.

[0135] In a fourth aspect, the present invention relates to a method for encapsulating one or more viruses or viral particles, comprising the steps of: providing a DNA-based nanostructure according to the present invention, and contacting said DNA-based nanostructure with a medium comprising, or suspected to comprise, said viruses or viral particles.

[0136] In an alternative aspect, the disclosure provides a method for encapsulating one or more viruses or viral particles, comprising the steps of: adding self-assembling DNA-based building blocks to a medium comprising, or suspected to comprise, said viruses or viral particles resulting in the in situ formation of DNA-based nanostructure according to the present invention encapsulating one or more of said viruses or viral particles.

[0137] In yet another aspect, the disclosure provides a method for encapsulating a cargo different from a virus or viral particle, such as a complex macromolecule, comprising the steps of: providing a DNA-based nanostructure according to the present invention, and contacting said DNA-based nanostructure with a medium comprising, or suspected to comprise, said cargo.

[0138] In an alternative aspect, the disclosure provides a method for encapsulating cargo different from a virus or viral particle, such as a complex macromolecule, comprising the steps of: adding self-assembling DNA-based building blocks to a medium comprising, or suspected to comprise, said cargo different from a virus or viral particle, such as a complex macromolecule, resulting in the in situ formation of DNA-based nanostructure according to the present invention encapsulating said cargo different from a virus or viral particle, such as a complex macromolecule . . .

TABLE 1

M13 8064 Template Sequence	
SEQ ID	Sequence
1	GGCAATGACCTGATAGCCTTTGTAGATCTCTCAAAATAGCTACCCCTCTCCGCATTAATTTATCAGCTAGAA CGGTTGAATATCATATTTGATGGTGATTGACTGTCTCCGGCCTTTCTCACCCCTTTTGAATCTTTTACCTACACA TTACTCAGGCATTGCATTAAATATATGAGGGTTCTAAAAATTTTATCCTTGCCTTGAAATAAAGGCTTCT CCCGCAAAGTATTACAGGGTCATAATGTTTTTGGTACAACCGATTAGCTTTATGCTCTGAGGCTTTATTGC TTAATTTTGCTAATTTCTTGCCCTTGCTGTATGATTTATTGGATGTTAATGCTACTACTATTAGTAGAATTGA TGCCACCTTTTTCAGCTCGCGCCCCAAATGAAATATAGCTAAACAGGTTATTGACCATTTGCGAAATGTATCT AATGGTCAAATAAATCTACTCGTTCGCAGAATTGGGAATCAACTGTTATATGGAATGAACTTCCAGACACC GTACTTTAGTTGCATATTTAAACATGTTGAGCTACAGCATTATTCAGCAATTAAAGCTCTAAGCCATCCGC AAAAATGACCTCTTATCAAAAGGAGCAATTAAGGTACTCTCTAATCCTGACCTGTTGGAGTTTGCTTCCGGT CTGGTTCGCTTTGAAGCTCGAATTAACACGCGATATTGAAGTCTTTCGGGCTTCTCTTAATCTTTTGTATG CAATCCGCTTTGCTTCTGACTATAATAGTCAGGGTAAAGACCTGATTTTGTATTATGGTCATTCTCGTTTTCT

TABLE 1-continued

M13 8064 Template Sequence	
SEQ ID NO:	Sequence
	TGAAC TGT TTTAAAGCATT TGAGGGG GATTCAATGAATATTTATGACGATTCCGCAGTATTGGACGCTATCCAG TCTAAACATTTTACTATTACCCCTCTGGCAAACTTCTTTGCAAAAGCCTCTCGCTATTTTGGTTTTTATC GTCGTC TGGTAAACGAGGGTTATGATAGTGTGTCTTACTATGCTCGTAATTCCTTTTGGCGTTATGTATC TGCATTAGTGTGAATGGTATTCTCAAATCTCAACTGATGAATCTTCTACCTGTAATAATGTTGTTCCGTTA GTTCTGTTTTATTAACGTAGATTTTCTTCCCAACGCTCTGACTGGTATAATGAGCCAGTTCTTAAATCGCAT AAGGTAATTCACATGATTAAAGTTGAAATTAAACCATCTCAAGCCCAATTTACTACTCGTTCTGGTGTCTCT CGTCAGGGCAAGCCTTATCTACTGAATGAGCAGCTTTGTTACGTTGATTTGGGTAATGAATATCCGGTCTTGG TCAAGATTACTCTTGATGAAGGTCAGCCAGCCTATGCGCCTGGTCTGTACACCGTTTCATCTGTCTCTTTCAA AGTTGGTCAGTTCGGTTCCCTTATGATTGACCGTCTGCGCTCGTTCCGGCTAAGTAACATGGAGCAGGTCGC GGATTTGACACAAATTTATCAGGCGATGATACAAATCTCCGTTGACTTTGTTTTCGCGCTTGGTATAATCGCT GGGGTCAAAAGATGAGTGTCTTGTAGTATTCTTTTGCCTCTTTGCTTTTAGGTTGGTGCTTCGTAGTGGCAT TACGTATTTTACCGTTTAAATGGAACCTTCTCATGAAAAGTCTTTAGTCCTCAAAGCCTCTGTAGCCGTTG CTACCCCTCGTTCCGATGCTGTCTTTGCTGCTGAGGGTGACGATCCCGCAAAAGCGGCCTTTAACTCCCTGCA AGCCTCAGCGACCGAATATATCGGTTATGCGTGGGCGATGGTGTGTGTCATTGTGCGCGCAACTATCGGTATC AAGCTGTTTAAAGAAATTCACCTCGAAAGCAAGCTGATAAACCGGATACAATTAAGGCTCCTTTTGGAGCCTTT TTTTTGGAGATTTTCAACGTGAAAAAATTATATTTCGAATTCCTTTAGTTGTTCTCTTCTATTCTCACTCCG CTGAAACTGTTGAAAGTTGTTAGCAAAATCCCATACAGAAAATTCATTTACTAACGCTCTGGAAGACGACAA AACTTTAGATCGTTACGCTAACTATGAGGGCTGTCTGTGGAATGCTACAGGCGTTGTAGTTGTACTGTGTGAC GAAACTCAGTGTACGGTACATGGGTTCCTATTGGGCTTGCTATCCCTGAAAAAGAGGGTGGTGGCTCTGAGG GTGGCGTTCTGAGGGTGGCGGTTCTGAGGGTGGCGGTACTAAACCTCCTGAGTACGGTGATACACCTATTCC GGGTATACTTATATCAACCTCTCGACGGCACTTATCCGCTGGTACTGAGCAAAACCCCGCTAATCCTAAT CCTTCTCTTGAGGAGTCTCAGCCTCTTAATACTTTTATGTTTTCAGATAATAGGTTCCGAAATAGGCAGGGGG CATTAACTGTTTATACGGGCACTGTACTCAAGGCACTGACCCCGTTAAACTTATTACAGTACACTCCTGT ATCATCAAAAGCCATGTATGACGCTTACTGGAACGGTAAATTCAGAGACTGCGCTTTCCATTCTGGCTTTAAT GAGGATTTATTTGTTTGTGAATATCAAGGCCAATCGCTGACCTGCCTCAACCTCCTGTCAATGCTGGCGGCG GCTCTGGTGGTGGTTCTGGTGGCGGCTCTGAGGGTGGTGGCTCTGAGGGTGGCGGTTCTGAGGGTGGCGGCTC TGAGGGAGGCGGTTCCGTTGGTGGCTCTGGTTCCGGTGATTTTGATTATGAAAAGATGGCAACCGCTAATAAG GGGCTATGACCGAAAAATGCCGATGAAAACGCGCTACAGCTCTGACGCTAAAGGCAAACTTGATTCTGTGCTA CTGATTACGGTGCTGCTATCGATGGTTTCTTGGTGACGTTTCCGGCTTCTGTAATGGTAATGGTGCTACTGG TGATTTTCTGGCTCTAATCCCAAATGGCTCAAGTCGGTGACGGTGATAATTCACCTTTAATGAATAATTTCT CGTCAATATTACCTTCCCTCCCTCAATCGGTTGAATGTGCGCCTTTTGTCTTTGGCGCTGGTAAACCATATG AATTTTCTATTGTTGATTGAGCAAAAATAAATTAATTCGCTGGTGTCTTTGCGTTTCTTTTATATGTTGCCACCTT TATGTATGATTTTCTAGCTTTGCTAACAATACTGCGTAATAAGGAGTCTTAATCATGCCAGTTCTTTTGGGTA TTCCGTTATTTATGCGTTTCTCGGTTTCTCTGTTAAGTTTGTTCGGCTATCTGCTTACTTTTCTTAAAAA GGGCTTCGGTAAAGTACGCTATTGCTATTTTCTTCTTCTGCTCTTATTATTTGGGCTTAACCTCAATCTTGTG GGTTATCTCTCTGATATTAGCGCTCAATTACCCCTCTGACTTTGTTTCAAGGGTGTTCAGTTAATCTCCCGCTCA ATGCGCTTCCCTGTTTTATGTATTCTCTCTGTAAAGGCTGCTATTTTCAATTTTGACGTTAAACAAAAAAT CGTTTCTATTGTTGATTGAGGATAAATAATATGGCTGTTTATTTGTAAGTGGCAAAATAGGCTCTGGAAGAC GCTCGTTAGCGTTGGTAAAGATTAGGATAAATTTAGCTGGGTGCAAAATAGCAACTAATCTTGATTAAAGG CTTCAAAACCTCCCGCAAGTCGGGAGGTTGCTAAAAACGCTCGCGTTCTTAGAATACCGGATAAGCCTTCTA TATCTGATTTGCTTGTGATTGGGCGCGGTAATGATTCTACGATGAAAAATAAAACGCGTTGCTTGTCTCGA TGAGTGCGGTACTTGGTTTAAATACCCGTTCTTGGAATGATAAGGAAAGACAGCCGATTATTGATTGGTTTCTA CATGCTCGTAATTAGGATGGGATATTATTTTCTTGTTCAGGACTATCTATTGTTGATAAACAGGCGCGTT CTGCAATCTGTAACATGTTGTTTATTTGTGCTGCTCTGGACAGAATTACTTTACCTTTTGTGCGTACTTTATA TTCTCTTATTACTGGCTCGAAAAATGCTCTGCTTAAATTACATGTTGGCGTTGTTAAATATGGCGATTCTCAA TTAAGCCCTACTGTTGAGCGTTGGCTTTTACTGGTAAGAAATTTGTATAACGCATATGATACTAAACAGGCTT TTTCTAGTAATTAAGATTGCTCCGTTGTTTATCTTATTTAACGCTTATTTATCACACGGTCGGTATTTCAAACC ATTAATTTAGGTGAGAGATGAAATTAATAAATAATTTTGAAGGTTTCTCGCGTTCTTTGCTTTGCG ATTGGATTTGCATCAGCATTTACATATAGTTATATAACCCAACTAAGCCGGAGGTTAAAAAGGTAGTCTCTC AGACCTATGATTTTGATAAAATTCATATTGACTCTTCTCAGCGCTTAATCTAAGCTATCGTATGTTTTCAA GGATTCTAAGGGAATAATTAATAGCGACGATTACAGAAGCAAGGTTATTCATCACATATATTGATTTA TGTAAGTTTCCATTAATAAAGGTAATTCAAATGAAATTTGTAATGTAATTAATTTGTTTTCTTGATGTTT GTTTCATCATCTTTTGTCTCAGGTAATGAAATGAATAATTCGCTCTGCGCGATTTTGTAAGTTGGTATT CAAAGCAATCAGGCGAATCCGTTATTGTTTCTCCCGATGTAAGGTAAGTACTGTTACTGTATATTCTGACGT TAAACCTGAAAACTACGCAATTTCTTTATTTCTGTTTTACGTGCAAAATAATTTGATATGGTAGGTTCTAAC CCTTCATCTATTAGAGATTAATCCCAAACAATCAGGATTATATTGATGAATGCGCATCATCTGATAATCAGG AATATGATGATAATTCGCTCCTTCTGGTGGTTTCTTTGTTCCGCAAAATGATAATGTTACTCAAACCTTTAA AATTAATAACGTTTCGGCAAGGATTTAATACGAGTTGTGCAATGTTTGTAAAGTCTAATACTTCTAAATCC TCAATGATTATTAAGATGATTCTGCGGCTCTAATCTATTAGTTGTTAGTGCTCCTAAGATATTTTAGATAACCTT CTCAATTCCTTTCAACTGTTGATTGCAACTGACAGATATTGATTGAGGGTTGATATTGAGGTTTCAGCA AGGTGATGCTTTAGATTTTCTTATTTGCTGCTGGCTCTCAGCGTGGCACTGTTGAGGCGGTTGTTAATCTGAC CGCTCAGCTCTGTTTATCTTCTGCTGGTGGTTTCGTTCCGTTATTTTAAATGAGCATGTTTGGGCTATCAG TTCGCGCATTAAGACTAATAGCCATTCAAAAATATTGCTGTGCCACGTATTCTTACGCTTTTCAGGTTCAGAA GGGTTCTATCTCTGTTGGCCAGAAATGCTCCCTTTTATTACTGGTCTGTGACTGGTGAATCTGCCAATGTAAT AATCCATTTAGAGCATGAGCGCTCAAAAATGTAAGTATTTCCATGAGCGTTTTTCTGTTGCAATGGCTGGCG GTAATATTGTTCTGGATATTACAGCAAGGCGGATAGTTTGAAGTTCTTCTACTCAGGCAAGTGATGTTATTAC TAATCAAGAAGTATTGCTACAACGGTTAATTTGCGTGATGGACAGACTCTTTTACTCGGTGGCCTCACTGAT TATAAAACACTTCTCAGGATTTCTGGCGTACCGTTCTGTCTAAAAATCCCTTTAATCGGCTCTGTTTAGCT CCCCTCTGATTTCAACGAGGAAAGCAGCTTATACGTGCTCGTCAAAGCAACCATAGTACGCGCCCTGTAGCG CGCATTAAGCGCGGGGGTGTGGTGGTTACGCGCAGCGTGACCGCTACACTTGCCAGCGCCCTAGCGCCCGC TCCTTTCGCTTTCTTCCCTTCTTCTCGCCACGTTTCGCGGCTTTCCCGCTCAAGCTCTAAATCGGGGCTC CCTTTAGGGTCCGATTTAGTGCTTTACGCGACCTCGACCCAAAAAACTTGATTGGGTGATGGTTCACGTA

TABLE 1-continued

M13 8064 Template Sequence	
SEQ ID NO:	Sequence
	GTGGGCCATCGCCCTGATAGACGGTTTTTCGCCCTTTGACGTTGGAGTCCACGTTCTTTAATAGTGGACTCTT GTTCCAAACTGGAACAACACTCAACCCATCTCGGGCTATTCTTTTGATTATAAGGGATTTTGCCGATTTCG GAACCCACATCAAACAGGATTTTCGCCCTGCTGGGGCAAACAGCGTGGACCGCTTGTGCAACTCTCTCAGGG CCAGGCGGTGAAGGGCAATCAGCTGTTGCCCGTCTCACTGGTGAAAAGAAAAACCCCTGGCGCCCAATACG CAAACCGCTCTCCCCGCGCTTGGCCGATTCAATTAATGCAGCTGGCACGACAGGTTTCCCGACTGGAAAGCG GGCAGTGAGCGCAACGAATTAATGTGAGTTAGCTCACTCATTAGGCACCCAGGCTTTACACTTTATGCTTC CGGCTCGTATGTTGTGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACAGCTATGACCATGATTACGA ATTCGAGCTCGGTACCCGGGGATCCTCAACTGTGAGGAGGCTCACGGACGCGAAGAACAGGCACGCGTGTGG CAGAAACCCCGGTATGACCGTGAACCGGCCCGCCGATCTTGGCCGCAGCACCCAGAGTGACAGGCGCG CAGTGACACTGCGCTGGATCGTCTGATGCAGGGGGCACCGGCACCGCTGGCTGCAGGTAACCCGGCATCTGAT GCCGTTAACGATTTGCTGAACACACAGGTGTAAGGGATGTTTATGACGAGCAAAGAAACCTTTACCCATTACC AGCCGCAAGGCAACAGTGACCCGGCTCATACCGCAACCGCGCCCGCGGATTGAGTGCGAAAGCGCTGCAAT GACCCCGCTGATGCTGGACACCTCCAGCCGTAAGCTGGTGGCTGGGATGGCACCCGACGGTGTGCGGTT GGCATTCTTGCGGTTGCTGTGACGACAGCAGCACCGCTGACGTTCTACAAGTCCGGCACGTTCCGTTATG AGGATGTGCTGTGGCCGAGGCTGCCAGCGACGAGAGCAAAAAACGACCCGCTTTGCCGGAACGGCAATCAG CATCGTTTAACCTTTACCCCTCATCACTAAAGGCGCCCTGTGCGGCTTTTTTTACGGGATTTTTTTATGTCGAT GTACAGAACCGCCCAACTGTGCGCGCAATGAGCAGAAATTTAAGTTTGATCCGCTGTTTCTGCGTCTCTTT TTCCGTGAGAGCTATCCCTTACCACGGAGAAAGTCTATCTCTCACAAATTCGGGACTGGTAAACATGGCGC TGTAAGTTTCGCGGATGTTTTCGCGTGAAGTTATCCGTTCCCGTGGCGGCTCCACCTCTGAAAGCTTGGCACT GGCCGTCGTTTTACAACGTCGTGACTGGGAAAAACCTGGCGTTACCAACTTAATCGCCTTGACGACATCCC CCTTTTCGCGAGCTGGCGTAATAGCGAAGAGGCCCGCACCGATCGCCCTTCCCAACAGTTGCGCAGCCTGAATG GCGAATGGCGGTTTTCGCTGTTTCCGGCACAGAGCGGTGCGGAAAGCTGGCTGGAGTGCGATCTTCTCTGA GGCCGACTACTGTCGTCGTCCTCAAAGTGGCAGATGCACGGTTACGATGCGCCCATCTACACCAACGTGACC TATCCCATTTACGGTCAATCCGCGTTTGTTCACGAGAGAAATCCGACGGGTTGTTACTCGCTCACATTTAATG TTGATGAAAGCTGGCTACAGGAAGGCCAGACGCAATATTTTTGATGGCGTTCCTATTGGTTAAAAATGAG CTGATTTAACAAAATTTAATGCGAATTTTAACAAAATATTAACGTTTACAATTTAAATATTTGCTTATACAA TCTTCCTGTTTTTTGGGCTTTTCTGATTATCAACCGGGGTACATATGATTGACATGCTAGTTTTACGATTACC GTTTCATCGATTCTCTTGTGCTCCAGACTCTCA

TABLE 2

Staple sequences used for the octahedron triangle (T octa) folding	
SEQ ID NO:	Core structure oligo sequences (5'-3')
2	CACGTTGAGGAATTGCGAATAATCAGATGATGAATATAC
3	AGGGAAGAAAGCGAAAGGAGCGGCTGCGGCTCGTTAGATAAAGGGA
4	CGGGGAAAGCCGCGAACGTGGCGGAGCTCGATTGCTTTG
5	CAGGCGCAGACGGTCAACGTAACGTGCAGCCTCCGGCCA
6	CCCTAAAGGGAGCCCCGATTATTCTCTGTGTGAAATT
7	AATTTCAACTTTAATCTTAATAAATTTTTCGAACTA
8	GGTTTTGAAGCCTTAAACGCTAATTTTGAGCGTC
9	ATGGAAACAGTACATATTAGATTAGGTGCTGGTAATTTTC
10	TAGAAAAATACATACATAAAGTGGGTATTCTAGAAGGTAA
11	ATAGCCGAACAAAATAGCTATCTTACCGAAGATGGAACAAATATTGATATA
12	CCAACAGGTGAGGATTAGAGAGTGACAGACTCATTCGAAGTAGATT
13	TCTATCAGGGCGTAATGAGTGTGCGAGCCCTTCACCCATTTTGAAAAACGCT
14	ATGGCCCAACGTGGACTCCAACGAGCACAGACAATATT
15	GACTCCTTATTACGAGTATGTTCTCCGACTTGCGGG
16	CGAAAGACTTCAAATATCGGTTGGGCTTGAGATGGTT
17	GAATAAGTTTATTGAGAATGACCATAAATCAAAAATCAG

TABLE 2-continued

Staple sequences used for the octahedron triangle (T octa) folding	
SEQ ID NO:	
18	AGAAACAATAAAATTAATTTTTCGTTGTAGCAGCCTGAGTAGAAGAAGCTC
19	GCGATTATAAATGGTCAATACCGCCAGCCAGTATCGGCC
20	CGGAGATTTGTATTACACGAAAGAGGCAAAAGACCTCCGGCTTAGGT
21	AAAGACTTTTTCATGAGGAAGAAAATACCACAAAATAGCGAGAGGCT
22	CCGACAATGAGCGACATTTTAAATCAAGTTTATCGGCATTTTCGGTCAT
23	ACTGCCCGCTTTCAGAGCAGTTGGGAAAAGAGACG
24	CATTCCAAGAACGCAACCATCCTAATTTACGAAAAGCCTGTTTAGTA
25	AGAACGCGCCTGTTTATATCCTGACCCAATCCATTAAGTGAACACCTT
26	AGGCGCATAGATAAGGCTTGCCAGCAAAGTACGTTAAT
27	CGGTCGTTTCGTGATAAATAAGGCGTTAAATAAGAATAAAGCCACGCATAACCGTGA
28	TTTGATTAGTGCCAAGCGAAACGTACAGCTTGAGAAGAGTCAATAGTG
29	TGCCAACGTTTTTTCAGCACCGATCAAAGTTAAATTTAATCGGCC
30	TCAGCGTAGACGCTGAAAACATAGCGATAGCAAT
31	AATTTATCTTTAGTGAGTCACCTGTTTAGCTCACGACGTGGTGGAGC
32	AAGGCACCTTTTACCTAAAAGAGGCTTTGAGGACTACGGAACA
33	AGCTTCAAAGCGAACCAGACCGGACTGACGAGAATATGCA
34	CGCCACGGGAACAATTCTTTTACTAATAGTACAAGGCAAAGAATTAGCA
35	CAAGAAAATTTTTAATATCCATGTTTCAGCTAATGCTTTCAGTA
36	TTACCTGAGCAGAGGCGAATTATTCAG
37	CTGCTCATTTGCCGCTCGGGAAGTTTTTCTAAAATCCTGTTTGATG
38	TCGTCGTATGTTTTTTAGTGTGTCCATCACGCACGACCGAGTA
39	AAAATCCCGACTTTCCTTAGAAAATCACTTATACTTC
40	AACTTTAATCCTTTTTATCGGAAAAGGTGGCATCGGATTTGGGG
41	TTAATTTTATACAGGGTAGCATTAACATCCAAACAGCTATATATT
42	ATCCTGAGGCTTGACAGGAGCCGAAACGTCGCTTGCTTATAGTTGCG
43	CAACATGTCGTTTTTCGCCCTTAGCGTCAGAAGTAACAGCAAGTTAC
44	TGTAAATGGTTTGAAATACCGACCTGACCTAAATTTAATGCTG
45	CAGAGGCATTTTTTTTATAATCAAGGTTTAAACATCGGG
46	GCTCTCACGGCGGTTGGCAGCAACCGCAAGAATTTTTTA
47	AAGGGATAGGCAACAGTTGCGCTCTAAGCCTGGGGTGCC
48	CCGTGGTGCAACAGGACGCTCAATAGTTGGCATCTAAA
49	GTGTTTACGAAATGCGGTGCGTGGTGCCATCCATTTTCAATTGAATTA
50	GGCCATCGCCTGATAACAAGACAAAGAATTTTTGCGAGAA
51	GCAACATCAGTTGACATTATTGAACGAGTAGTAAATTTTAAATTCGGGGGTAAT
52	CAGCGAAAACATAACGCCAAAAGGTAACCTCCACCATCAAATGCCGG
53	AAAGGTATTAAACCAACAACAGTAGGGCTTTTTTAATTGAGAATCGAGCCA

TABLE 2-continued

Staple sequences used for the octahedron triangle (T octa) folding	
SEQ ID NO:	
54	ACAATAACAGCCAGCCTAATTAGGCGTTTTAGCGAACAGCAAACGCGCTAATA
55	AAAGGTAATTTGTTTAACGTCAAACATAAAAATATTCACA
56	ACACGTTAACGGCATCAATAACCTTGCTTTTTCTGTAAA
57	CAGAAGTATTGGGAACGCGCGTCATGGTCATAGCTGTGAGCTTGACTTATAAA
58	TTTCACCATTTACATTGGTTTTTAGATTACCCGGTTTGC
59	CTGATTGCAAGCGGTCTAAGAATACGTGGAAGGAAGGTT
60	GCGAACGTATAACAAAGATTGTTTCATATGTACCCCGTTGATAAT
61	GCTCCATGAAATGCAACATAAAGCTTTTTTAATCGGTTGTACCAAAA
62	AGATTTTTTTAGGAAATCTACGATTGTGAATTACCTTAAGAAGCAA
63	AGCGGTGCAGTCACACTCCAGAACATTTTTATTACCGCCAGCCATTG
64	TTTGCCATCAACATGTTTTAAACACCAACAGGTAGTTACTTAG
65	GCAAGCCGAGAACCACTTTTCATATTTTTCAAATCACCGGAACCA
66	GGTAATTCGATTGAGGGAGGAGAACGCGTGCCAGTTTTTTTATT
67	CGTTCGGGACCCCCAAAAATCATACCGGAAACAATCGGCTT
68	AGGCTCCAAAGGACCATCAAGAGAAGGATACCGCCAC
69	TTATCCGCTCACAATTGCCAGCTGTTTTTATTAATG
70	ATAATACATTTGAGGAAGCAGCAATATTAATTTTTTAGACAGGAATAC
71	GTCACCGAAAATTGTCACAATCAATAGAAAATGCCCGTAAACCTATT
72	TATCACCTATTTTCGGTAAACAGTTAATGCCTTCTAGCT
73	AGGTGAATAATTAGAGCATAGTTAGACGTTAGTAAATTTTA
74	ATATCTTTAGGAGCACTTCTGACCATGGATTAGTGAGACG
75	GGCAAAAGCGCGTACTATGGATTGTAAGGAGAGGCGGTGCCC
76	GAGTGTGGGGTTCGAGGTGCCGTAAAGCACTAAATCGGA
77	TTCCAGTTTGGAACAATCTTTAATTAGAACCCTAACAACTTTTAATAGA
78	TATTAAGCTACGTGAACCATCACCCAAATCACCGAAGCAGCAGGCG
79	CGCCTTTTTGGGTGCCTGTCGTCCACACAACATACGAGAGTTTTTT
80	CGACCTTTTGATAAGAGGTCATTTCATGTCAAATAAGCAA
81	TGCTGAATAGAGAATCTGCCTGAGGAGTAACATGTTAA
82	AGTAAATGTTTTTTTTGACTGGATAGCGTTTTGCTTTTTAAAGAAGT
83	CCAATACTAGCGGATTGCATCAAAAAGATTAAGAGGAAGC
84	GCGGAATCGTCATAAATGAGAGATGAGAAAGGCATTAAATTTTTGTGAGC
85	GAATCCCCGTCTTTACCCTGACTATTATAGTCTGCGATTTTAAAAACC
86	CTCAAATGCTTTAAACGATAAATTATATGATAGTCTGGCC
87	AGTTCAGAAAGAGCAACACTATCAAATTACGAAAGCCTTCCCTGTAA
88	TTTTGCGGAGCCAGTAAACAGGAGGCCGATATCAGAG
89	AACAAAGATTGAGTAACAACCTCGTTATTAGACTTTACAACGAGAGGG

TABLE 2-continued

Staple sequences used for the octahedron triangle (T octa) folding	
SEQ ID NO:	
90	GACAAAAGAGTTTTAACCACCAGAGGTTAGTTAGGAT
91	TCAGAGAGATAATTTTCCACAAGAATTGAGAACATTTTAGTCAGAG
92	GTTAAGCCTAACGGAATACCCAAAGAACTGGCATGATTA
93	CAATAATAAGAGCAAGTCCAGTAAGGCAGGTCCACCGTATTTTTCAGGA
94	AATTAAGCAATAGGGTTTCCAGTATATTTTCATAACCTCAGGCTCG
95	AGCCCTTTTTTACAGAGAGAATAAAATGAAAACCGCCACCACCGGAAC
96	TTAGGCAAGTGTAGCGGTACGCGGCGGTTCATGAGAGCC
97	TCAAAAGAATAGTTTTTCCGAGATAGGGTTGTGGTTTTTTCGAAATC
98	TTTGAATGGCTATTAGGAGTCCACTTTGCCCATAAAGTG
99	ATTGCCTGAGAGTCTGGAGCAACAATAATGCTGTAGCGAG
100	TGTTGGGAAGGGCGTTTTTTCGGTGCAGGGGATGTGCTG
101	ATGATACAGGAGTGTACTGGTAATAGCGACATTCAACGAG
102	AAGCGCAGTCTCTGAATTTACCGTAAACAATGCGCATTAGACGGGAGA
103	CGAACCACGACCAGTATTAGAGCCAACCT
104	AAACATCGCCATTAAAAATACCGAAGCGCCGCTACAGGTCC
105	CGTTAATAGATGAACGGTAATCGTAAACTAGTTTGC GGATGGAAGTT
106	TAATGTGTAGGTAAAGATTTTTTCAAAGGGTCTACAAAGGCTATCAGGT
107	AAGGCGATTAAAGTTCAGCGCATTAAATTTTACCGTCG
108	TTCAACCGGCGGGAGGGCATAGTCTTTTGC GGATCGTC
109	GTATTAAGCGGGTCAAGTGCCTTGAGTAACAG
110	GCCGCCGCCAGCATTGATTTTTTAGGAGTTGAGCGTCATACATGGCTTTT
111	AGACGATTTTCAAGCCAACGATTTAGTAATCTGTCCAGACGACG
112	AATCCTCACTCAGAGTAGCAGCCATAAGAGAATATAAAG
113	TGCCACGCGTATTACACCGCCTTTAAAGCC
114	TCTAAAGCCAGCAGAAGATAAAACAGAGGTGATGCGCGTATGCTTTCC
115	ATAAAAGGGACATTCTGTTTTTCCAACAGAGAGCGGAAGTATAGCCCT
116	ATACCTAGCCTGGCCGAGAGATAGTAAAAAAAAGAACG
117	GACGACAGCTTTCCGGCACCGTAACGCCAAGGAAAGATACA
118	ATTTCAACACCAATATTCCTGTAGCCAGCTTTCATCAA
119	GCGTAACGACAACTACGCCACCCTCAGAGCCTAATTCGC
120	TTAAAAGTAACCACCAGAAGGAAAGAAATTGCGGTTCGTACGCCAGAA
121	ATCGCACTGAATTTGTCCAATTCTACTGACCAACTTTCTAACTCACAT
122	CCTCAGAGTTTTGTAAAAATTTTATTGTAAAACGGTGTCTGGCTTAG
123	TTTTGTGAAAGGCCGCTTGAGCCATTGGG
124	ATTGGCGCTCCAGTAGCACCATTACCATTAGCAAGGTTTCTTTCCA
125	TTGCCATAGGCTGAGACTCCAGACATGAAAGGAAATTA

TABLE 2-continued

Staple sequences used for the octahedron triangle (T octa) folding	
SEQ ID NO:	
126	TCAGCTCATTTTTTAGCAAGGATAGTCAAATGTTTACCA
127	GGAACGCCATCAAAAATACGCCAGTACTTTTAATAAATC
128	TGGTAATAATCACCTTGCTGACGTATGAAAAGTATAACGACCACCAC
129	AGCGGGGTTTTGCTCCGCCACCCGGCCTTGACAGGGAAG
130	GCCCCCTTATTATATCGGTTTATCAACCAATGACAACCATCCACCGGA
131	AACTATCGGCCTTGCGTAGATTTTCGTGAGGCCATTGCGCTAACAAAA
132	AGTACCAAGTATAGCCCCGAATAGGTGTAT
133	AGAATAGAAAGGAATTTTAACTAAAAATCTCCAAAAA
134	AATCCTGATTGTTTTTTTGATTATAATATCAAAATTATT
135	AACCTACCTTCTGAATAATGGACATTCGCCGACGGCCA
136	GCACGTAAAACAGAGAAACCTCAAATATCAGTCAATAG
137	AATCAATATCTGGTCCGTCTGAATGAAAGCGCACGCTGG
138	AAATCAACAGTTGAAAAGGGTTAGCATGGAAGTAATAAC
139	ATAGGTCATCAGGAAGAAACAGGGTTGATTC
140	CGTTGGTGGATTGACCGTAATGGGGGAATTGAAGCCCCAA
141	CCTCAGAACCGCCACCATTCTGAAACAGCCCTCCAGCA
142	CTCAGAACCAACGCCCTGTAGCATTGCCGAATTTTCTGTATGCGGAGTG
143	CCCAATAGGAACCCATGTAGCTAAACAACCACCCATCTAAAGCCCTGCC
144	TTTCGTCAACAGTCATTTTCAGGATAGCAAG
145	GGCGGATAAGTGCCGTAATTTTTTCGCCTCCCTGTAGCG
146	GATTCTCCGTGGGAACATATTTAAGAGGGGACTAGTTTGACCATTATC
147	AAACGGCGTAGACCGTGCATCTGCGGGCTTCTGGTGCCGGTGCGCAAC
148	CAATTGACATTATCAGCAACAGCGGGAGCT
149	TTTAGAAGATTAAATCCTTTGCCCAATAGCGGAATTATCAATCAATAT
150	TATTTTTTTTTATATCTTACCATCAAGATTAGTTGCTCGCAATAA
151	AAAACACTGCGAAACAAAGTACAACCGGAACGCGGACCTACTAAAGT
152	AAATAAGAACCACCTCATTTTTAGCCGCCACACAAAATA
153	TTCAACTATTAGAACCCTTTTTTATATATTTTAAAGATTCT
154	AAATCACATCTAGGAATCTAGAAGGCTTATCCGCAACATATGCGCCAAA
155	CTGGCGAAGGGCCTCTTCGCTATACTTTCAAATTTCTT
156	TCAAAGGGCTGAGAGAGAGGAAAGAGGACAGATTGACAAG
157	TTTCGCACCAAGCCATCTTTCAAACGCGGTCCGTTTTTTAAAT
158	TAATGGACTTGTTTAAACGATGCTGATTGCGAGCACATAGGCGGCCGGAACCGA
159	TGGTCGTCTCGTCGAAAGCTGCTCAATTT
160	TCTCTTCAGTGAGCTGGCTGACCTTCATCAACCCAAATCATCATAAG
161	TATAAACCGGATATTCATTAGAGTAATCTGAACGGTACCTTTAATTGCTAAAACCG

TABLE 2-continued

Staple sequences used for the octahedron triangle (T octa) folding	
SEQ ID NO:	
162	TCCGCCATGTTTACCATGAAGGGTAGCCGCACCCTCATAACGGAACGTGCCAATT
163	CGCTACCGTAATCAGTACAAAACCATCGAAACCAATCT
164	TTCATATGTTTATTAAATCAGATAATTACCGCGCCCAT
165	TATACGTAATGCCACTACGGGGTTATATAACTATA
166	TCGGCTGTCTTTCCTTATTTTCATCGGAGAACAAATATTGAC
167	CCGGAGACAAAATTTATGCAGATGACAGCATCCATTAAACGT
168	TATACCAGTCAGGACGTTGGGAAGTTTCGGAACGAGGGTA
169	AGAGGGTAGCTATTTTTATTTCATTGACGACGATAAGAACTGGT
170	GTTTACCAAAAAGAAACGCAAGACACCACGGCAAGCAA
171	TTTTTTAAGTCCTGAACATATGCGTTATACAAATCTTACCATTTCAA
172	AATTTAGGAAAATCGCGCAAAAGAAGATGATGAAACAAACGCTGGTAATGGTTTTT
173	TTTTTCACTCTGTGGTGCCTAGGGCGCTAGAAAAGTAAGCAG
174	TTTTTGTAAGGTTTCGCGCGCCTGTGTTTTT
175	TTTTTAGCTACAATTTTCAACAATAGATTTTTT
176	AAATAGCAGTTACCAGAAGGAAACCGAGGAAAATTTTGACCCCTTTTT
H1 design oligo sequences (5'-3')	
177	GCAGTAGAGTAGGTAGAGATTAGGCACAATACTATTAAATCTGGTCATGTACATCGACATAAATAATTGCG
178	GCAGTAGAGTAGGTAGAGATTAGGCATCAGATGTAAAACATTCAGGCAAACAGGCAAAGCGC
179	GCAGTAGAGTAGGTAGAGATTAGGCAAGAGACTACCTTTTTAAATACACTCGCGAGCTATTGTGTC
180	GCAGTAGAGTAGGTAGAGATTAGGCAATGCATTCAAATATATTTTAGACCCCTCAGCATTATGA
181	GCAGTAGAGTAGGTAGAGATTAGGCATACCGTCGAGGTGCAGTTTCAGGGATTTCCGTAACACTGAG
182	GCAGTAGAGTAGGTAGAGATTAGGCAATCATAATTACTAGAAGCATGTAGATAGCAGCGTACCGCATATAA
183	GCAGTAGAGTAGGTAGAGATTAGGCACCATATTTAACAACGCTACCGACAGAGCCACCCTCAGAAC
184	GCAGTAGAGTAGGTAGAGATTAGGCATTTTCGTCAGATGGCAATTCTCATATTCCTGATTAT
185	GCAGTAGAGTAGGTAGAGATTAGGCATTAATTACATTTAACACACGCAACAAAGAGTCAGATGCCG
H2 design oligo sequences (5'-3')	
186	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTCAATACTATTAAATCTGGTCATGTACATCGACATAAATAATTGCG
187	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTCAGATGTAAAACATTCAGGCAAACAGGCAAAGCGC
188	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTAGAGACTACCTTTTTAAATACACTCGCGAGCTATTGTGTC
189	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTATGCATTCAAATATATTTTAGACCCCTCAGCATTATGA
190	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTACCGTCGAGGTGCAGTTTCAGGGATTTCCGTAACACTGAG
191	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTATCATAATTACTAGAAGCATGTAGATAGCAGCGTACCGCA
192	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTATAACCATATTTAACAACGCTACCGACAGAGCCACCTCAGAAC

TABLE 2-continued

Staple sequences used for the octahedron triangle (T octa) folding	
SEQ ID NO:	
193	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTCTTTTCGTCAGATGGCAATTCTCATATTCCTGATTAT
194	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTAAATTACATTTAACACACGCAACAAAGAGTCAGATGCCG
H3 design oligo sequences (5'-3')	
195	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCACAATACTATTAAATTCTGTGTCATGTACATCGACATAAATAATTGCG
196	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATCAGATGTAAACATTCAGGCAAACCAAGGCAAAGCGC
197	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAAGAGACTACCTTTTAAATACACTCGCGAGCTATTGTGTC
198	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAATGCATTCAAATATATTTAGACCCCTCAGCATTATGA
199	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATACCGTCGAGGTGCAGTTTCAGGGATTTCCGTAACACTGAG
200	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAATCATAATTACTAGAAGCATGTAGATAGCAGCGTACCGCA
201	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATATAACCATATTTAACAACGCTACCGACAGAGCCACCCTCAGAAC
202	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCACCTTTTCGTCAGATGGCAATTCTCATATTCCTGATTAT
203	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATTAATTACATTTAACACACGCAACAAAGAGTCAGATGCCG

TABLE 3

Staple sequences used for the T1 pentamer triangle folding	
SEQ ID NO:	
Core structure oligo sequences (5'-3')	
204	TGCGGCGGCCGGGTCAGTCCAGCATCAGCTCGATAACGGA
205	TGCAGCAAGCCTGGGGTGCCCTAATGAGTGAGCTTTTAACTCACATTAATTG
206	ATTTTCATTGAATTACCTTTTTTAAGAAGATGATTCATT
207	GGAGGTCAATAACCTTTTTTTTATAGTAGTAGTTTTTATTAATAGAT
208	CTCACAGTGTTCTGCACAATAAAGGATTTA
209	AAAAGAATTAAGTACTATGCCGTACTGGTAATAAGTTTAACTTGCGTA
210	TACCTACATTTTGAAAGGGACGAATGGCTCGCTTAGGAGCACTACAGCACG
211	AAATAGTTTGACCATCATCCAATAA
212	GTTAGAACTCAAACCTACCTGAAAGC
213	TATGACCCTGTAATACTTTATAAAGCCTCAGA
214	GCGCGAACTGATAGAAGATAATGCTGAACCTCAATTTTAA
215	ACCACCAGAGCCGTGATATTTGATACAGGAGTGAGTAA

TABLE 3-continued

Staple sequences used for the T1 pentamer triangle folding	
SEQ ID NO:	
216	GCATAAAGCTAAAAGAAGCCTGTGAGAAAGGCCGATTGA
217	CGGAACAAAGAAACCAAGCCCGGATCAAGTTTCCGGTTTTGCTCAGT
218	CATTTTGTATCATCATATGGGTCGAGGTTTTTCCGTTTCATTTGC
219	AATATTTGCATTAAATTGTTTAGACTGTTTTATAGGCATCGTA
220	GTGTACAGACCAGGCGCATAGGCTATGCCACTGAGGCGAAAACAGCT
221	CAGCTTTCATCAACATTAAATGTGAGCGAGTCAGCTCAT
222	GCCGGAACGTACCCCGTTGATATATAAGCA
223	TACTCAGCCCTCAGAACAGAGAGATAATTTTCCACACGCCAGG
224	GAGGTTTAGTACCGCCAAAACAGGGGGCGCGCTTTAAAA
225	TTTTTAACTATTTTGTTGCTTTAAATTCACC
226	ACAGCGCCACCCGAAACAATCGGAAGGTGCCGTCGAGAGTATCACCG
227	GTGCCAGCTGCATTAATGAATCGGCCAACGCCAGGTTGGTTTTCTT
228	CGCTTTCAGTCGGGAGTTAACGGTGGTGGTTACAAGAGTCCACAATCCGCCGGGC
229	CCTGTAGCCAAAAATAATTCGCGTAAATTAATCCTTTGCAACATTAT
230	CAGAACGATCAACTTTAATCATTTGAGCTCAACAGCTTCAA
231	TCACGACGTTGAGCGCTAATATCCGCAAGTCAGATTGAATA
232	GGAATTTGTGAGAGATCAGATGATGGCAATTCACAGAAGGGGGAAAGTTTGCCA
233	ACGGGTAAAATACGTAGGCTGACCACGTTAATGACGGTCA
234	ACGAAGGCGCGCCGACAATGACAAGCTCCAAAAGGAGCGCAATGAATT
235	AGAGCCACCACCTCAACCTCAGATAGCTATAGCTAGCAAGGACCAT
236	TGAGAATCGCTTTTATATTTAACTACCTTTTTTTTAACTCCGG
237	CGCAACTGCGCAGAGGGAATTAAGTGAACACCAAATAGCAAACCGCCA
238	CCAGCGGTGCTTTTGGTGCCCCCGGTATTTTTTGGGTAAAGGTTT
239	TGCATCAGACGATCCATGTAAGCGGTCCACGAATCATGG
240	CATGAGGAAGTTTTTTTCCATTAAATAATTTTTCGATATATTCGGT
241	AATAGCAAGCAAAATCAGATATAGTCAAATATATCCCAATCCAAAGAT
242	ATCGTAGGAATCATTATATAAAGCGCCAGTTAATAGCAGCCTTAGACGCTGAGAAG
243	AGGCGGTTAGAAACCAATCAATACTAATTTACGAGCATG
244	CTGATGCATTAAAAATCTTACCAGCCGCGCCCGCTTAAATCAAGATT
245	CTTAGGTTAACAGTAGGGCTTAATAGCCGTTTCCAGCTACAA
246	CTTTGCTCGCCGGTTACCTGCAGCGTTGCGCCTGAGAGAGT
247	TTCTAAGAAGAGGACAAGAGGCAACCGCGACCTACAACGGGGCTATCA
248	AAGGCTTAGCGAACCTCCCGACTTACATTCAACAAACGTA
249	TTGCGGGAACGAGGGTAGCAACGAGTGAATAGATTTTAAGAAGTGGC
250	ATCGGCTGTCTTTCCTTATCATTTCTTAGGCAGTAAGTCCTGTGAATTT
251	GTGAATAACCTTGCTTTGTAATGAAATGAAACAAAATAAAAGGTGGC

TABLE 3-continued

Staple sequences used for the T1 pentamer triangle folding	
SEQ ID NO:	
252	CGTCGCTACAATAACGGATTGCGCCTATTACGCCAGCTGG
253	TTAATTAATTTTCCCTTTTTTAGAATCCTTGAAAACCATAGGTCTGAGAGAC
254	CGGTGGTGCTGGTCTGGTCAGCAGTAGCTCTCACGGAAAA
255	CCATCCCACGCAACCATTTTTCTTACGGCTGGAGGTCTGTTGCCCTGCGGCT
256	AGCCGGAAGCATAAAGGCGCAGTGAATTCTGTGTATGAGGCCGTTTT
257	GAATAATACAGTTTCAGCGGAGTGGGAACAAACGGCGGAT
258	ATTTTTTCACGTTGAATTTTATCTCCAAAAAAGCAACCATCGCCACGC
259	TTCTGTATTAGGTACGTTGGTGTAGATGGGCCAGGCAA
260	TTTCCAGATCCAGCCATCACCAGTAAACAAGAGGTCATTG
261	GATGAATATACAGTAAGCTGCAAGGCGATTAAGTTGGGTACGAAACGT
262	ACGTGCCGAGCGGATCAAACCTAAATTTCTGCCTGGCCTT
263	AGGCTTGCCCTGACGAGAAACCCGAAAGACCACATTCAACTAATTC
264	GATCGCACCGTTAGTACTGTAGCATTCCACAGGCGATTAT
265	GCTTTCGGTCAATCATATGTCCAATACTGCGGA
266	GTAAACGACGGCCATTTTTTGCCAAGCTTTTCAAGTTTTCCAG
267	ACAAAGCTGCTCATTCGCTACAGAGAAAGATTAGTAAGAGCAATGCTTTCGAGGTG
268	TAAGTGATTGTTGGATTTCAGGGCGATGGCAGCTTGACAGCGAAT
269	TGTACATCGACATAATTTTAAATCCCGTAAACGCCAGCAGT
270	AACGCCAACATGTAATCAAGAACGAATCTTACGAACAAGA
271	AGGGGACGACGACAGTTTTTATCGGCCTCAGGAAACCGTGTCATC
272	TTAGACGGGAGAACCCGAAGCCCTTCCTTATTTGCAGCCA
273	GTCGGGGTCATTGCAGGCGTTTTTTTTTCGCACTCTA
274	GTTTTTTCACGGTCAGGCCAGAACGCCTGTGCACTCTGTTTCCACAC
275	TGCCGTTCCGGCAAACCTTTAGTTCGACAACTCGTAGCACTAAATCGGA
276	AATTTCTTAAACGAACATAATTTTGGAACAACATTATCCAGTCAGTCAACGTA
277	TGATACGCAACCTTTAATTGTATCGGTTTTTTTATCAGCTCAC
278	AGATTTGTAACACTCATAGTTAGCGTAATTATGAAACA
279	ACCAAGTTACCAACCTAAACGAAGATGAACGACTGACCAACTTTGAA
280	AGTCAATACAACGCTAACGATTTTTCGTCTTCCAGAGTTTGCAGTTATTTTC
281	AATATAAAATTTATTTGCACGTGCGATATTTTCTTAGATTATAC
282	CTGTCCATAATGGAAGGTTAGGGAACGGAACCGGCGATAAAATTGAGTTAAGC
283	GACGACGACAAAGCCCGAGATAGGTTAATGCGGAGAAAGG
284	GTTGAGTAAAGGGCGAAAAACCGTCTAATAAACGTGGC
285	GCGGTTGCCTGGTTTGCCCTTTTTCGAGGCGAAAATCGCCTGGCCTCACTGCC
286	TGTTCTGCAGATACATAAGAAAGACTGAGAATGA

TABLE 3-continued

Staple sequences used for the T1 pentamer triangle folding	
SEQ ID NO:	
287	CGCCAAAGACGATAAAAACAAAATAGACGCAGAAAAC
288	CAAAGAACCCTTAAGAAACGATTTATTAAGACTTTTAAGA
289	CCTAATTTCAACGCTCGGGTTATATACTATACTGTAAATAGAGAGAA
290	TTCAGCTAATGCAGAAAGTAATCTGAAACAGAAGGATTACCACCGG
291	TATGCAACTAACAGTTGAAGCGAACGAAGCCC
292	AAGGAGCGCGTAACCACCACACCCACGTATAAGGCAAAATGTGAGACG
293	CATATCCAGAACAAATTTTTTTTACGATAGAACCTTTTTTCTGATCGG
294	TACAGGTAGGCTTTGACTTGCAGGGAGTTAAAGGAATTGCTATCATAA
295	AGAAAAATCTTTCATCAAGAGTAATCTTGACATTTTGAACCGGATATTCAT
296	TTTTATCTGGGTATTAAACCAAGTACCGCACTTTTTCATCGAGACAAGCA
297	CCGGAATCATAATTACATTTAATGAACTTTT
298	TGTGAAATTGTTATCCATCTGGTCGAAGGTTAGTGAGGCGACAGACAA
299	CGCTGAGGGGACTAAAGACTTTTTTACCCTAAAGACGTTGGGA
300	TGATAAGAGGTCTTTTTTTTTTTCGGATGGTCATTATA
301	AATATGATGAGAGGGTAACGCAAGCAAAGAATTAGCAAA
302	CCAGCGCCCGAAATTCGCAGTCTCTTTTGAATTTACCGTTCAG
303	ATGGTTTAAACATATAAAAGAAACGTTTTTAAAGACAC
304	TTTGTCAATCTTTTATAGAAAATTCATAGTTGCTA
305	CTGTTTGACATCAGATGTCATAAACATCCCTTAGCACCGTTTAAAGAA
306	GGGGTCAGAATGCCCCAAATAAATCTCAGAGCCACCACC
307	TGAGGCCAGTTGCTTTGTAATAACATCACGCCCCGCCAGCATTGACAGGAG
308	CTGAGAAGATTAACCGAGTGCCACGTTTTTTGAGAGCC
309	AAGGGATTTTAGTTTTTTCAGGAACGGTACGCCTTCACC
310	AGTTGGCAATGAAAAATCTAAGCATCACCTAACAGAG
311	CTTAGAGCTCCAACAGGTGAGGATTTTTTTGAGAGTAC
312	CTTTAATTTAGTCAGAAGCTTTTAAAGCGATTGCATACCCTG
313	TAATGCCGATTCAACCTGTGTAGGTAAAGATTCAAAGGTTATTTC
314	AGCTATTTTTGACAGAAATTGTGGCGTTTTATCCGGTA
315	TATGTTAGCCGGAGACAGTCAAATCACCATCACGCGAGTCGAAAT
316	CACGGAATAACCGAGGAAATTTTGCAATAATAACGAGTTACC
317	AAACAGTTTGCCTTGATAAGCGTCATACATGGCTTTTGACACAAAC
318	CTGCCTATTTTCGCGCACAAATGTTGGGCGCGCGGGGAG
319	GAATCAGAGTTTTTGGGAGCTAAACAGGAGGGCAAGTGTAG
320	TCTAAAAATATCTGTTGCGTCCGTGTTAATTGTAGTAAATTGGGCTGCTCACA
321	TTAATTCGATGATATCAAACCTCAATCAATTGAGATGGAGCCTC
322	AAGCAAACCTAATTGCGTCTGGAATATTTTAATTTTTTGAATGC

TABLE 3-continued

Staple sequences used for the T1 pentamer triangle folding	
SEQ ID NO:	
323	CCGCTACATTGTTGCCTGAGTAGAGAGGCAGGCATTTTCGGT
324	GGTCACGCTGCGGGCGCTATTTTGGCGCTATAGATAA
325	TCAAATATCCAGAACGAGTAGATTAATACCGATCGTCTGAAAT
326	ACTATTATCTGGAGCACAACTAATAGCGCGAAACAAAGTGCTCCAT
327	GAATCGATTCTACTAAGCTATATTTTCATTTAAGATTGATCAGAA
328	CCGAACAAGAATACCCAAAAGAACCATACATAACAGCCATGTTTGTAA
329	AGAAGGACTGAGACTATATCAAAGTACCGACAAAAGGTAACGCGCCT
330	GCAACGACCAAGTAATAACGCTCAAACGAACCAATTAGTCTTTAAT
331	CAGGTCTTTCAAAAAGATTAAAGAGCAGACCGGGAGATTACCTTATGC
332	CTAGCATGCAGCAAGCCCAATAGCGAACGATCTAAAGTT
333	CCAATAATAAGAGCAAGAGCAGATAGAAACAGGGAACGTCAA
334	TTCCCAATTCTGCGCAGCCCTAAAACATCGCCATT
335	CCCTCAAATAAAATTCAAATTGTAAACGTTAATTTAAAAG
336	ATCGTCATAAATATTCATTCAAAAATTACCAGACAGGAATTA
337	CGATAGCAGCACTTTTGTAAAACCGCCTCTTTTCTCAGAAATC
338	TCAGTAGCGACAGAAATAGGTGGGTTGAT
339	TCTTTTAATAGCCCCCTTATTAGCGGCCTTTAGCGTCA
340	CAGGAAAAACATTTACAAACAATGATGAAGACGCCAT
341	AGTCACAATTTATTACATTGGCAGACGCTCATGGAAA
342	CATCAATGAACGGTAATCGACCATGTACCGTATCATCG
343	GCACCATTACCATTGAAAAGGTGGATTAAGCAACGGAGATCTACAAA
344	ACCAGTAAAAGTAAACAATGCTGAACACACCCTCGGCGATC
345	CAGAGCCAGGATTAGCGGGCGCTTCTGAA
346	GGTCATAATCAAAATCTTTCATCGTCAGACGACGCCACCAGAACC
347	CCTTGCTAAGGAAGATTAGCCACTACTGATTATAGACTTT
348	CCAACAGACGCCAGCCATTGCAATTTGAGTCCGAACG
349	CGTCACCGACTTGGGGTCGGTTGTACCAAAAACAT
350	ACATTTCCCATAAATGAATCCCAAAAGAGTTAAATAACAACC
351	ATCATACCCTGAGAGTGATAAATGTTACTTAGGAACCGA
352	AGGCAAGGGATAAAAATTTTGTAAACCTTTTTCATGTTTCATT
353	TTTGGGAATTAGAGCCAGCAAGCCGCCACC
354	CTCAGAATTAAGAGGGCCCGTATGTTTATCAATCCCATC
355	CCGCCACCCCTCATTAAGCCAGAATGTTTTTAAAGATTCATTA
356	GTAAGAAAGCCGTCAGTTGAAAGCCCGGGTAGTTTCCTGAACATACG
357	TATTTTATCTGGGAAGTATTAGACGTTATGCTGATCGTGCC

TABLE 3-continued

Staple sequences used for the T1 pentamer triangle folding	
SEQ ID NO:	
358	CTGAGTAAGTTCTAGCGCTCCTTTATCATAAGCCGGAAC
359	AAGGTGAAAATATTGAAAAGACAAAAGGGCGGCGGAG
360	GTCAGTATTACACCGCCTTTTTCAACTTGTAGCAATACTTCT
361	GACTGTAGCGCGTACCGAACCTCAGAGCGGGGAACCTATTATT
362	CCATATAAAGTACGGTTGAATATAATGCTGT
363	AGAGTCTGTCCATTGATTAGACGAGCGCCGCGC
364	GGGAGGGAAGGTATTATCACGAAAATATGGCATG
365	TCACGCAATGTTTTTATAATCAGTTCACCACCTTATAAATC
366	AGCAGCAAAATCAACAGCCGATTATCATAGCTCCGAGCTCTCACTGCG
367	TTTGTTTAAGCGCATATGTGACCAAGTTAAGTACT
368	TTATACATTTTCATCTTCTGACCTAATAGAAAAATCGCAAGACAATTACC
369	TGTTAAATAAGGACCCCCAACAGCCCTGAGTTTCG
370	ATTATTTATTTTAGAATCCAAGCCTGTTTAGTATCATT
371	AAGAATACACTAAAGAGTTTGAAATACCGACCGTGTGATAAAT
372	TCAATATGAGCAAAATGGAAACCAAAATCGTTGGGAAG
373	TACATTTAACATTGTCGTCAGCGCCATCTTCTGGT
374	GCGAACACTCATCTTTAATAAAACAAACATCAAGAAAAACAAAT
375	TTAATTCAGCAAAATCAACCTGTGCGCAACAGCTGATTGCCAGAATCCCTCGTTA
376	TATGCCAGTAATAAGAGATCAAAATATAAAAAACAGAAATAA
377	AGGTGGAGTAACGTCAAGAAATTGCGTAGATTAATT
378	GTAACAGTAAGTTTATAAAATAATACAATAGAAGGCATTTTCGAGCGTT
379	TGGGCGGTTGATCAAGTTTGTCCGTGAACCAAGAATT
380	ACCCTAAAGGAGCCCCGAAAGCGACCAACGTGTTGTTCCGTGGGCT
381	TTTAATGCCAACGCGCACTGGTAGTTTGACCGAAATCCGTGCTTT
382	TATATTCAGGTTCCGCCACGAACCTACCCCTCAAGATGAAAGTA
383	AAAGCCGCACATCCTCTCGCTGGCAGCCTCCGTTTT
384	TGCCAGTTTGGTAATAGTAAATTTTAGTTTTGCAAGCCAT
385	TGAATTAGGAATACAGCATCGGTGTCACCCCTCAGCAGCAGT
386	TTGCGCCAGAGCACAGGCGGCGGTCCTACATTTGTAGATTAGTACGTGGC
387	TACGGGTGCTGCTACCGGGGTGAGGATCGAATTGAG
388	TTAAGAACAATAAAGGCCGCTTCGAGGCATCATCAGTT
H1 design oligo sequences (5'-3')	
389	GCAGTAGAGTAGGTAGAGATTAGGCAAGAGACGTTTTTAGAAACGACTTGTAGAATTTTGTGACGCGTGGTG
390	GCAGTAGAGTAGGTAGAGATTAGGCACGAAAGGTTTTTGATGTGAGTACCTTTTTTTTCATCGGGAGAAA
391	GCAGTAGAGTAGGTAGAGATTAGGCACGTCGGATTCTCCGTGAGAATAGACAGAGGGGCCCTCGTT
392	GCAGTAGAGTAGGTAGAGATTAGGCACTCCGTGGTGAAGGACAACCGCATCACCCAACGTGGACT

TABLE 3-continued

Staple sequences used for the T1 pentamer triangle folding	
SEQ ID NO:	
393	GCAGTAGAGTAGGTAGAGATTAGGCAATAAGTATATCAATATAATCCCTATGTTTACCAGTCCC
394	GCAGTAGAGTAGGTAGAGATTAGGCATTATTAATCAATAGGAGGTAAAGTTAACGAGAGGCTTTTG
395	GCAGTAGAGTAGGTAGAGATTAGGCATGACCGTTTTTTATGGGAGGGATTTTGCTTTTTTAACAACTTTCAA
396	GCAGTAGAGTAGGTAGAGATTAGGCAGGTGCGGGCCTCTCGTGATTGCTGGGTAATTTAACATAA
397	GCAGTAGAGTAGGTAGAGATTAGGCAAAGCCCCATTTTCAGGGATACCGTCGCCATTGAGGCTG
H2 design oligo sequences (5'-3')	
398	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTGTAGAGACGTTTTTAGAAACGACTTGTA GAATTTTTGTGTCAGCGTGGTG
399	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTCGAAAGGTTTTTGATGTCAGTACCT TTTTTTTCATCGGGAGAAA
400	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTCGTCGGATTCTCCGTGAGAATAGACA GAGGGGCCCTCGTT
401	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTCCTCCGTGGTGAAGGGACAACCGCATC ACCCAACGTGGACT
402	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTATAAGTATATCAATATAATCCCTATG TTTACCAGTCCC
403	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTATTAATCAATAGGAGGTAAAGTTAA CGAGAGGCTTTTG
404	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTGACCGTTTTTTATGGGAGGGATTTTG CTTTTTTAACAACTTTCAA
405	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTGGTGCGGGCCTCTCGTGATTGCTGG GTAATTTAACATAA
406	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTAAGCCCCATTTTCAGGGATACCGTCG CCATTCAGGCTG
H3 design oligo sequences (5'-3')	
407	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAAGAGACGTTTTTAGAA ACGACTTGTAAGATTTTTGTGTCAGCGTGGTG
408	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCACGAAAGGTTTTTGAT GTCAGTACCTTTTTTTTCATCGGGAGAAA
409	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCACGTCGGATTCTCCGTG AGAATAGACAGAGGGGCCCTCGTT
410	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCACTCCGTGGTGAAGGA CAACCGCATCACCAACGTGGACT
411	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAATAAGTATATCAATATA ATCCCTATGTTTACCAGTCCC
412	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATTATTAATCAATAGGAG GTAAAGTTAACGAGAGGCTTTTG
413	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATGACCGTTTTTTATGG GAGGGATTTGCTTTTTTAACAACTTTCAA
414	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAGGTGCGGGCCTCTCG TGATTGCTGGGTAATTTAACATAA
415	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAAAGCCCCATTTTCAGG GATACCGTCGCCATTGAGGCTG

TABLE 4

Staple sequences used for the T1 ring triangle folding	
SEQ ID NO:	Core structure oligo sequences (5'-3')
416	GCAAAGACAAGGTGGCAACATATATGGTGATGGTGGTTCCGAATAGCC
417	TTCCATTATAAATTGGGTCAGGACACAGGTAGAGGTCTTTA
418	ATGCGCCGCTACAGGGAACGTGCTGGAGGCCG
419	TCTCACGGTACATCGACATGGAGAGGGTAGCT
420	GGGATAGCGGAACGCCTCTGGAGCAAACAAGAAAAGGCCG
421	AGAGAGAAATTGAGTTAAGCCCAAGAGATAAC
422	TTTTTCAAATTTAAGACGCTAATTTTC
423	CCTGACTATTATAGTCAGCTTCAATGAATTACCAACAGTT
424	CGGAACAAGATTTACACCAGAAACAAGAAAACGAAAGCGCG
425	CGGTGCCCTCGTTAACGCGCATCACCGGACAGCGGTTTGTTA
426	GCGGTGCTGTCACTCGGGCGCCAGCATCCGCCAG
427	AACAACATGTTTCAGCTCGAGCATGTTTTTAACCAAATATCCTAAAGCA
428	TCTGACCTAATATATTTGCAAATCCAATCGC
429	GAAATACAAATGCTTTAAAGATTAAGAGGCGCGAGAAAAC
430	ATCAGTTGACATTATTGTTGGGAAGAAAAATGGGAGTT
431	GGAATACCACATTTACGAGCCGGAACCTGTCTGCCAAAACGAACTAA
432	GCGCGCCTGTGCAAATAAGAGAATAACAATAGATAAGACCTGCAGCCA
433	AAATAAGGCGTTATATATTAATTGAGAAGAGAACCTACCACAAAGAA
434	GCGGAATCGAGAATGACCATAAATCAATTTTTAATCAAAGATTCT
435	CAATACTTTGATAAGAGCGAACCATTTTCTGTGACGCGACGAATAAT
436	TTACGGGAATCAACGTACGAGTAGAACGGGTAAAAGGCCG
437	TGAAATTGGCCTGGGGTGCCCTAATGAGTTTTTGAGCAGACGATCCAGCGCAG
438	TCCTGTGCCGCTGGGTATTGGGGGACGACGCCAGCTT
439	CATGTAATAAGGTAAAGTAATCTGTCTTTTAGACTTTCATCT
440	ACGCCAACCGCACTCTAGAACCTGAAAAATAAACCCCTC
441	AGAATAACTTTTTCAAGAAAGCTTTGATTGCTATTTATTTA
442	ACGTTATACAATAAGAACCATAGACGTTAGCCCTCATCCTTTAAT
443	ACAAACAATTGACAACCTCGTATCTGGCCAAAATTATTTGCACGAAG
444	ATACATTTGAGGATTTAGAAGTATGAAAGCGT
445	TTAATGCGCGAACTGACTAAAATAAATTCATCCTGAACCT
446	GGCGAAAGGGGATGTGCTGCAAGAACCAATA
447	GTCTGGCCTTTTTTCTGTAGCCAAAGGTTTTTTATCAGGTCATTG
448	GCTTTCATCAACATTAGCGCAACTGAATTTGTGGAAGATC
449	GTTGGGTAAACGCCAGGGTTTTCTGATAATCATCAAATTAATCTG
450	ACCGCCACCCTTTTTCAGAACCGCACGGTTTTTGTGAGTGCCTTGA

TABLE 4-continued

Staple sequences used for the T1 ring triangle folding	
SEQ ID NO:	
451	CACCCCTCAGAGCCACCAATCTCCAGCAACGGACAACTTT
452	ATAGCAAGCCCAATAGAGGAATTGGTGAGAATAGAAAGGA
453	TTTGAATGGTTTTTTATTAGTCTAGTATTTTAAAGAACTCAAAC
454	AAAACGACGTTGGCAAATCAACAGAATCAATATCTGGTCA
455	CTTGCTTTCGAGGTGAATTTCTTATACTCAGG
456	AATTTTTTCACGTTGAACCCCTCATTTTGCTAAATGATACAAACGCCTG
457	CCTGAGAGATCAAAAATAATTGCGCGCCAGCTCGCCATGTTT
458	TACGGTGTATTTTATCCATTACCAGGCGCTAGGGCGCTGCGCGCTTA
459	GCAAAATCTTAGCTATATTTATAACCACGG
460	CGAGATATCCACTATTATTTTCGTCTCTTTTTTCGCGCAA
461	AAAACGCTCTATAAAACAGAAATACCTTAGAATGAATTAC
462	TACCGCCAAGAACCCTTCTGACCTTAGACTTTACCACCAGAAGGAGCG
463	ATCGGCCTGTGGCACAGACAATATAATAGATACCTGATTATC
464	GCGCATTAGATCGCGCAGAGGCGTAGATACCAAG
465	TTTGCAACCAGCTACAGAGGTTTAAATTACATAAAAAATTACTAGAAAA
466	CATCAATATTTTGTATATTCAACCGTTCTCGTGAGAGATCTACA
467	TAAAGGTGAAATCCGCGACCTGCATTGATAAATCCGCCTCC
468	CCAAAAGAACAACGCGGTCCGTTAAGGATTGCCGTGTACCA
469	GTCAAAGGCAGTTTGGGTAGAACGGTAGGGGGTTTCTGCC
470	GACGATTGTTTTTCTTGATATTCACAACCTGGTAATAAGTTTTTA
471	AGTTGATACCATTAGATACTCCATGTTATTTTTTTAGTTGACGGA
472	GTAACAGTTACCGCCACCCTCAGATTTATCAGGACAGCATCG
473	AATCAGTGTTTTTGGCCACCGAGTAAAAAACATCACTTGCCTG
474	TTAACGTCTTTTTAAATGAAGGGAGCCCCCTTTTATTAGAGCTTGGTTTTTAT
475	CTCAGAGCATAAAGCTTAAGAAAAGTAAGCAGATAGCCG
476	TAATTTGCCAGTTACAAAATAAAAAGGAGCGTTAGATTGCGCTGATT
477	GTCATAGCTGTGTGACCCCGCGTGGCTGACTTACCCAATAGCGTC
478	TTTCATCGGCATTTTCGAGCGCCAAAGACAAAAGGGCGAC
479	GAGAAAGGACAGGAACGGTACGCAACAATATTTTTTCAGG
480	GTAATTGAAAGTTTTTCAAGCAAGACCAAGTA
481	ACAAATAACTCTGAATTTATTTTTCGTTCCAGTTCAAGGTTGAG
482	ATCCTATCAGGGCGATGGCGGGTAAAGTTAAACCGGACTTAACAAGAGGGGTTGAG
483	TTCCGGCAGTAAAAAAATGCCAATTACGGCTAGCTGTT
484	TACTCATTTGGGGCGCGACGGAGATTTGTATTGACCAACTTAGTTTGTCCAATT
485	TTATCACCGTCACCGTTTTTCTTGAGCCATTTGGAATTATTCAT
486	GAATTAGAAGTAGCGATAAAGCCATAGCAGCAAGTTTCGTACAGACAGTAAATGA

TABLE 4-continued

Staple sequences used for the T1 ring triangle folding	
SEQ ID NO:	
487	AATCACCAGTAGCTCAACATGGAACGAGGCGCAG
488	GGTAAATAGCGTCTTTCCGTAATCGCCAGCAAACCATCGATTGCGGATGCTCCTT
489	AAAATCCCTGGCATGATTAAGACTTTTTCCTTATTACGCAGTAACGGAATAC
490	ATTCTACTAAAATACACGAAAATCCTGTTTCAGCCTCCGGCCA
491	TACAAGCTGATGAACGTCAGTGAAACACAATAATCGCCAAAAGGAA
492	AGCGCAGTATCCTCATCAGAATCAAGTTGCCCGATTGAGGGAGGGAA
493	ATAACGGCAATTTTCATTTCCCTTGAACCTCCGACCGTGTGAT
494	GTAACAGTCAGTACATATCGTCGCTTTAACA
495	AATATACACGGGAGAATTAAGTGTTTTACACCTGAACAAAAACAGGGAA
496	AGCAATACATGTGAGAAGTACGGGAAAGCCGGCGAAACGATTTTTTGT
497	AATAAATAAATAATCCTGTTTTTTGTTTGGATTATATCATATT
498	ATTAACACCGCTTCTGCTCATTTGCAGCGGGGTCTGGTC
499	ATTTTGATGAGAGATAGACTTTTTTCTCCGTGGTGAAACGTACAG
500	GAACAAAACATCCAATAAATCTGATATTTTATTAATGCCAAAGAGACAGT
501	ATTCAACTTTAGCGTCAGTTTTTCTGTAGCGCGTTGCAGGTCA
502	GGCTTTTGCTACAGAGGCTTTTTTTGAGGACTAAAGACCAGCGAAA
503	GTAACGATCCAGTCACACGACCGTAACACTGAGCGGTCCCAATGAA
504	GCGCTAATGCGCCCAATAGCACCTGAGCAAAAGA
505	TCACCTTGCCGAACGAACCACCAGAGGACGCAAATTAACCGTTG
506	CCACGCTGAGAGCCAGGTGAGGCGGTATTAACCGTTTTTGTAGGGCT
507	ACACCGGAATCATAGAAGTTTGTGCTGAATCCC
508	ATGGGTAAGTGGTGCCATCTTTTTCACGCAACCAGCCGGCAGC
509	AACCGTGCATCTGCCATGGGATAAGCTGATTGCAAGCGG
510	CCTACGGGGTTGCCCTGATAGCTGCATTAATGAATCGGCCTAACCGA
511	AACAAAGATTAGCAAAATTTTTAAGCAATAAAGCCAAATCAC
512	AGATAAGGCACCAACCTCTGCTCATGTGTACAGAGCAACACTATCATGAGG
513	GAGGAAGTCACTAAAACACAAGCGTCATACAT
514	TTTTTCATAGGTTTAGGCCCGTATAAACAGTTTTTGACAGGTCTTTGAC
515	TATGGGATTTTCAGGGCAAACACGAGTGTA
516	CCACGCAGTGCCGAAACCAGGCTCCGGCACCGCTTCTG
517	GATACCGATAGTTGCGCCGACAAAGGCTGAGTAATGC
518	ATAATGGATTTAACGTCAGTTCTTTGATTAGT
519	ACTTCTGAAAGAATACTGCTGGTAATATCCAGCAGAATCC
520	AGATGATGGCTCTTTAGGAGCACTAACAACATTTTTTAGATTAGAGCCGTC
521	ACCAGTCCCGTTGGGAAGGGCGATCGGTGCGTTTTTGCCTCTTCGCTATTA
522	GAACGAGGGTAAAAAAAGGCTCCAAAAGGAGTTTTTCTTTAATTGTATCGG

TABLE 4-continued

Staple sequences used for the T1 ring triangle folding	
SEQ ID NO:	
523	GAGTAACATTATCATTAAAGACAAATTAGATTAATGGTTT
524	AGGGTTAGTCAATAGTAATGCTGATTAGTTAAGACGACAA
525	CTCCGGCTTAGGTTTTTTGGGTTATATAACGAATTATC
526	ATCCCATCATCGGCTGACCGACAATTAGGCAGAGGCATT
527	TCAGCAAACCTGCATCTAACTCACATTTTTTAATTGCGTTGCGCTC
528	TGGTGTGTGGGTCAGTGTGCCCTGTTTTGGCTGGTA
529	TTGCTCGTCATATTTTTACATCCCTTACACTCGGCGAA
530	CGGTTTGCAGGTTTCTGCACTCCAGACAGTATCGGCCTCATCTCCGTGG
531	AACGCGCGTGGTTTTTAGTGTAATTATCCGCTCACAAT
532	TTATACCAGCTTGAGATGGTTTAATTTTTTCAACTTT
533	CTTATGCGATTTTTTTAAGAACTGGCTCAACCCCTCAG
534	ATCGCGTTAGCAAACCTCAGAAAACGTCATAAATATTCAT
535	TATATGTAGAATTTATCAAAATCATTTTTTGGTCTGAG
536	AGACTACCGAGTGAATAACTTTTTTGTCTCTGTAAAAATCAA
537	AATCAATACTAATTTAAATGCAGAACGCGCCTGTTTATCATAAAGT
538	TCTTTCCTTATCGCTCAACAGTGGCCAAGCTACGTTGT
539	TTGCACTGTTTCTCTGAACAAGAAAAATAATGGCCAGT
540	TTTGAAAGAGTTAACCCTCGTTTACCAGACGACGAT
541	CGCCAGGGGGGAGAGGACTGCCCGCTTTCAGTCGGGAAAGCATAA
542	CTTTTACCAGTGGCCGCATCGTTATATTCGACCATCGC
543	AATCATTGATCTTGACAAGTTTTTACCGGATATTCACTTCATC
544	AGACCGGATTAATTCGAGAAGCAAAGCGGATTGCATCAAAACAGTT
545	CCAACAGGTCAGATATTTTGTGAAAAGTTTTTGCCCGA
546	GCGATAGCGAAAAGCCCGAAAGACTTCAAATTAATTTT
547	GGCATTCCAAGAACGGGTCAGTACCATCACCCTAATC
548	TCATTGCAAGTCTCTGTGGTGTGCGGCCAGAATGCCAT
549	GGAGGTGTGGTTGCGGACGCAGAAAACGGATA
550	AACGATTAGAGAGTAAGTTAGCAACGTCAGAGCGGGAG
551	TATATGTATCGAGAATGGGTCGCAGAAGATAAACAGAGCAGCAA
552	ACCGTCGCCCTGAGATAGCATTACGGCGGATTGACCGTAAGTTTGAG
553	AAGAGTAAGGTCATTGAATGGAATAGCATTCCACCAGTA
554	TGGCTTAGGGTAATAGCCAAAATAGCGAGAGTTCATTCACTAAAG
555	GAGCACATCCTCATAACGACCGCAAGAGCCGACCAAGTTGGG
556	GCTTTTGCTTAGAATATAATGCTGTAGCACCTGAATC
557	ACCGAACCATCGCCTTCGCAAATAGAAAATTCATATGGTTTACC
558	ACGGTCAATCATAAGGGAGCATAGGCATTATACCAAGAGGCA

TABLE 4-continued

Staple sequences used for the T1 ring triangle folding	
SEQ ID NO:	
559	ACAAAATTGAAGCCTTACCTCCCGACTTGCGGCTGGAAGT
560	AGATGATGAAACAAACATAATGGAAAACCTTTTATGCGTAGA
561	ATGCGTTATACAAATAGAACGCTAGAAGGCTTATCCG
562	AGCACGCGTGCCTTTTGTTCATTTCGTAATTTTTATGGTGGCGG
563	AAAATAAAATGTTTTTTAGACTAGGCATAGTAAGACCAGGC
564	GCCGTTTAGCAGCAGAACGTGCGATGCTAACGTGGAGCTATC
565	AGCCTGTTTAGTTTTTTTCATTAAATTGAGATTTTTTCGCCAAATA
566	CTCAACAATTTTCATCGTAGCGCTACGTGA
567	TCACGGTCATACCTAAAGCCAACGTTTCGAGCC
568	GGGTACCTCCACGCTGGTTACAGAAAAGGTTTGGTGT
569	GAGCTCGATTTCGCGTCCGTGAGCATCAAAGAAATCG
570	ATCCCCTCCACACATAAGAGACGGGCAACGGTCACGGGCATCA
571	CAGAGGGAGCTTAATTGCTACAAGGCCGGA
572	TTTTTAAACAGGAAGATTAAC TAGCATGTCAATCATATTTTT
573	TTTTTCAATGCCTGAGTAATGTGTCTTTAGTGATGAACCATGGTGCTG
574	TTTTTAGCACTAAATCGGAACCTAAAAATAGCAGCCTTTAC
575	TTTTTCATCGCCATTAGAGTCTGTCCATCTGCCGTAATTTTT
576	TTGAAAGGAATTGAGGAAGGTTATTAGCCCTAAAAATTTTT
577	TTTTTTGTACCCCGGTCAGTCACGTTT CAGAGTGAGCCGGATGCCGGCAATCCGC
578	GCGAAAAACCGTAGATAATAAGAGCAAGAAACAATGAATATTTTAAATGTTTTT
579	TTTTTTAAAGGGTGAGGAATCGATCGGTTGTGAAAAAGAGTATGAGCC
580	GATTCAAATACTTTTGAAATATTTAAATTGTAAACGTTTTTT
581	GAACGGTAAATCAGCTAAATTCGCATTAAATTAGAAAAGCCCAATTTTT
582	TTTTTTAATATTTTGTTACATTTTTTGCATTAAACCTCACCGGAACAA
583	TTTTTCGCAAGGATAAAACCCCTCATAATAGCAATACTCCAAC
584	ATCGTAAGTATAAGCCGGGAGAAGCCTTTATTTCAATTTTT
585	AGGCGGCAGGTAAAAAATAGAAATTTTACATTATGACCTGTTTTT
586	TTTTTGCCACCACCCTCAGAGCCGCCTCTGAAACATGTTTTT
587	AAAGCGCCATTTCGCCATT CAGGCTAATGTGAGCGATTTTT
588	TTTTTGTAACAACCGTAGCATACAGGTTTTT
589	TTTTTCAAGGCAAAGATTACCAGAAGGAAACCGAGGAAACAAAGAAAC
590	TTTTTAAAGTATTAAGTGACAACAGTCGCTGAGGCTTGCACTACGTTAACGAGAAA
591	CACCACCCTCAGAGAGAGCCACCACCGGAACCCCTTATTAGCGTTTTT
592	AGAGCCGCCATAATCATTGCTCAGTACCAGGCGGATATTTTT
593	TTTTTAGGATTAGGATTACCTATTATACCAGAACAAACAAAG
594	TTTTTACCCCGCCACCCACAATCAATGGTCAATAACCTGTGAGTAGAT

TABLE 4-continued

Staple sequences used for the T1 ring triangle folding	
SEQ ID NO:	
595	CTGCCTATAATAGGTGGGGTTGATATAAGTATACTCCTCAAGAGATTTTT
596	AAATACGAGCCCGGTTCCGAAGCGGGTTAAATCACCGGATTTTT
597	TTTTTTTTGCCATCTTTTCGCCAGCAAATGCCCCAAAGAATA
598	TTTTTAGTGCCGTCGAGATATCACCGAACAGCTTCTTTTGCGGGATCGTC
599	TTGCGGAATATCAACAGAGATGCCATTGTTTTGT
600	AAGAAATCATCGGGGATTTTAGAAGGGAATGGTT
601	CTAAACATTCTCGTTAGAATCAACGCTGCGCGT
602	TCTTCCGTACCAGTAATAAAATACATTGG
603	CTCAATCGTCTGAAATTACCTACACAACAGGAATTAAAGGAGAAACA
604	TATTCACATGGAAAGGATTATTGGGACATTTAAATCCT
605	TCCACACCCGCGCAAGTGTCAGAGCCTTACCAAC
606	TGACGAGCAGTATCGCGTACTGAAAGCGACAGCCATA
H1 design oligo sequences (5'-3')	
607	GCAGTAGAGTAGGTAGAGATTAGGCACCACAAGATTAAGCAAATCAGATAGAGCGTTTTTTTAGCG AAAATCAA
608	GCAGTAGAGTAGGTAGAGATTAGGCATAATATGTTAGCAAACGTAGAATAGTAGGAGTTGCAGCCCTT CA
609	GCAGTAGAGTAGGTAGAGATTAGGCAAATAAGTTTATTTTGTCTCAGAACCGCCACCCTCAGATTTTT
610	GCAGTAGAGTAGGTAGAGATTAGGCATCCCAATCCAAATAAGAACGTGGCTTACAAAA
611	GCAGTAGAGTAGGTAGAGATTAGGCACTGCGAACCCTTATAACTCCTCACAGGTGCCCCAGCAGG
612	GCAGTAGAGTAGGTAGAGATTAGGCATTACCGAAGCCCTTTTAAATCGGT
613	GCAGTAGAGTAGGTAGAGATTAGGCAGCTAACGACCGTTTTAAATATGCACATATAAC
614	GCAGTAGAGTAGGTAGAGATTAGGCAGATAATTATTCATTTTTTCAATAACATAAGTCAGAGG
615	GCAGTAGAGTAGGTAGAGATTAGGCATGTTGTTCTGATTCTATCTTACCATCAGGAATCATTACC
H2 design oligo sequences (5'-3')	
616	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTCCACAAGATTAAGCAAATCAGATA GAGGCGTTTTTTTAGCGAAAATCAA
617	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTAAATATGTTAGCAAACGTAGAATA GTAGGAGTTGCAGCCCTTCA
618	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTAAATAAGTTATTTGTCTCAGAACC GCCACCCTCAGATTTTT
619	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTCCCAATCCAAATAAGAACGTGGC TTACAAAA
620	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTCTGCGAACCETTATAACTCCTCAC AGGTGCCCCAGCAGG
621	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTACCGAAGCCCTTTTAAATCGGT
622	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTGCTAACGACCGTTTTAAATATGCA CATATAAC
623	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTTGATAATTATTCATTTTTTCAATAA CATAAGTCAGAGG

TABLE 4-continued

Staple sequences used for the T1 ring triangle folding	
SEQ ID NO:	
624	GCAGTAGAGTAGGTAGAGATTAGGCATTTTTTTTTTTTTTTTTTTGTTGTTTCGTATTCTATCTTACCAT CAGGAATCATTACC
H3 design oligo sequences (5'-3')	
625	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCACCAAGATTAAG CAAATCAGATAGAGCGTTTTTTTTTAGCGAAAATCAA
626	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATAATATGTTAGCAA ACGTAGAAATAGTAGGAGTTGCAGCCCTTCA
627	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATAATAAGTTTATTT TGCTCAGAACCGCCACCCTCAGATTTTT
628	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATCCCAATCCAAAT AAGAACGTGGCTTACAAAA
629	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCACTGCGAACCCCTTA TAACTCCTCACAGTGCCCCAGCAGG
630	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATTACCGAAGCCCT TTTAAATCGGT
631	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAGCTAACGACCGTT TTAAATATGCACATATAAC
632	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCAGATAATTATTCATT TTTTTCAATAACATAAGTCAGAGG
633	GCAGTAGAGTAGGTAGAGATTAGGCATTGCAGTAGAGTAGGTAGAGATTAGGCATGTTGTTTCGTATT TATCTTACCATCAGGAATCATTACC

TABLE 5

Staple sequences used for the T3 triangle 1 folding.	
SEQ ID NO:	
Core structure oligo sequences (5'-3')	
634	TATCAGGGACGGGAAAGCACTAAAGCGGGAGTAGACAGG
635	AGAACGGGATCCGGTAAGCAGCCTTAACGTCAGTGAGAGA
636	AAAGTGTCTGCCCGCTTTCCAGCGGTGCCGGGGTTTCTGCCAGC
637	TATTAGCTGGTAATATAATACCTATCCTCGTTCAGGGCGC
638	GGAAACAATCGGCGAAGATAGCTCAAGAAACACGGAATTT
639	AACAAAATTAATTACATAAGTATACCACCCTCAGACTCCT
640	GCAGAGGCATAACGGATGAGTTTCTCCACAGACCTTGAGCAAATAA
641	CCACCCTCTAGGTGTATCACCGTAAACAAACTTACCTGA
642	TAACAGTGCCCGTATAGGTCAGTGCAGCCCTCATAGGAGCGTAGATT
643	CATACTGGTAATAAGTTTTAACGGAACAGTTACATGTACGGATAGC
644	TTCTAAGAGAGAAACAGAATTATTAGAGCCAGCCCGGAA
645	AATAATATCGTTTTAGCGAACCTCACGATTTTTCTTACCAACG
646	ATCGTCGCGATGCAAAATCCTTTTATCGCAAGACAAAGATGATG
647	AGCGGTGCTCGGGAAGATCCCCGACACAAACAGCAGCAAGCGG
648	AATAAGAACCAGCTTGCGGGAGGTAATAGATAAGTCCTGA
649	CAAGTACCGCACTCATTAGGAATCCTCACATTGTGTCACT
650	TTCCGCTGTTCAGGTTTTACAGAGCGGCAATAAAGAGCTCACGGAATTCTCCGT
651	GTAGAAGAACTCAAAATCTTTGATCAAATTAACCGTTGTA
652	GAATGCGGCGGCGGAGGCGGATGTGTTTTTCTTCTGAAACATGA
653	CCGCGCTTAATGCGCGCTAAGAATCAGATCGGAACATCCGCTC
654	CCAGAACAACCGGTACTGAGGCCAGCTCGAATGTACATACCCCTGCAT
655	CCAAGCTTTCAGAGGTGGAGCCGCGCATTTTTGGGAACGGATA
656	CAAGTTACTGCGGCCAGCGCGCTGTGCACTCTTTAACAAGCATGTA
657	GCAATAGCTATCTTACACCTCACCGGAAATTTTTTATTCATTAAA
658	AAGTATTAATTAGCGGTGAGAGGGTTGATATGTGGTGC
659	CAGACGATCCAGCGCAAAATGCGTATGGTCATAGCTTACAAGCCCAA
660	GTACATAAATCAATATATGTGAGTGATTTTTTAACCTTG
661	AATGAGTGGAAGTTTCTGTGTGCTGAGAAACGTGCTT

TABLE 5-continued

Staple sequences used for the T3 triangle 1 folding.	
SEQ ID NO:	
662	CTTTCACAATTACCGTTTTTTTCAGTAAGCGTCATACATGG
663	AAGTGCCGGGTTTTGCTCAGTACCGCAACTCTATCGGCATAATCAG
664	CAGTTGAGCCTGTCTAGAGAGTTTACGAGCCCCGTGGTGGTCCGAA
665	AAAATCGCGCAAGCAATCAGTAAATTACCGCGCCCAATAGAAACCAA
666	GATGAATATGTAGCATGTCACCAGTACAACTGCCACCTCATTTCAA
667	AGAGCCGCAAGAGAAGGATTAGGAGAGGCTGATTTTCAGCGTAACAC
668	AAGGTAAATATTGTTTACCAGTCCATGAAATACGTCAAAATGAAAT
669	GAGATAAAGAAATGACAGTATATAGGAACCTCGGAACC
670	TTTAGCACACGCCCTACAGTAACAGTAGAATTGTTTAA
671	ATTCACAAACCACCAGAACCATTTTTCACCAGAGCCGC
672	TTTGATTTTTAGCCTTAAATCATTATTTATTTTCCCAATCCA
673	CCGATTACTTCTGAATTTCTGCTCATTGCGCCAGC
674	TAACGAGCGTTAGATTTTGACCTTTTGTAGCTACAGAACGCGA
675	CGAAGCCCTTTTTATTTTGAAGTAAGCAGATAGCCGA
676	GTAACGATAGCAAAATTTATTTGCGCATTAGAAGAATAACGATAACCC
677	AGCGGATCAGGGAAGCACGTAAACAGAACCCGTACTATGTAATCCTGTCAGATGA
678	ATCAAGAAGAATTACCTTTTTTAATGGAACAACAAGAAAGCAAAAGA
679	CCCTCAGAACCGTCAGACTTTTTTAGCGCG
680	GATACAGGAGTGTGACGCTCACCTGATTAATTGTTGGATCTGGCA
681	ATAATGGAAGGTTACATCAATAGTTGCTTT
682	GGGATTTTGTAGTAAATGAATTGAACCTACCATATACTAAAGAGACGCAGAAAC
683	TTTGGGTCGAGGTGGGAAGCATACAATTCC
684	CTTCTGTAACAGAACGCGCCTGTTTTTTTATCAAC
685	AGGGCGAAAAACCGTCAGTTGGGCAAGGAGC
686	GTCACTTTTTACTTGAGACGTACAGCGCCATGA
687	ATCCCGTAGAAGATTTTTAAGCGGTTGTGTACATCGAC
688	TTAGAAGTGCCATTGGTGAGGCGGTGCCCTG
689	AGGAATACCGAACGAACCACCGCCACGCTGCACCTTGCTGAACCT
690	TCATATTATCGTCTGATTTTTATGGATTACAACAGGAAATTTTACGCTCATGGA
691	AGAACCCTTTACATTGGCAGATACCAGAAGATCATTTGCGGAACAATGCTGAT
692	CATCGCCATTAAACAAACTGATAGTGGCTATTAGTCTTTA
693	CAGCAGGCTTTTTAAATCCTGTTTGATAAAGCCGGCGAACGTGG
694	TGGTCAGCAGCAACCGAGCACATCAATTTTAAAGTTAACCACACACC
695	ACGTATTCTCATAACTGCCGTTACGGGTAAAGTTAAACGAGTTTGAG
696	AATCATACTGACCATTTTCTGCGAGCTTGCCC
697	GTTACGCAAAATCGTTAACTTTTGCATCAGATGCCGGGTTACCTGCAGCC
698	GGTACCGACCGAGTAAAGAGCAGTTGTTTTAAGGAATTGA
699	TCCTCACTGTTCTTTTTTGCGTCCGTGAGCCGGGTCAGTGT
700	CCAACGGCAGCACCCAGCCCGAGGAGTCCACTATTAAAG
701	GCTCATGCAAAATAGATTTAAAGCATAGAGCCAGCAGCAATGGGCGCGA
702	CTTATAAATCGCTGATTGCCCTTTTTTACCAGCCTGGCCCTG
703	TTTTAATGACAGTATCGGCCTCAGGAAGATAATATTC
704	CTAACGGAACAACATTATTACATTTCACTTTAATCCAGCGATT
705	GCCAGCTGCATTAATGATTTTTTCGGCCAACGCGCGG
706	ATATTATTTTTCGCCAATTAGACTGGAAGGTTAATCCGCTGCCCTGC
707	CCGATTATTTTAGGGATTCTAAACTTTTGGAGGTACAAAC
708	GAACCTGACCACTTTGCCGAATAACCTGTTTAGGTGGCATC
709	AGACAATATCTGGCCATAAGAATACAATAGAAACAAC
710	GCTATATTTTCATTTGCAATCATAAACGTAAACAAAGCT
711	AGGTCTTTCAAAGGATGTTTTAAACAGTTGATTTCCCAA
712	AAAAAAGCCGCACAGCGGCCCTTAGTGATGACGGCAAC
713	TGTTACTTGTCATTGCAGGCGCTTCAACCAGCTTGAGGATAACTCGTA
714	CTTTTGATGCTCAACACAGTTTCACTGACGAGAAACACGATTTCATTA
715	AATCAATATCTGGTCAGTTGTTTTTCAAATACGCGTGC
716	AAAACGAGAATACGTACGCGTGGTTAATTGCAGAGCCGT
717	AGATACATCCCAATCAGGGAACCTTTGTATCATCGCCTGATAAATTG
718	TCAGCGGGAGCCGGAATCCGCGACCTGCTCCAAGGTTTCT
719	GAGGAAAGCCAGCTTTCGCAATACACTATCATAACCTCTCGT
720	GGAACGTGCCGACTTGTAGAGACATTTTGCGCGCATCGGATAGGTC
721	ATCGCGGAGGCAAGAGTACCAAAATGCTGTAAGAGGTCCTATAAT
722	TTGCTCGTGGCTGGTAATGGGTAAACAAATATCTATCTTTA
723	TGTCGAAACGAGGCGCAGACGGTGAAAAATCATTTAGTT
724	AGTGATAGCGTTTTTTCACGCTGCTTTTTTCGTCGAGG
725	GGGACGACTAACCGTGCATCTGCCGTAGATGGGGATGGCTCACACGAC
726	CAGATGAAATACCAAGCGCAAAACAAAGTACAACGGAGA
727	ACCGCCTGCAACAGTGAGAAGATAGGAGCACTTAATACATTTACGGCT
728	TAAAAAAACGTGGACTTTTTCCAACGTCAAATCGGCAAAATC
729	CAATCTGTCCATCAGTAGTAATAACTTTTTCACTTCAG
730	AAACAGAGTTCTGACCTGAAAGCGACAGAGATTTAAATCCTTTGAGC
731	AAGAGCAACTGCGGAATCGTCATACGCACTCCGCCCGAAA
732	GGGCGCGGGTTTGGAGTGAGACGGGCAACAAAAGAAAT
733	TGCTGGTCGGAGGTGTCATCAGTGAATAAGACGAGTAGTGAATATA
734	CATAAACATCCCTTACACTGGTGTGGAGAGGCTTGCGGTATGAGCC

TABLE 5-continued

Staple sequences used for the T3 triangle 1 folding.	
SEQ ID NO:	
735	AACATAAAAGGGACATTTTTTGAACCCATAAACGTGGCAC
736	ACCCTGACTATTATAGTTTTTTAGAAAGCCACATTCATTTTTCTAA
737	TCAAAAGGATTTCGAGTAAACGCTCAGCAGAATCATAGAAGAGTCA
738	ATATGCTTTTTACTAAAGTACGGAGAGTACTTTTTTTAATTGCTC
739	CTCCGGCTAACATAGCGATAGCTTAGATTAAGTTAATTGATTGAAAT
740	ATTCCGCTCTGGCAACAGCTTGGCGGGATCGTCCCGTGTGATAAA
741	GGCGATCGGTGCGGCGACAATGAACGCCATTACGCTCAAAGCCTC
742	TCTTTGACCCCATTTGTGAATTTTTTACCTTATGCGATAATAAACGA
743	ACCGAACCTTAATATTTTGCCCTGAAGATCTACAAAGGCTACGCCACCC
744	TCCTCATTAAGCCCCAGACGATCGGTCTAATCAAATACCTAAAT
745	TTCTAGCTGAAGTAGTAGTTTTTATTAACATCCAATAGCTGAAAA
746	GTGAGAAAGGCCGTTGAGGCAGGTAGAATGGAAGCACCGTTACCATTA
747	ATAATTACCTTTCCAGAGCCTTTTGACATTCAACCGATTGAGGGAGG
748	CAAAATAAACAGCCATAAGATTAGTTGCTAAAAACATGTTACGCTAATG
749	ACGACGACTTAATTTCCCGAATCTTTTCATCATCAAGTTTGCC
750	CGCCAGCATCAGAACCCCGCCTCCCTCAGAGCTTTTAACTTAATGGT
751	AATCGGTTATTAGCAAAATTTTTAAGCAATATTTTTTAAACAGGAAG
752	AAGAGAATAGCAAAATATTCACCGATGTGTAGGTAAAGATTGCAATGC
753	CAGTAATTATGACCCGTGAATTTTTCTTTTGGCGGAGAGCATAAAGCT
754	GAAACCGAAAAAGGGCTGTCAAGCGACAGAGGCATTTTGGCCTTGA
755	AAAACCTTTTCAAATATAAATGCTTATTAATTAATTTCCCTTAGAAT
756	GCTACAGAGGCTTTGACCGATATAAAGTTTTTTTTATTCCATAT
757	AACGTGTCATGCCATTGCGCCATTACGGGGGATGTGCTGCAACGCCAGC
758	TCAGTTGTTTTTAGATTTCTACGTTTTTAAGAACTGGCTCACGAAAGAG
759	CGGTGTTTTTACAGACCAGGGCAAAAGAATACACTAAAACATC
760	AAAACCAAAAGAAAAATAGGAATACTGGCTGACCTTCATTTTTTAAGAGTAATC
761	TCGTGTGCCGGAACCATATAATAGTAAAAATGTTTCGACGATA
762	GCCAGTTAACAAAGTTACCAGAAGGGCCAGTGGGTGAATT
763	ACAAACGGCGGATTTTTTACCGTAATGGTCGAGCTT
764	CGCATAGGGGACTAAAGCATGTCAATCATTTTTATGTACC
765	CTGAGGCTTGCAGGGAAACGAGGGATTGTATACGATGAAC
766	ATTCTACTAATTAAGTAAATCGTAAACTAGACTTTTTT
767	GTAAAGGTTGTTAAACAAAAATACTGAGTAAGAAGCC
768	TTGCTTTCGAGGGTTGCAAGGCCCTAAAGGACGGAGTGA
769	CACCCCTTCCTGTAGCGCTTCGGTTTATCAGC
770	TTCGCAAAATGGTCGTTGATATTTTTTCAGAAAAGCCCCAAAA
771	GAGTCTGGAGCAAACTTTATCAACGAAAGACAGCATCGG
772	AATAACAAAGCCTGTTTAGTATCTTAAATTT
773	TACCGCGCCAAAAACAGTCACGACGTTGTAAAAACGAC
774	GTTATATACACGCCACCACCGAAGCCACCCTCAGAGCC
775	ATTAATGCCGGAGAGGGTAGCTAATATGATATTTAAAT
776	AGCACCATAATCAGTATCAATAGATAAATAAGTATACAA
777	CAGCTTTCATCAAGGTCAGGATTACCAATAGGACAACAAC
778	TTGCCATCAGGGAGACAGTCAAATCACCATCATTTTTTGAG
779	TTAAATTTCCGCTTTTATACCGATAGTTGCGCGCCTTTC
780	TCTAAAGTTTTGTTTTGGGAATTAGATTTTTCCAGCAAAA
781	TACCAGAAGACTTTTTTGGATAGCGTCGCACCCG
782	ATTAGCGTTGATAGCAAGAGAACCCTCATATATTTTAAA
783	ATAGTGAAGTAAATACGTAATGCAGTTTCCATTAAACGG
784	TTGACAGGTTTTTCATAGCCCCCTTTAAGCGTAAATTCAT
785	ATCTTCTGCACCGGAACAGACCTCAGAACCGCC
786	AGGAACAAGGAACGTGATAAAAATTTTTCGCAGTCTCTG
787	AGTTTCAGATTGCGAATAATAATTTTTTACCTCACCAGT
788	TTTATTTCCCGTGGGAGCGAGTAACAACCGTGCCTTT
789	TGGTGTTTTTATTTTCATCGCGAGAACAGGGTGCCTCACCCAAA
790	TTTTACGCTCGTAATCTGCGCTCAAAGCCTGAGCAAGCCGGTT
791	ATTGCTTTAGCATATAGAAGGCTTTATTAACACAAGAAATTGACAAT
792	AATTTTATCCTCCTTTTACATCGGACGCGAGGCCATCCTCGGCTGTGCGCT
793	TTTGTTCCTTATCATTCATCAATAATAATTTACGTTTCATTT
794	TCCTGTAAAGCGAGAATTAACTGAACACCCATCAGAGAATAAAAAAC
795	TTTGGTAATTGAGCGCTAATTGAACAAATGAACCATAAACTTAA
796	GCCAGAATAAATTTGTTCCATAAGGTCAGTTTCACTACGGTCAGAGGTTCT
797	AAAGAGGATTGACAAGATGGTTTAAAGGTAGAGTAAT
798	GAGAAAGGATCGCTGGCAGCCTTGGAAACAAATAGGGTTGAGTGT
799	TTTGAGAACCCGATAACAGAGTAAGAGTTTATGCAGATA
800	CCACGCTGGTTTGGCTCGGTGGTGCCTTTTACCTATTGGGCGCCAGT
801	CAAAGCGAATATCGCGCAAAAATCATTGAATCCCT
802	TCAGTTCCGGCCAGCAAGAATGAATTCGAC
803	TTTTTCATCCCAGTCGCACTCATCTAAAAAACCCCTCTATTAAC
804	TATGCTTTAAATTACGAGGCATAGTCATAACGC
805	TAGCAACGCATGAGGACACTACGAAGGCACCAACCTAAAATTATACCAT
806	TACGTTGGGAATAGCGAACGCATAAAAAGCGGATTGTGGGAA
807	GTGCTGGTTTCGGTCGCATCGCCCGAGGCTTTTGT

TABLE 5-continued

Staple sequences used for the T3 triangle 1 folding.	
SEQ ID NO:	
808	TAAGTTTTGCCAGAGGGGGGCAAAGCCAAAAGATTA
809	TAGTAGGGCACGCTGAGGCTCTGAGAGACTACCTCAGGTCA
810	TACATAACAACGCCAACCCAGTATAAGCCAACGT
811	TCACATATAAAAGAAATGTGAATTTCTTAATGAAACCATTGTTAAA
812	AATTGTAATTGGCGATTAAGTTGGGTAAACGCCAGATACATAAT
813	AGTAATTATATAAAGTACCGACAGAATCGCCATATTAATT
814	AAAGGTAAATTCTTACATGTAATTTGGCATGATTAAAGT
815	TAGCAGAGAAAAATACGGTTTTCCAAGGCTCCAAAAGGA
816	TCATCAGTAATAAGAGACTGTCCAGCCTTGAATAGGTTGG
817	ATGTTAGCCACCACGAAAAGAACTAGGCAGAGGCATTTTCGT
818	TCGGAATACCCGAATAAGTATATGCGTAATAAAACA
819	CGCAAGAAAAACGTACTCCTTATTACGCAGTGGAAACGCAATAAAGTT
820	CTCAGGTTTTTGGTTTAGTACCGCACTATATGTATTTTAG
H1 design oligo sequences (5'-3')	
821	GCAGTAGAGTAGGTAGAGATTAGGCATGGCAATTTTCTGTATCTTTTGATGACTTAGC
822	GCAGTAGAGTAGGTAGAGATTAGGCAGGTGAAGGCCAGCTCTTTCCAGACGCTAAACA
823	GCAGTAGAGTAGGTAGAGATTAGGCAGGGCGCTAGGGCGTATAGAGCTTGCGATGGCC
824	GCAGTAGAGTAGGTAGAGATTAGGCAACGTTGGTAAGAAACCTCACCAGTTAGCCCGA
825	GCAGTAGAGTAGGTAGAGATTAGGCAGCGGTCCGGCGTTGAGTAACATTGAGCGGAATTATC
826	GCAGTAGAGTAGGTAGAGATTAGGCAGACTTCAAACCGACCTTTTGAAGCAAACCTCCAGCTGCGC
827	GCAGTAGAGTAGGTAGAGATTAGGCAGAAATAGAACGGATTCTAACGCAAG
828	GCAGTAGAGTAGGTAGAGATTAGGCATGGCGAAAACACATTAAATGTGA
829	GCAGTAGAGTAGGTAGAGATTAGGCAATCACGTTGAAAATCTCCAAAGAC

TABLE 6

Staple sequences used for the T3 triangle 2 folding.	
SEQ ID NO:	
Core structure oligo sequences (5'-3')	
830	GTGCCCCCTGCCCCGTAGAGAGTTCGGGGGTCCGTGGTGGTTCCGAA
831	AGAAAAACAAGAAGATGATGAACCCCTGCCTGTACAGGGATACAGAGCCAGA
832	GAGCAAAATTATTTGACGTATAATGATATAACTGAATTT
833	ACGGATTTCTTTGCTCGGCTGGTAATGGGAAACATCAGAAAAATATGCTTTGA
834	GAATTATTAGTACCTTTTCAGGTTTAACATTGAAGCCT
835	TAGCAAGCAATAAAAGCCGTTTTTTTTTATTTCTGCTGATGC
836	ATTTGAATTACCTTTTTTAATGGAAATTTTGTAGTACATA
837	ACCCCTATAAATGGAAGGGTTAGAATGAAAAACGATTATTATTTA
838	GCACGCGAGCATTAAATTGCGTTCCGGGCGAGCAAAATCGTTAACGG
839	CGCGCTTAATGCGCCGCTACGAATCAGAATCGGAACCGGCGGGC
840	TGCGTAGAGCCACCTACCGTACTCAGGAGGTAGCGGGGTTTACAAA
841	ACATGTGGTGCTGCGGCTGAGAAGCGTGCTTT
842	TTTACATCACAGAAATCAAATAAGATAGCAGCTCTTACCA
843	TAAAAGAAACGGGGATGTGCTGCAACATAAAATAAATCAAGATTAGTT
844	CAGTACCTGGTGTATCCAGAACCGTCATTAACGCCACC
845	TACCGTAAAGCAAGCCCAATAGGATACTTCTGAATACGCACGGCCAGTGCCAAGC
846	ATCGTAGGAATCGTCAGATGAATAATCGGCTGGATAAGTCTGAACAA
847	GCCATTACGGCTGCGCAACTGTTGGGTTTTTAGGGCGATCGG
848	TTTCAGAGAACGTCAAAACCTACCATATCCCTACTATGGATCAGATGGGAATTAT
849	TGCCGTACGGGGTCAGTGCCTTGCAATACTTTATCGGCCTAATCAGT
850	CATGAATTTTTGTATTAAGAGGCTTAACCTCCAAGAACG
851	TCTTCGCTATTACGCCACGACGTTAGAGAATAAGGCGATT
852	CCGATTAAATATAATCGCCACGGGAACGGATAACCTCAC
853	AGGGCGAAAAACCGTCCGGAAACAAAGGAGCG
854	ATTGGCCTTGATATTCACAAACAACGAGTCTGTATAGCTGCCGTC
855	TATCAGGGGGGGGAAAAGCACTAAGCGGGAGCAGACAGGA
856	CGATGGCCTTTGGGGTCGAGGTGTTCTGCCACGTTTCA
857	ACAGGGAAGCGCATTATGCGGGCCAGACACTTTTTCACGGAATAAGT
858	CCCTCAGAACCGTTTGCCATCTTTTTTTTCATAATCAAAA
859	ATGGAAGATAAATCCCCACCTCAGAAAGCCACGTAA
860	ACCGTCTGGTAATATCATACCTACCTCGTTAAGGGCGCG
861	CAGAAACAACCGTACGGAGGCCACGCGCAGTGGTGTGTTCCGGTTGCG
862	GACGGGAGAAATTAATTTTTGAACACCTGAACAAAGTCA
863	TCAATATTTTTAAATTCAGCTGGCGAAAGGCAA
864	GACTTGCGCAAGAACGGGTATTAATAATGCAGAACGCGC
865	ATACATGGAGTTTTAATAACAGTTAATGCCTAAAGGTTGCGCTGAT

TABLE 6-continued

Staple sequences used for the T3 triangle 2 folding.	
SEQ ID NO:	
866	TAGAAGAACTCAAACCTTTGATTAAATTAACCGTTGTAGTTACTG
867	CTGATTGTTGGATTTCCTGATTTTGCTTTG
868	ACGCTAACGAGCGTCTACAGCCATTTTGTTCCTGAGC
869	AATCAATATACAGTAAATACCAAGTTTGCTCAATTCGGA
870	ACCAATTTTGTACCGCACTCCGTTTGTAGTTTGAACCTCCC
871	GGATTAGGATTGAGCCAGCTTTTAAATCAC
872	CAACAATATCTTCTTATCATTCGGAGGTTTACCGCGCCCA
873	TAGCCCCACCTATTATTCTGAAATCAATTACTCGCGCAG
874	TTATTAGGAGTGTACTGGTAATACTTTTGATCGGATAAGCCGGAATA
875	AATCAATATCGACAATAAACAACTTTTTGTTCAG
876	ATGTGAGTTTGGGTATATTTTTACTATATGTAATATTCATT
877	TCAATCCGGCGCTCACTGCATCAGCGGTATAGCAGCAAGCGG
878	GTATGAGCCGGTCAAGGATCCCCGCCTGTGCATCATAGAGAGGGT
879	TCGTCACCCCACCAGAGCCTTTTCCGCGCAGATTGACAGGAG
880	AGGTCAGACGATTTTGACGCTCAAGAAGGAGCATGGCAATTATGGCAA
881	CTGAGCAAAAATTAATTACATTAAACAATTTCTGTTTATAGCGCAAT
882	CGGAAGCATAAAGTGTCATAGCTGACAGTTGATGTTGCC
883	TCGTATAAACATCCAGTAACAGTGTTTTATTGTCCAGTAAGCGTC
884	AATGCATATAACAGTTTTCGCAAAAAAGATGGTCTTACAAAATA
885	AATAGGAATGTAGCCAGCTTTCATCAACATTGGCATAGT
886	AGACAATCTGGCCAAAAGAATAACCTAACAACTATCTAACGGCCAGA
887	CTATTAGTCTTTGTGGCAC
888	CTGGAGGTGTCCAGCATCTTTTTCGGGGTCATTGCAGGCGCTTTCGCAC
889	CCATCCACGCAACAGCTTACGGTAATGAATCAGCGTGGCCAGCG
890	CCACTACGAATACACTAAACACGCGCACGCTTAGAGAGT
891	TTGCTCCTCGCAGACGACCCCAATTAAACGGGTAAAAACGTAATG
892	ATCCTGTCCATCAGCAGTAATAACATTTTCACTTGGAA
893	AGTACAACGGACTAAAGACTTTTTCATGAGGAAGTTTC
894	CGATTAAATTTTGGGATTTTAAACATTTTGTAGGCTTTGAGGA
895	GCGATTATACCAAGCGTCCATTTTTCGGGATTAGCTCAAC
896	GGCTTAGAGCTTAATTTTATCTTTGTCAATCATAGGG
897	TTCCAGTCGGGAAACCTTTTTTTCGTGCCAGCTGCAT
898	AGAGGCGGCGGTGGTGCAAGAAATGCCAACGGCTCTAAAGCAGGAATTG
899	TTGCGGAACAAAGATTTTACCACCATCGTCTGAA
900	CGCCATCAAAAATAATATTTTGTATTATAGTACACGACC
901	AAAAATCATAAGAGGATTTACCAGGACAGATGAACGGTGTGAACGAGG
902	ACGAAAGAGCAGATCCTCATACGCGCTTTTTCAGAGCCTATTAGAC
903	AACCGACTCAATATCTTGAGAGCCACACCGCTGCAACAGTGTGAATA
904	TAGAACGTGGCCAAACAGTGAGACGGGCAACAAAAGAAT
905	ATTGCATCATGTAAAGGGACATTATTTTGAATGG
906	AAGTTTCAACCTTTAATCCAACAGTGAAAGAG
907	TTCATAGCCCTAAAAACATCGCCATAGTATTAAGCAGCAAATGAAAAA
908	CCGAACGATCTGACCTGAAAGCGTCAGAGATATTACAACAAATAG
909	CGAGTAGTAAATTGGGCTTGATCAAGAGTAATCTTGGGCTTTGA
910	AAACGATGCTGATCCAGCCGAGGAGTCCACTATTAAAG
911	ACGATCCACGAGTAAAGAGTAATATCTTTTGGTCAGTTGG
912	CTTATAAATCGCTGATTGCCCTTTTTTACCAGCTGGCCCTG
913	CAGCAGGCTTTTAAATCCTGTTTGATAAGCCGGCAGACGTGGC
914	GTGCCGGCCGGTTTTTTTCTGTCAGTGTGGTCTGGTCAGC
915	GCTGAACCTCAAAATATCAAATTTTCTCTCACATCAGAT
916	TCGTCTCGCTTTAGTGATGAAGGTTGGCGGATTAAATCCTTTGAACCAACACACCCG
917	TGGATTATAACAGGAAATTTTTCGCTCATGGAA
918	CCACGCTGGTTTGTGCCGTTCCGGCATTTCACCGCGCGGGG
919	ATCACCTTGATAAAACAGAGGTGAGGCGGTCTAAAAATAAGGAAGTTAATAGA
920	TATTACTTTTTGCCAGATAATACACAAATCAACGGACTTGAGCAACCG
921	CGCCATGAACGTGGACTTTTTCCAACGTCAAATCGGCAAAATC
922	ACGACGATAAAAAAAGAGCCGCAAGCGGTAGGAGCAAAATGCGCG
923	GAACCTTTTACATTGGCAGATTATCATTAATTTTAAAGTTTGCTCACGGA
924	ACAGGCGGTGCTGGCAGCACTGACCAACTTGTGAGGAT
925	TCCCGGAATTTGTGAGAGATAGACTTTCCTCGCGCAGAAA
926	GTGTAGCGGTTTTTTCAGCTGCGCAACTTAAATTTGGG
927	TTGTGTACAAAAGAGATGGTGAAGGATAGCTTCGCGCTGGCCTTCC
928	GTCAATAGCCATTGCACCAACAGCACTGA
929	TTTACCAGAAGAAATTTTAGCGAATCGGCGAAACGTACA
930	CTTTTGCATACCAAGTCAGCCGCAACAGCCGGAAGCAAAAC
931	AAAATCTACAAAAGGAATTACGAAATGTGATACTGCGG
932	GCGAGTAACAACCTAACGCGTTAATAAACGAACTA
933	ATCGACATAAAAAATCCCGTAACCCCTGACTTAAATCAGTTAAATTT
934	ACTTTAATTTTTTTCATTCTGACGAGCCAAATCAACGTAAACAGCAGCGAA
935	GCGCATCGTAACCGTAATTGCGACGATGAACAGAGTCTGGAGTAGA
936	AAAGAAGTTTGGCAGATTTTGGGGTACCTTATGCGTTTTTTTTT
937	TTTAATTCAAAAAGGCTTTTTCAGTTGAAAAATAGTAAATGAGATGG
938	CGATTGAGCAGTATGTCTTTCAGCTGGTGCCGGAACACGAGCAAAG

TABLE 6-continued

Staple sequences used for the T3 triangle 2 folding.	
SEQ ID NO:	
939	ACCAGAGCCATCGATAGGCCGAATGGTTTGA
940	GCTTAGATCAGGGAGTTAAAGGCCCGGTGCTGAGGCTTG
941	CTTTTTCAGTAATCAGTAGCGACTCCTCAAGAGA
942	AATCCAATCGCAAGACGGCTTAGGGAATAACCTTGCTTCTGTAATCG
943	ATCAGGTCGGGTTTATAAGGAACAATAAGGGCATCTGC
944	GTAAACGTTTTTTAATATTTTGCTCATTTTTTATATTTCATTGAATC
945	TCCTCGTGGGAACAAACAGGAATACCACATTCAACATTATT
946	AACAGTACGGGAGGGAAGACAGCCCTTACAACG
947	CATTCAACATCACCCTATGATTAAATACCGAGAAAAAG
948	AGTACAAATCATAGTTAGCGTAACGATCTCAAAGGGCGA
949	AGTAAGTCAATAACCTGTTTATTTTCTATATTTTCTTTAGTTTGACCA
950	CGCGGAGCGCAAAATATCCAAAACAGGAAGATTTTCT
951	TAATATCAGTAGAAAAAGAACTGGCCACCGACTCCATTACCAGAGCCGCG
952	GAGCCACCACCCTTGGTTTACCAGCGTTTTTCAAAGACAA
953	TCACGGGACGTTTTTATGCTTTAGCGTCAGAAATCAAATTTAGTTAA
954	TAGATACATGATTCCTCAATTTTCTGCGAACGAGCAACCGGTTCTA
955	TTCTAAGAACGCGAGGATCGAGAACAGCGAAAAATTCGTCCAGACGA
956	TAAAACCTAGCATACGAGAATAGACAGCTTGCTTTACCTAAATTTAA
957	TACCCCGGTTGATACAGTTTCAAAAAGAGAAATATAATAAT
958	TCCAAAAGGAGCCTTACCAGTAGGCTGATAAAATGTGTA
959	AATTGTATATTGCTCGGTAATCGAATCATAATTTGGG
960	GCCTTAAATCAGATATAGAAAAATACATACATAAAGGTGGCAACAT
961	AGACTTTTTTAAATATCGCGTAACGAGAAATTTTGACCATAAATC
962	CGGAACAACATAATTTTTTGCAGATACACGTCGGA
963	TTAGCAAAAAGGCTGTAGCTAGGTGAATTTGAGAAGATAGCGATA
964	CATTATGACCTGCACTATTTTGTACGGTGTCTGGTAATGCTG
965	ACGTACCTCATTAACGCACTAATAGTAGTAGCATTAACT
966	GGTAAAGTTAAACACCGGAATCATCCAGGCTT
967	CGACAAAACGAGAAAAATAAAGCAGTAGCATGAGCCATTTGG
968	ACAATGACAACAACCATCTCCAAAGAGCTTTTTTAAAGCGAAC
969	TGTTTTAAATACTGGGTAAGATTCAAAGGCATAACC
970	TTCTGCGATTTTTGAGCCCTCATAACGCAAGGATAAAAAATCTGTAGCG
971	TCAGAGCCACCACCCTCCCTCATTAGCAAGCAGCACCAATATATT
972	ATCCGGTAGAGGTAATTGAGCGCCGCTTCTTATTTTG
973	CATAGGTCTAATTAATTTTCCCTTAGAATCTGTATAAAGTTTCATC
974	GCCTGATATCGCCACGGTGAGAAAGGCTTTTGGAGACA
975	TGGGATAGTATCGGCCTCAGGAAGGAGGGGAC
976	ATTCCACAGGTAATATGGCATCAATTCTCACCAGAACCA
977	TAATACTTTTGCGGGAGAAGCCTTCGGTTGTCGGAGAGG
978	GTATGGGTTTATCGCACTCCAGCCAGCTTCCGGGTTAAGCCACTTTC
979	ACGGCTACAGAACAGAATCTTTTGGATATTCATTACAAACACCAGA
980	GGAGATTTTTTGTATCATCAGACAGCATCGGAACGAGGGTAGC
981	ACAGGTAGGGCTTGCCGTGAATTAAATTGTGTCGAATCTTTTGCAGCTGCT
982	GAAATATAGATCTACATTAAGCAATAAACCCGGAACCGCTCAGAGCGGTGAATT
983	TACCTTTTGAGACAGAATCAAGTTTCGGCATTTTCGGTC
984	AATGAAACCACGCCTCAGAGCATAAAGCTAAATTATTTCA
985	TTTGATAAGAGGTAATCACTTTTATCAATATGATATTTCAA
986	CCCTCAATGCTTGTACGTTGGTGTTTTGACCGTAA
987	GCCTGAGTATTAATGCACCAAAAACAGGCAAGGCAAGAAATCCAATA
988	TATTTTAAATGCAATTAAGACGCTCTTAAACAGCTTGAT
989	ATATCCCATCCTACTCTTTCCTGTGTGAATTTGTTATCCGCTCACCTGT
990	TGCCAATTCACACAACCTAGAAACCGCTATTTTAAAGAAATAAGTTGGG
991	GGGAGAACTCATTACGAGCATGATACGAGCTCCCAATCGCACCCAGCTACAATTT
992	TTAGATTTTATCCTGAACTTTACAGGTAAAACGAGGGTTT
993	TTTTTAACTCACTCGAATTCGTAATCATGGTAAAGCCTGCTTCGCGTCAACCAAA
994	TCACTGCGGGGTACCGTGCCTGTTGGGTGCCTAATGAGTGAGCTCAT
995	TTTGGCCAGTTACAAAATAATTCCAGAGTGAACCATGTGGAGCC
996	CCAGAAATCCAGAATGCCTAAAGGTCAAGTTTCACTACGCCTAATTTTCT
997	TCTTGGCTGGCTCTTAGCCGACAGACCAATAATTTCAAAGAACTG
998	CGAAACAACCATGTTAGACCTTCAGATGGTTGGCGCATATCCT
999	TGAGATCATAACCTCGAGGACGTTGGGAAGAGCTCATTA
1000	GGCAAAAGAAGGCACCAACCTAAAAGCACCGTTTTGCGTATTGAAGT
1001	TAAACGCGCCAGGGTGGTTTTTCAACGCGGTGAACGTGCCAGTTGAA
1002	AGAAAGGAAGTCTCATTTGCTTGAACAAATAGGGTTGAGTGTGACCT
1003	TAACTTCCAGTTGCCAGCAGGTAAAGTTTTTAGAAG
1004	TGCTGTAAATTTAGGCATAAAGTACTCGCTATTGAGAGAC
1005	GAGAGATAATAATAACAGAGGCCATATTTAACACGCAACAAGGT
1006	TCAATTACGGAATACAATTTACTACCGTGTGA
1007	GACTCCTTTGAAACGCAACCCACAAGCAGATAGCCGAACAAAGAAGT
1008	GAGAAATAGAGGCATTTTCGAGCCCCAACGCTCAACAAACT
1009	TCTCGTAGGGCTTAATTTGATCATATGCGTTATT
1010	AGTAATAACCTGTTTAGAGAATCGAAACGAGTACCGAAGCCCTTTGGT
1011	TAGGTTTAAAGAAAGTAAGAATTGACACCGCTTACGTTAGTAAATGAA

TABLE 6-continued

Staple sequences used for the T3 triangle 2 folding.	
SEQ ID NO:	
1012	TTCTTACCATGAAAACAGTCAATAGTGAATTTTTTTAGAA
1013	TAAACAATGAAATAGCAATAGCTATCTATTTTCAGCGGAGTG
1014	TTGGTCAATCATATGCAGATTTTGCTAAACACAATAAAGT
1015	TTGCGCGGATATATTGCTTTTGGCGGATCGTCACCTCAAAGCTGCTT
1016	TAGCTCAGTGAATAAAAAGATTTCATCTAAT
1017	TTGAGATTTGGCGGATTAGACTGGATAGTTTTTGTCCAA
Hi design oligo sequences (5'-3')	
1018	GCAGTAGAGTAGGTAGAGATTAGGCACATCATATAACCCATGGTTGAGGCACGCCGCC
1019	GCAGTAGAGTAGGTAGAGATTAGGCATCCAGTCATACATTTTCAGGGATCACTGAGT
1020	GCAGTAGAGTAGGTAGAGATTAGGCAGGCGCTAGGGCGCTCAGAGCTTGA
1021	GCAGTAGAGTAGGTAGAGATTAGGCACGCAATTAAGTAACATCACCAGTCCAGTCGACAACCTCGT
1022	GCAGTAGAGTAGGTAGAGATTAGGCACAGCGGATCGTCCCGAACGTTATT
1023	GCAGTAGAGTAGGTAGAGATTAGGCAATCGTCATAAAACCGCAGAGGAAGAGCAACACT
1024	GCAGTAGAGTAGGTAGAGATTAGGCACCTGTAGCTTGTAATGAAAAGG
1025	GCAGTAGAGTAGGTAGAGATTAGGCAGACGACAGTAAATCAGAAAAGCCTTAAATT
1026	GCAGTAGAGTAGGTAGAGATTAGGCATCAAAGTTTGTCTGTAGCAAAC

TABLE 7

Staple sequences used for the T3 triangle 3 folding.	
SEQ ID NO:	
Core structure oligo sequences (5'-3')	
1027	AAAATCCTGTTTGAACAGTGTAGCGGTACG
1028	AACAAAATTTTCAATTTTTCAGGGACACTGAGATGATACAGGAGGT
1029	ATCAAGAAACGCGAGGCGTTTGCTATCCGTATTCTAAGAGCACTCAT
1030	GCGCATCGGGAGAAATTTTAGACACCTCAGTGAGTAAC
1031	TTCCCTTAGAATCCTTGAAAAATAGTTTTTGATAGCTT
1032	ATCAATATATGTGAGTTTGTCTCACCCCTCATTTCGGAA
1033	TTGAATACGTACCGTAATAGCAAGCCCAATAGGGAGGTTTTTAAACA
1034	AAACAAACGTTTTCACGCTGCGGCAGAAATGCGAATAACCCAGTACC
1035	AGAAGGATATTAAGAGGCTGAGACGCTGGTAACATGGAAATCTTTGA
1036	CACCGTACTCAGCAGCACCTTTTTTAATCAG
1037	TCGATAGAACCCTCAAGTTACAAAATTGCAATAACAT
1038	GGAGCGGGAAGAAATTCAGTTGGGCGGTTGTGTACATCG
1039	GCCACGGGAACGGATAACCTCACCGGTTTTTAACAATCGGCG
1040	AAACAGCCATAACCTAACGAGCGTTTTCTTCCGTTTGAAAT
1041	CGCCTGTAGCATTATTATTCTAATAATTTTGTGAATTAT
1042	AAAAGAAGTGCTAGCGAACCTCCATTACCGCGCCCAATAGCAGGT
1043	CCGGGGGTTCCCTAAGAGCAATACTACCGGTATAAACAGT
1044	GCGTAGATTTTCAGGTCAAAATTGCCGATTA
1045	TTCTGCCATTCTGTGGTGGGAATGAGCTAACTCACATTACCCAAA
1046	GGTAAATTTTATGACGGTCCCGGAATTGCAA
1047	ACCTGAGCGATTTCGCCGGAAGATTTTGAGCGCTAATATCAACAGCGGATCTCACGG
1048	AGACGTACAAATAAATCCTTTTTCATTAAAGCCAGAATGGAA
1049	CTCTGAATTTACCATAAAAGGGACATGGAAGGGTAAACAGAAAGGGCGC
1050	GTAGTTCCAGTAAGCGTCATACATTAAGTTTTCAGCCACCACCTCA
1051	GCTCATTTCAAAGTCACGTACCTTTTAAAAACAGGAGATTGACGTTAGAAC
1052	AGCGCCATGTTTACCAACGCGAAGAGAGATATTCTCCGT
1053	AGCATTGACCACCTCAGAACTTTTGGCACCCCTCAGA
1054	AGGAGTGTACTGGTAAGGCTTTGTTCGTACCAGGGACAATAACG
1055	AGTGCTACATTTTGACACGACCATACGCCAGGGGAGCTA
1056	AGCGCCAAAGATGAGAGATAGACTACCCACAAAAAAGAGGGAAGCGC
1057	AGGGCGAAAAACCGTCACATAAAACGCGCTTA
1058	TATCAGGGCGCTGGCAGTGGCGAGGAAGTTAGAGTCTG
1059	CGATGGCCTGACGGGGAAGCGGGGTTGCCCGGAAGC
1060	GAATTGAGTTAAGCCCAAACGTACAAGGGCTTTTGACATTCAACCG
1061	AATAATAAGAGCAATTTTAAACAATGAAATAGCAATAGC
1062	CGCTCAATTCCATCACTTAGTAATATAGCTGTGCACGCGTCAGTGTC
1063	TAATGCCCATGAAAGTTAGGATTAGCGGGGTGGCGGGCC
1064	AACAGGAAAAACGCTTATCCAGATCAAACTATCGGCCTT
1065	TTCCCTGTTAGAATCAGAGCAATCTGAAAAGGAAGCATACGAG
1066	CTCCCTCACGGATAAGTGCCGTGCTGGAAACATTTTCATT
1067	GAGCGCCCTATTATTCTGAAACCTGCTAGAACCGCCACCTCAT
1068	TAATGAGACCTGTGTCGACGCTCCAGCGGCTGTTCTTCGCGTC
1069	TAGGAATCGACTTGCGATTAGACGTGATTGCTGGTGAAGG

TABLE 7-continued

Staple sequences used for the T3 triangle 3 folding.	
SEQ ID NO:	
1070	ACCCTGAAAATTGCGTATGCCGCTCACAAATTACCGTTGTAGGAACGG
1071	CAGCCTTTGCTATTTTGACCCAGTCAATAATCGGCTGTC
1072	AGTTTTGTTTCATAGTTAGCGTAATTTAACGTCAGATAGCTCAAACCTAAATTTCT
1073	GTACATAAGTAAATCGTCGCTATTAAATTAATTTTCTTAGAATTACC
1074	TGAATCGGAGAGAGTTTAAAGCCTGCGTGGTGGTTCCGAA
1075	TCAGACGATGCATTAAATTCGTAAATAAAGTGGCAGCAAGCGG
1076	GGAGGTTTATTATTCATAATTACAAGTACCGCGTACCAGG
1077	CTACATTTTATTTTATCCTGCGTCAAAATTTTTGAAAATAG
1078	AGATTAAGATTTACGAGCATGTATTTTAAACCAA
1079	CGCTGAGATCATCTTCTGATTTTCTAAATTTAATGTTTTTAA
1080	CTGCGCGCCTGTGCACGCTTTCCATGAAATTTGTATATGGAACCGCC
1081	TCATTCCATAAATCAAGATTAGTTACAGAGAGCAGTTACAAAA
1082	CGTCAGCGTGGTGCTGGTCTGGAAGGAAGCAAGCGCATCGGATAGGTC
1083	CTGCGAACGAGTAGATGCGACCTGTCAAGAGTAATCTT
1084	GTTTGGATTATACTTTTTCTGAATAATTCTGGCC
1085	GACAAGCGCTTCGACAACAACAGTTTCAATATCTGGTCAGTTAGTTTG
1086	TTTGGGGCGTGCTCGTGTTTTAAATATTACCCA
1087	GTAATATGGTTTTTTGCTTTGACGCTCCGGCCAGACCA
1088	GGCAACAGTGCCACGCTGAGAGCACCTCAAGAAAGGAATTGAGGA
1089	TTTCGCACCTGATAAGTACAACGGAGATTGCATCCCTT
1090	TGCCATCCACGCCCCAGCCCCGAGGAGTCCACTATTAAG
1091	CAAAGCGATCCCTTTGATCGTCATAATCAACGTAAACAAAGGCTGGCT
1092	TCAGATGCCGGGTTACCTTTTTTCAGCCAGCGGTGCCGGTCCCCCTGCA
1093	AAATATTTCATTGTGTCAGCAGCAACGGATTAGACTCGTATT
1094	GTGTTTCAGCAAATCGTTAACGGCAGCGCCAGGGTTGCCCTTCCCCG
1095	CCACGCTGGTTTGAACCAAGCTTACGGCTGGAGGTGGTTGCGG
1096	ACACTGGTCTTTGCTCGTCATAAAAGGTTATCGAAGTATT
1097	CTTATAAATCGCTGATTGCCCTTTTTTACCGCTGGCCCTG
1098	CGCAAGAAGTCATTGTCAGGAACCGGATATTGATGCAACTGAGTACCT
1099	GAAACAAAAATTGTGTCGAAATCCTGGCAATAAAGTACG
1100	GCAGCACCTGCCGGACTTATCATCATATTACGTATAACGTGC
1101	TGAACGGTGTGGTCAATGCAAAAGAATACACTAAACACTC
1102	TAAAAATATTTGAATGCCTAAACAAATCCTCAAAACA
1103	CCAACGCGCGGGAGAGTTTTTCGGTTTTCGCTATTGG
1104	AAAACAGATGAAAAGCGAGGCAAGTTAATTGCACCAGACCTCCCCCT
1105	TATCATCGTCAATCCGCGGGCGCGTCCAGCAACGTTATTGCGGAACA
1106	AGAGGGTTTTTTGATATAAGTATATATTTTTTAAGCGCT
1107	TTAAACAGACTGCGGAATAAGAGGGCTGTAGCTCAACATG
1108	TTATGAGTAGAAGAACACAATATTACTTTTTGCCAGCTCA
1109	AACTTTTTTATAGACGTGAGCCGAGCTCGA
1110	AAATCTACGTTAATAAAACGATAAATTGGGCTTGAGACGAAAAG
1111	TATTATCTGGTCCGGAATTTGCCAGATACATAACGCCAAAA
1112	TTAGTGATGAAGGGTAAAGTTAAACGATGCTGCGTCTCGT
1113	AATTTTAAATTATTTATCTAAAGCACATTG
1114	CAGCAGCAAGACTTTATTGCCCGATCAGCGGTGCCAAG
1115	AATGAAAATGCGCAACTGATAGCGCTATTAGAAGAAACCATCA
1116	AGGATTTATAAAATATCCTTGCTGAACCTCAAAATATCAA
1117	AAGTTTCAGACCTTCACTCCATGTACCCCGCGATTATACCAAGCGC
1118	TTAACACCGCTGAGCGGTGAGGCACACCAGCAGAGAAGAT
1119	CGTCTGTTTTTAAAGGAAGTTTGATACATTTGTATGAGCCAAAGGTTT
1120	GGTACCTCTCACATTTTTTTGAGGAGCGGCTGG
1121	GAGGCCATTTTTCGAGTAAATTATATTTTTTCAGTGTAACATT
1122	ACAGAGATACATTGGCAGTTTTTTTCACCAGTCA
1123	GGCGGCCTCCACATTTTCCCGCAATCCCGTAAAAAAA
1124	TTGTAGAACGTTTTTTATTGCCGTTCGGCAATCGGCCTCAGGAAGA
1125	CCGCACAAACGTGGACTTTTTCCAACGTCAATCGGCAAAATC
1126	GCACACAGACAATATTCCGAACGAGGTGAGTAATCGCCAT
1127	TCTTTAAAGAACCTTCTGACCCCTGATTTGATGGCAATTATCAGCGGGT
1128	ACACCAGAGGCGCATAGCTCATGAACAACATAGGAATACCACAAAT
1129	TCATGGTCAACATCACTTGCCGAGCCGTTTTTCAATAGATAA
1130	AGGGGACGACGACGTGTAGATGGACTCCAACATAAGAATA
1131	GAATAAGTAACATAAGCGGCCAGTGCCAAAGCTTTCAGAG
1132	TCATTTTTTTTTTGCGGATGGCTTCGCGTTTTTTTTTAATTCGAGCTT
1133	CCAACCTAAATGGTTTAATTTTTTCAACTTTAATCGTTGGGAAGA
1134	AAGAGGAAGCCCGGATCGGTGCGGGGTCAACTGTT
1135	GCTCCAACGGGGAATTAACAACTTTATTTCT
1136	CCGACTTGAAGGCCGGAAGAAACTACAAATTTTAAACA
1137	GGGAAGGGTAAGTTGGGTAACGCCTGTGCTGC
1138	CAGCTTTCATCAAGACTTCAAAATACCAATAGGTGAATTC
1139	GTAAATGACAACAGTTTCAGCGGAGTGAGGAACACCGTCA
1140	CGTGGAAGAATTAGCAAAATTTTTTAGCAATAAAGCATTACATCCAA
1141	CGGATTTGTTTTCCGTAATGGTAACCGTGCAATGCAATCAAAAGAT
1142	AAATCATATGGCATCAATTTTTTTTACTAATATTTTTTAAACAGGAAG

TABLE 7-continued

Staple sequences used for the T3 triangle 3 folding.	
SEQ ID NO:	
1143	CACCATTATTGCATTATGACCCTGTATTTTAATCAAAGGTTTGTAGAA
1144	ACCAGTAGTTGTTAAAGTGAGAAAGCCGCCAGTAAAGATATGCAATGCCGTGAGTA
1145	GAGGCAGGTCAGACACCACCGTTTGCCTTTTTCGGTCCGGAATC
1146	GCATAAGCCGTGGGAGCGAGTAACAACCCGTATTTT
1147	TTAAATTTATAACCGACGGTTTATCAGCTTGCTGGCGAAA
1148	TTATATAATTGAGGACTAAAGACTAACGGCTACAGAGGCT
1149	TTTTTAAGTCACAATCAACATATAAACGTCAGAATCAAGAGCCGCCGC
1150	TCCTGAACATAAAGTATAAAGCTAGCGACACCAATGAAACCA
1151	ACGATTTTTTGTGTTAAATCTTACCAACGAGTTAATATCCCATCCTAA
1152	TTCCATATAACAGGTTGATATTTTTCAGAAAAGCCCCAAA
1153	ATAAGAATATCTTACCGAAGCCCGTGGAGCCATTGAGGG
1154	AAGAAAAAAGGGCTTAATTGAGAAGGTATCCAA
1155	TTCTAGCTGATAACCTGTTTTTTAGCTATATTTTCAACCATTAG
1156	TTTGCTAAGAGCCAGCGTTGTACCAAAAAATATTACAAA
1157	TCAGGTCACGCGTTTTTCGGAGACAGTCAAATCACCATCATTTTGTAG
1158	GAGTCTGGAGCAAACCTATATGTCAGGGAGTTAAAGGCC
1159	AATGACAACAACCATCGCTTTTGCATTGTATACGATGAAC
1160	TTATTACTTTTGTAGGTAGTCAGGACATTGTGAATTACCTTAACGGGTAA
1161	GCCCACGCTTGTAAACAAAAATAGAAGCCTCCTCAGA
1162	CAACGCTCATTATTTATCCCAGGCAATAGAAAAATCATATGGTTTAC
1163	AAACCTTCTGTAGCGGGGAAAATCTCCAAAAAAG
1164	AGCGCCATTGCGCATTCAAAAAGCGAGAGGCTGAGGCATA
1165	GCCTTAGCGTTTGCCACCACCACCGGAACCG
1166	CCCTCATATAACTTATAAAAAAT
1167	CACCCCTCAGCAGCGAACGATAGTTAATTGCTTTTGAATATAAT
1168	ATGTGTAGCCAGAACCGATTGGCCCCATTAGCAGCCATTT
1169	AGCCTGCTTTAATATTTGCTGAAGATCTACAAAGGCTAATCACCGG
1170	AATAAACACATAGCCCCCTTACGGAATAGGTGTA
1171	CCGACCGTGTGATAAAAGTTAATTAGAGTCAATAGTGAATTTATCAAA
1172	AAGAGAATAGCAAAATATTCAACCGTTATTCAACGCAAGGTTGCGGGA
1173	AGAACGCGGCTGAGAGACTACCTTTTAACTACCGACATAGAAAA
1174	CGCTATTACGCCAGCTTTCGAGGAACGCCATTGAGTCAGTAGTAG
1175	TTCAGAAAACGAGAAATGTTTTCCATAACATCAGTTGTTTTGATT
1176	ATTAATGCCGAGAGGGTAGCTAATATGATATTTAAATT
1177	CATAATTTTTGGAACCGAACAATACGTAATGCCACTACGAAGGC
1178	TGACCAACAGACAGCAGCATGTCAATCATTTTATGTACCGCGCAGA
1179	GTAAGAGCTTATACAGAAAGATTTTGAAGAGGACAG
1180	CAATGGTCAATAAGGTAATCGTAAACATATCGGAACG
1181	GAATTACTTTGCTTTTAAAGAAAGTTCCAGGCA
1182	ATTCGCAGTAAACGGAGGCTTGAAATGCTGAGGTTGGG
1183	GCCACCACAACAGAGTCTTTTCATAATCAAACAAGACAAATAATTAC
1184	CCTCAGAGCATCGGCATAGCGTCAATGCGTTAGCAAAAGAC
1185	AGAGCCTAATTGCGCAGAGGCGATGAAGCCTAGAACGGGAGCAAGCCAGGT
1186	TCATGTTTTTATTTTCATCGCGAGAACAATATAAAGTTGCTTCT
1187	ACTACAAGATGAATATACAGTAAGAGGGTAAAGTGAACCGAGGTGCTCGT
1188	TCCCCGTAAGCACTAAATCTTTGGAGCAAAATCAGGGCT
1189	TTTGATATAGAAGGCTTCACTGCCCTCTGTGGTGGTCATA
1190	TTTGCCCCGATTCAAGTTTGGAACCTTGAACCATGCCGCCAG
1191	GCAAATTACCACACAAGGAAGAAATTAGAGCTCACTACGAAAGGAGTTCT
1192	TACGAGAAATCTTTGTACTTAGCCGGAACGACCGTTGATTCCTAATTATACATTCT
1193	ACAGACCAACGAGTAGACTAACGTCAAGTGAATAAGGCTTT
1194	TGGATAGCGTCTTCAACTAATGCAGAGGGGTAATAGTACCGCTTAGTCAGA
1195	CCAGTTTGTGCGACTCCAGCCAGCTTTCGGCAAAATGTTT
1196	CTTAGGAGCACTAATAATGGGTGGGTCACTGTGGTTTTTCT
1197	TCCAGTGAGACGGGCAACAAAAGAAATCAGCAGGC
1198	TGCGCGTAAGCACATCCTCATATGGAACAAATAGGGTTGAGTGT
1199	TCAGTTACGGAACGGTTCGGTGGATCATTTT
1200	GGGATCGTAGGGTAGCTTTTCATGAGGAAGTTTCCATTATGCGATTTT
1201	TCTGGCTCAAACACTATCTTGATACATCAAAAAACACCTCTT
1202	TAGAGCTTGCCTCGACTTAAACAGCATAACCTCT
1203	TCAGACGACGATAAAAACAGGCTGCGCTTTACCTGA
1204	TTAATAAGAGAATATAAAGTCCGGCTTATGCAAAATCCAATCG
1205	ATTCGCGTCTGGATCAATTGTATTATATTCGGGCATTTTCGT
1206	TACGCAGTATAGCAGAGTCGTTAGTATCATGACTGTAG
1207	TCACGTTGGAAGGGTTTTCCAGTCACGACGTTGTAAGACTCT
1208	TCAACAAGCTAATGCAGAACGCGAAGGTAAGTAAATTT
1209	TTGTCCAGACGATGTAATTGTTAGCAACCAAGGAGCCTTT
1210	CCTGTTTAAACGCCAACACGACAATAGGAAACCT
1211	TTACGAAGCATGATTAAACGAGAATTGCGAATAATA
1212	TACATCATAGATAAGATCATAGAGAAAACCTTTTCAAA
1213	ATACCAAAAAATACATTACCAGAAAAACAACAT
1214	TCCGAACAAAGTACATAAATCGCCATACTTACCAG
1215	AACGTAGAAAGAACTACGCAATAATACGGAAAAAGTAAGCAGAACT

TABLE 7-continued

Staple sequences used for the T3 triangle 3 folding.	
SEQ ID NO:	H1 design oligo sequences (5'-3')
1216	GCAGTAGAGTAGGTAGAGATTAGGCACTACCATACGATCTAAAGCGCAGTAAGTACAA
1217	GCAGTAGAGTAGGTAGAGATTAGGCAAAAAAGAGGAACACAGACAGCCCCGTCCTTC
1218	GCAGTAGAGTAGGTAGAGATTAGGCAATGCGCCGCTACAATACGCTAGGG
1219	GCAGTAGAGTAGGTAGAGATTAGGCAACGTTGGTAATATAATTGAAAGCGAGGGAAGGAGCGGAA
1220	GCAGTAGAGTAGGTAGAGATTAGGCACGCTGGCAGAGCCTGATTATCAGA
1221	GCAGTAGAGTAGGTAGAGATTAGGCAAGCAAAGCGGATTCTGCAAAATGCT
1222	GCAGTAGAGTAGGTAGAGATTAGGCATATGGGATCGGATTCTCTAAATCG
1223	GCAGTAGAGTAGGTAGAGATTAGGCAAGGCGATAAACATTAAATGTGAACAAACG
1224	GCAGTAGAGTAGGTAGAGATTAGGCAAGGAATAGAAAGGAACCTATTTTG

TABLE 8

Staple sequences used for the T3 triangle 4 folding.	
SEQ ID NO:	Core structure oligo sequences (5'-3')
1225	GTGAATAACCTTGCTTCTGTAAATCGTTTTTCGCTATTA
1226	CCGGGGTAGGCGGATTGTTTTTATTGTTCTGAAACATGA
1227	ATATTATTGAGGCGTTTTAGCGAATAGATAAGTCCTGAAC
1228	AACGGGTAAAATCAGATTAACTCAACAGTACGGTGAAGG
1229	TATAGAAGTGATTGCTGAGCAAAACAGAGCCAGCCGGAA
1230	GAGCAGGTTTAACTGACGTATAATAGGAACCTCGGAACC
1231	TATCAGGGGGGGAAAAGCACTAAGCGGGAGCAGACAGGA
1232	AATAGCAATAGCTATTTTTCTTACCGAAGCCCTTTTAAAG
1233	CCGATTTAGAACCTACCAGTTGGGCGGTTGTGTACATCG
1234	CGATGGCCTTTGGGGTCGAGGTGGGGTGCCCCGAAGC
1235	CCACCCTCTAGGTGTATCACCGTAATTAATTATGAAACAA
1236	CAAGTTACTATACAGTAAAAATGAAAACACAAGAATTGAGACAGCGGATCTCACGG
1237	AAGTATTAATTAGCGGTCGAGAGGTTGATAGCGGGCC
1238	TAGAAGAACTCAAACCTTTGATTAAATTAACCGTTGTAG
1239	GAGCTCGATGAATCGGAGAGAGTTTAAAGCCTCCGTGGTGG
1240	AATTGCGTGCACCGCTCACAATTCTGAGAAGCGTGCTTT
1241	TATTACTGGTAATATCATACCTACCCTCGTTAAGGGCGCG
1242	TTCCGAAAGGGCGAAAAACCGTCACATAAAAAAGGAGCG
1243	ATTCACAAACCACAGAACCATTTTTTACCAGAGCCGC
1244	TTTAGCACAACGCCTCGGGAGAAACAACCTCCAAATA
1245	GCCACGGGAACGGATAACCTCACCGGTTTTTAACAATCGGCG
1246	ATCCTGAATCATATTAAATCAAGTTTTTTTAGTTAATATATTT
1247	CTTTCAACAATTTACCGTTTTTTTCAGTAAGCGTCATACATGG
1248	TAATATCCGGTATTCTAAGAACGCTATCCCAAAGCTACAATTT
1249	TCAGACGATGCATTAAATTCGTAAATAAAGTGGCAGCAAGCGG

TABLE 8-continued

Staple sequences used for the T3 triangle 4 folding.	
SEQ ID NO:	
1250	CCTCCTTTTTCGACTTGC GGGTTACAAATTTTAAACAGCC
1251	GCTCATTTAGAGAGAATGCGTAGATTTTCCCTACTATGGCTTCTGAAATAATCCT
1252	CTGCGCGCCTGTGCACGCTTCCATGAAATTGTTATGAGAAGCCCAA
1253	TAACAGTGCCCGTATAGGTCAGTGCAGCCCTCATAGAGCCAGATGAA
1254	AGCGCCATGTTTACCAACGCAGAATTAAGCCCTTCTCCGT
1255	CCCTTAGAAAAGAACGCGATTTTAAACTTTTCAAAAACAA
1256	AATTTCAATTTGAATTATAAGTATACCACCCTCAGACTCCT
1257	GGGATTTGTTAGTAAATGAATTTGCACGTAAAACAAGCTCAAACCTAAATTTCT
1258	ATTAATTTTGAACGCGCTGTTTTTTTTTCAACAA
1259	CGCGCTTAATGCGCGCTACGAATCAGAATCGGAACCATACGAG
1260	GTCACCTTTTACTTGAGGTCCCGGAATTTGTGA
1261	AGAGCCGCAAGAGAAGGATTAGGAGAGGCTGATTTTCAGCGTAACAC
1262	CCCTCAGAACCCTCAGACTTTTTTTAGCGCG
1263	TTCTGCCATTCTGTGGTCGGGAATGAGCTAACTCACATTCACCCAAA
1264	TTCATTTCAATCATTACCGCGCTTTATTTTCATCGTAGGAACCAATC
1265	TAATGAGACCTGTCGTGCCAGCTCCAGCGCCTGTTCTTCGCGTC
1266	AATTACCTTTGAATACTGAGTTTCTCCACAGACCTTGAGCAAATAA
1267	ATTACTGGTAATAAGTTTAAACGGAACAGTTACATGTACGGATAGC
1268	GAGCAAGAAACAATGAAAACGTACCGGAAATTTTTTATTCAATAA
1269	GATACAGGAGTGTGTTGACGCTCAATCATCAATTAATGGAAGGGTGGCAA
1270	CATATCAAAATTATTGGATTATATTGCTTTG
1271	GTAACGAGATGAAATAAAGAAATTAACATAA
1272	GCGAATTAGTTTTTACGCTGCGGCCAGAATGCCCTTTTTTCATGTAGA
1273	AAGTGCCGGGTTTTGCTCAGTACCCAATACTTTATCGGCCTAATCAGT
1274	CTTTTACATGTAGCATGTACCCAGTACAACTGCCACCCTGAAGATGA
1275	CAGAACAAACGGTACGGAGGCCACATAGCTGTGCACGCGTCAGTGTCA
1276	AAGGTAATATTGAGAGATAGACTAATAATAAGAAACGATTTTTTGT
1277	CATTTAACCACTACATAAATCAATATATGTGAAAGAAAAACATCAAG
1278	TGGATTATAACAGGAAAATTTTCGCTCATGGAA
1279	GCAGCAAGGTCAGTAAGAGCCGTTATTAGACTCAGCGG
1280	CAATTCGACCATTGCGCCTGCAACACCTGA
1281	AAAGAGGATTGACAAGGATTTTAACGCCAAAGTAAATTGTAGACTGAGACGACG
1282	ACATCGCCGCAAAGAATACCAAAATGCTGTAGAGAGGTCAAAAAGAA
1283	TTTCGCACAGCCGGAATCCGCGACCTGCTCCACATCCCTT
1284	GAACCTTTTACATTGGCAGATTCATATTCACCAGAAGGAGCGGAACGCGGTC
1285	ACATAAAAGGACATTAAATGCGCGTACCGAACGTGGCACAGACAATTTTT

TABLE 8-continued

Staple sequences used for the T3 triangle 4 folding.	
SEQ ID NO:	
1286	TCAGATGCCGGTTACCTTTTTTCAGCCAGCGGTGCCGGTGCCCCCTGCA
1287	TGCCAACGGCAGCACCTGCCGGACTATCATTTTGCGAACCACCACACCCG
1288	ATCATACAGACCATTATCTGCGAAGCTTGCCC
1289	TGTCGAAACGAGGCGCAGACGGTCTGAACCTTTTAGTTT
1290	GTGTAGCGGTTTTTCACGCTGCGGCCTCCGGCCAGAGGG
1291	TGAATGGCTATTAGTCTTTCTGGCCAAAAGAATACTTAGAAGCAATAGA
1292	GCTCATCGCTAATACATCAATATCCTAAAGCATCACCTTGGGCGCGAG
1293	CTATATTTTCATTTGGCAATCATAAACGTAACAAAGCT
1294	CTTATAAATCGCTGATTGCCCTTTTTTACCGCCTGGCCCTG
1295	AATCAACAGTTGAAAGGAATTTTGTAGGACGTGAGCC
1296	TAACAAC TAGTTGGCACCACGCTGAGAGCCA
1297	ACCCTGACTATTATAGCACCGCTTCGTTTTAA
1298	CGTCAGCGTGGTGCTGGTCTGTGCTTTTGC GCGCATCGTATAGGTCA
1299	CCACGCTGGTTTGAACCAGCTTACGGCTGGAGGTGGTTGCGG
1300	ACACTGGTCTTTGCTCGTCATAAAATCTGGTCAATAGATT
1301	TGTTACTTTCAATCCGCCGGGCGCGTCCAGCATTTACAAAGAACGTTA
1302	TCATGGTCCGAGTAAAGAGTTTATCTTTTTTAAATATCTTT
1303	TATTACTTTTTGCCAGCAACTCGTAGGAGCACTATGAGCCAAGGTTT
1304	GAACCACCAGCAGAAGATTAAAAAACTGATAGCCCTAAA
1305	CCAACGCGCGGGGAGAGTTTTTCGGTTTGCGTATTGG
1306	AAAATCCTGTTTGATAAGCCGGCGAACGTGGC
1307	CCGCACAAACGTGGACTTTTTCCAACGTCAAATCGGCAAAATC
1308	CGATTAAATTTTGGGATTTTAAACATTTTGAGGCATTAAATC
1309	CAGGAAGAAGGGGACGACGACGTTAGATGGGGATGGCTTACACGACC
1310	AACCGGATAACGAGTAAGGAATTACGTTTACCGATAGCGT
1311	AGTAACATTTGTAGAACGTTTTTTATTGCCGTTCCGGCAAATCGGCCT
1312	AGAGGGAAATAAAACGTTTTTAAACAGTTGATTCCCAAT
1313	CTGGTGCCGGAATCTTTCCAATACTGCGGAATCGT
1314	GATACATTCCCAAATCAGGGAACCTTTGTATCATCGCCTGATAAATTG
1315	GGTACCTCCTCACATTTTTTTGAGGAGCGGCTGG
1316	CAGATGAAATACCAAGCGCGAAACAAAGTACCCTTATGC
1317	TTAGTGATGAAGGGTAAAGTTAAACGATGCTGCGTCTCGT
1318	CTGATTATCAGATGTTTTTTGGCAATTCGTCTGAA
1319	TTAACACCTCTGACCTGAAAGCGTCAGAGATATTAATTTTAAAGCT
1320	TCAACTAATGCAGATACATAAGAAGCTGGCTCATTATCAGCGATT
1321	CAACTTTGCGGATAACCTGTTTAGGTGGCATCA

TABLE 8-continued

Staple sequences used for the T3 triangle 4 folding.	
SEQ ID NO:	
1322	GGCAAGATAAAACAGAGGTGAGGCTGAAAAATAAACCCCTCAATCAAT
1323	AGGCTGTCCATCACGCAGTAATAACATTTTCACTTGGTG
1324	CGCAAGAAGTCATTGCAGGTCAGTGAATAAGCGAGTAGAGAATATAA
1325	TGCCATCCCACGCCCCAGCCGAGGAGTCCACTATTAAAG
1326	CGAGAGGCTTTGTGAGCAGCAACTAATTGCTTTGAGGAT
1327	GTGTTGAGCAAAATCGTTAACGGCAGCGCCAGGGTTGCCCTTCCCCG
1328	GGCGGCCTAAGAATTTTGTAGCGAAAAATCCCGTAAAAAAA
1329	CGCTATTACGCCAGCCGACAATGACGCCATCCAGCTCATAGCCTCA
1330	TAACTATAAGACGCTGAGAAGAGTCAATAGTGCCGACAAAAATAAGG
1331	TCCTCATTAAGCCGAGCAGATCGGTCATAATCAAAATTACCGACC
1332	GCAAAGCGGATTGCATCTTTTAAAAGAACTATCATATTTTCCCT
1333	ATAAATAAAATCTTTTAAAAATCAGGCCAGGCA
1334	TCTAGCTGATGTAGTAGCTTTTAAACATCCAATAACTGAAAAG
1335	TAAATTTTCCGCTTTTATACCGATAGTTGCGCTGGCGAAA
1336	TACCATTATACCGCGCCAAAAACGGCCAGTGCCAAGCTTTCAGAG
1337	TTAGTATCTTACCAACGCTAATTTGACATTCAACCGATTGAGGGAGG
1338	CAAAGGGTTTCGATTAAACGCTCTCAGCAGCTTTTAAAAAATCAT
1339	TTGCGCTCTGGCCAAACAGCTTGGCGGGATCGTCAATAAGAATAAA
1340	AGAGAATCGCAATATTCAACCGTTGTGTAGGTAAAGATTGCAATGCC
1341	GACGACAAATTTAATGAAGCCTGTTTTTCATCATCAAGTTTGCC
1342	AGAGCCTAATTTGCCAAGGTTTGAAGCCTGCATGTTGAGCTAATGCA
1343	CGGTGTTTTTACAGACCAGGGCAAAAGAATACACTAAACACTC
1344	AGGAACAAGGAAACGTATAAAATTTTACGCAG
1345	AAGCAAACTCCAACGATCGGTGCGGGCCCCAAGTGT
1346	GGGAAGGGTAAGTTGGGTAACGCCTGTGCTGC
1347	AAACGAACGGCTTTGACCGATATAAGTTTCTTTTTTCCATATA
1348	CGTTAACCTAATATTTTGCCTGAGGATCTACAAAGGCTATCGCCACCC
1349	GTTAAAGGTGTTAAATAAAAAATAATGAGTAAGAAGCCT
1350	CGCCAGCATCAGAACCCCGCCTCCCTCAGAGCGGGTTATAGTGTGATA
1351	TCTTTGACCCACCAGTCATTTTGTACGTTGGGAAGAGAATACCACA
1352	TTCTACTAATAAAAGTAATCGTAAACTAG
1353	TCGCAATGGTCATTGATAATTTTTCAGAAAAGCCCCAAAAA
1354	ATCGGTTGTTAGCAAAATTTTGTAGCAATAATTTTAACCGAAGA
1355	CTCAGGTTTTTGGTTTAGTACCGCAATCCAAT
1356	AGTTAATTTTCATCTTCGCAAGACATCCTTGAAAACATAGCGATAGCT
1357	CCGAACAAAAAGGGCTGTCAACAGCGACAGGCATTTTGGCCTTGA

TABLE 8-continued

Staple sequences used for the T3 triangle 4 folding.	
SEQ ID NO:	
1358	TTATTTACGTGGGAACGAGTAACAACCCGTCGCCTTT
1359	CGAGGCATTTTTTAGTAAGATTTAGAAAACTACGTTAATACGAAAGAG
1360	AGTAATTATGACCTGTAATATTTTTTTTGC GGAGAGCATAAAGCTA
1361	TGTCTGGATTTCGGTCGCATCGCCACGCATAATTAAGAGGAAGCCTCTT
1362	AGCGCCATTTCGCCATTCAAACGAGAATGACCATTTCATTGA
1363	AATTGTAGGAAGGTTTTCCAGTCACGACGTTGAACGCAATTGCTT
1364	TCTCTGAGTTTCAGATTGCGAATAATAATT
1365	ATCCCCCTATCAGTTGAGAGCAACCTGGCTGACCTTCATTTTTTAAGAGTAATC
1366	TCGAGGGTTGCAAGGCCCTAAAGGACGGAGTGA
1367	TGACCTAATAAACAACGTTATACAAATCTTATTACGAGC
1368	TGAAACCATGTTAAATGAGAAAGGCCGTTGAGGCAGGTAGAATGGAAGCACCGT
1369	TTAATGCCGAGAGGGTAGCTATTATGATATTTAAATTG
1370	ATTAGCGTTCGATAGCAAGGAACCTCATATATTTTAAAT
1371	AGCTTTCATCAACGTCAGGATTAGCAATAGGAACAACAAC
1372	TATGCATTTTTCTAAAGTACGGAGAGTACCTTTTTTAATTGCTCC
1373	CGCATAGGGGACTAAACATGTCAATCATTTTTTTGTACCC
1374	CGGATTGATTTTTTCGTAATGGGAACCGTGATCGCGAACCAGACCG
1375	TTGCCATCAGGAGACAGTCAAATCACCATCAATTTTGAGA
1376	CTGATGCACACGCCACCACCGGAAGCCACCTCAGAGCC
1377	GTTTGAAACACCGGAACCAGACCTCAGAACCGCC
1378	TCTAAAGTTTTGTTTTGGGAATTAGATTTTCCAGCAAAA
1379	TTTTCACCTCACCAGTAGCACCATAATCAGTATCAATAGA
1380	TAGCAACGCATGAGGACACTACGAAGGCACCAACCTAAAA
1381	AGTCTGGAGCAAACAGAGACTACCGAAGACAGCATCGG
1382	AGGTCTGAGTAAATACGTAATGCAGTTTCATTAAACGG
1383	CTGAGGCTTGCAGGGAACGAGGGTTGTATAAGATGAACG
1384	GTCTTTCAAAAGTAAGCAGATAGGTGGAGCCGGTGAATT
1385	TTGACAGTTTTTCATAGCCCCCTTACC GGAAAAATTCAT
1386	GCTATTTTGCATAACGGATTGCGCGCTTATCCCATCCTAAGCTGTT
1387	TTTATCATTTCCAAGAATAATCGTTTACGAGAATGGAAA
1388	AAAATCGCTGCCCCAATAGCAAGCTTAAACCATAATATCAT
1389	AGATAACCCAGGGAAGCGCATTAGACGGAGGTAAGCCTTTAC
1390	TCAAGCCGTTTCACTGCCCTCTGTGGTGGTCATA
1391	AAATAGCATTGAGCGCAGTACCGCACTCATCGAGT
1392	TCTGAACAAAGTCAGGAGAATTTGAACCATGCCGCCAG
1393	CCAGAATCCACACAACCTAAAGGTCAAGTTTCACTACGAACTGT

TABLE 8-continued

Staple sequences used for the T3 triangle 4 folding.	
SEQ ID NO:	
1394	TTTAAATCATTGTGAATTAACGGAGAGAACTGAC
1395	TTTGTACTCAACATCAAAATAGTGACGAGAATGGTTTAATT
1396	TAAATGTTGGCTTGAGAACACCAGATTCATTA
1397	CAGTTTGTGCGACTCCAGCCAGCTTCCGGTCAGGGTT
1398	TAATGGGTGGGTCCTGTGGTTTTTCT
1399	TCCAGTGAGACGGGCAACAAAAGAATCAGCAGGC
1400	AGAAAGGAAGCACATCCTCATATGGAACAAATAGGGTTGAGTGT
1401	TCAGTTACGGAACGGTCGGTGGCTTTGCCC
1402	TTAAACAGTAGGGCTTAGTCCAGACTAGATTATGTAAATG
1403	AGTTACCACACCACGCAACATATCATGTAATAAGCCAACGCTCAGAAT
1404	TGTTTGGGAATAAGTCCAGTATAACTAGAAA
1405	CGCAAAGAGAAGGAAGAAAATACATACATAAAGGTACT
1406	TCATAATTTTAGGCATAATTCTATTGAGAATCGCCATATTTAGT
1407	TAGCAACTTCGAGCCAGACTCCTATGTGAATTTCTTAA
1408	AGGTAAAGGAGGCATTAAACGCCAAAAAGAAATATTACGCAGTATATTT
1409	TCTTGTGTAGCAAACGTAACCGAGGATAAAACGAAAGGCTCCAAAAGGA
1410	TATAAAGTAAATTTATCCCTCCGGCTTAGGTTACAGTCAT
1411	TGAGCCAAAAGAACTGGCATGATTAAGTAATAAGAGTTCT
1412	TGTAGCCGGGTCGGTTTTATCAGCAATAACGGAAT
1413	GACTTTTTGCTACAGATAACGGAACAT
1414	TTGTTTATTACAGGTAGAAAGATTCCAAATGCTTTATAGT
1415	TAGTTCAGAAGGCTGCGGAAAGACTTCATTTTATATCG
H1 design oligo sequences (5'-3')	
1416	GCAGTAGAGTAGGTAGAGATTAGGCAGATTGTTTTCTGTATCTTTTGATACGTTAGC
1417	GCAGTAGAGTAGGTAGAGATTAGGCAAAAAGAGCCACGTCTTTCCA- GACTGCTAAACA
1418	GCAGTAGAGTAGGTAGAGATTAGGCAGGCGCTAGGGCGCTTAGAGCTTGA
1419	GCAGTAGAGTAGGTAGAGATTAGGCACGTTGGTGATTATCATCACCAGTCAGAGTTTG
1420	GCAGTAGAGTAGGTAGAGATTAGGCACGCTGGCACGTGAACAAAGAAACC
1421	GCAGTAGAGTAGGTAGAGATTAGGCATTTCGAGCTTCAAATGCGTTTTGCC
1422	GCAGTAGAGTAGGTAGAGATTAGGCAGAATAGAAGATTCTCACGCAAGGCACCTTCC
1423	GCAGTAGAGTAGGTAGAGATTAGGCAAAGGCGATCAGATTAAATGTGAGCAAAACGG
1424	GCAGTAGAGTAGGTAGAGATTAGGCAATCACGTTGAAAATCTCCAAAGAC

TABLE 9

Staple sequences used for the T3 triangle 5 folding.	
SEQ ID NO:	Core structure oligo sequences (5'-3')
1425	TACCGCTGGTAATATCATACCTACCCCTCGTTAAGGGCGCG
1426	GCGAAAGGGGATGTGCAGTGCCATAGCAGCCCCAGGGTT
1427	CGTCCGTGTCATTAATTGCGTTGCCGGGTGATGCCGGGTACCTG
1428	CAGGGAAGCGCATTTTTTTGACGGGAGAATTAAGTGAACA
1429	CGCGCTTAATGCGCCGTACGAATCAGAATCGGAACCATACCGG
1430	AATCGTCGTGATGCAAATCTTTTAATCGCAAGACAAAGATGAT
1431	CGATGGCCTTTGGGGTCGAGGTGGTCTTCGGGGTTTCT
1432	CCAGAATCTTCACGGTCCTAAAGGTCAAGTTTCACTACG
1433	AGTACATAAATCAATATATGTGAGTGTTTTTATAACCTT
1434	TGAACCATATAACCTCTAATGGAAGGGTTAGATCAATATTGCTTTG
1435	GAGAATAACATAAAAAGCCAGCTGATAAAATTTTGAACGCAAAGA
1436	GCAAGTTTTTCCGTTTTTATTGAGGTTTTTTTTTAAGCCTTAA
1437	CACCCAAAGAGATAAAGAAATTGACGTATAATTGATATATCTGAATT
1438	TGGTGTGTTTCAGCAAGAGTAACATGTTTTATTGTTCCAGTAAGCGT
1439	CCCTGCGGCTGGTAATCGAATTCGTGCGGCCAGAATATACGAGAGGG
1440	AGAGCCACCACCCAATAGAAAATTCATTTTATGGTTTAC
1441	AAATAATAATTAAACCAAGTACCGTAGTTGCTATATAGAAGGC
1442	CACCTCCACAAAATTATTTGCATTTGTTTACCCAATCCAAAATAAA
1443	CGCAGAGGATAACGGAAGGTGATTTCAGAACCAACAACGCCACC
1444	CATACATGAAGTTTTTAATAAACAGTTAATGCTAAACATC
1445	CCGATTTACTTCTGAAACCGGAAACAATCGGCGAAACGT
1446	ATTTTCATTAAACAAAATTAATTACCCCTGCCAGTACCAGATGATACA
1447	TCCGAAAGGGCGAAAAACCGTCACAGCGCCAAGGAGCG
1448	ACTTGATTTAGTACTACAGTAACAGTACAGATTTTGCA
1449	AAAGGTGGCAAAAGTTGGGTAAAGTTTACAGACCCAGCTACAATTTTA
1450	AGCAAGCAAAATCCTTTTACATCGGGAACGGTTCCTATCCTAATTTAC
1451	CGGTATGAGCGCTCACGTGTCACTGCCAGCACGCAGCAAGCGG
1452	GTGCCCCTACGGGGTCAGTGCCCTTCAATACCTTATCGGCCTAATCAGT
1453	CAGAACAAACGGTACGGAGGCCAGTGCCTCCGGCATCACACTGTTG
1454	GCTTCTGTAGCAGAACGCGCTGTTTTTTTATCAA
1455	TTCGTCAACCCGCCACCAGATTTTTCCACCACCAGAGCCGCCGC
1456	TATCCGGTATCTATCATTACCGCTTTTTCCCAATAAGAACGCG
1457	ATCAAGATCACTCATCGAGAACAAATAGATAAGTCCTG
1458	AGGATTAGGATGCCATTTGTTTTTGAATTAG
1459	CATTCCAAGAGAAACACGAATTATTTTTGCTCTATTTCCG

TABLE 9-continued

Staple sequences used for the T3 triangle 5 folding.	
SEQ ID NO:	
1460	TTTATTTTTTTTGTCACACTGCAAGGCGATTCAT
1461	CCAAGTTACCTTACACGTTTCTTTGCTCGTCAATTTACAGAGCATGT
1462	GATGAATACGCCACCCACCGTACTCAGGAGGTAGCGGGTCATTTCA
1463	CGGAAGCATAAAGTGTAATTGTTAACCAGAGCTGGGTAAAG
1464	TTGGGAAGGGCGATCGGTGCGGGCCTTTTTTTTCGCTATTAC
1465	TATCAGGGGGGAAAAGCACTAAGCGGGAGCAGACAGGA
1466	CCCTCAGAGCTAGCCCCCTTATTTTTTTGCGTTTGCCATC
1467	TAGAAGAACTCAAACCTTTGATTAAATTAACCGTTGTAGATCGTTAA
1468	CATCAAGATGAATTACCTTTTTTAATGGAACAACAAGAAAGCAAAAG
1469	CGGCATTTAACCTATTATTCTGAAGAAACAAAATTACCTG
1470	CCAGCGCATGCCCGCTAGAGAGTTGCGTGCCTCCGTGGTGGT
1471	GTACCGTATAGCAAGCCCAATAGAACCTACCATATCGACGAGGTGGAGCCGCCAC
1472	ACAGGAGGTTGAGTTGACGCTCAACTGATTATTTGTTTGGATTGGCAA
1473	GGGAACGGAACGATTTTCGTAAACAGAACCTACTATGGAATCCTGACAGATGAT
1474	TCGGTCAGGAGTGTACTGGTAATGCTTTTGGCGGATAACCCGGAATTTGCCTG
1475	CGTGCTTTATTGCAGGTCAGACGATTGGCCTTAGCCAGAAAGTATAGGTGCCGT
1476	ATAAATCCTCATTAAGATATTCGCCACCCCTCAGAAGCCGTAGATTATTGCTT
1477	CTTATAAATCGCTGATTGCCCTTTTTTTACCGCCTGGCCCTG
1478	TAGAAGTACCATTGCTGAGGCGGTCCCTGA
1479	GACGATGGTGCCGGTTTTTGGCCCCCTCCGCAAGAATGCCAAC
1480	TCTAAAATAACCCCTCAATTAACACCGCCTGC
1481	GGGGTCATTGCAGGCGCTTTTTTTCGCACTCAATCCGCCGGGCGCGGTTG
1482	GAACCGATAATAGATTAAAGCATCGAGCCAGCAGCAAATGAACATGTT
1483	GTGTAGCGGTTTTTTCACGCTGCGTTGCCGCCAGCAGGGG
1484	AACAGTGCGAAGATAAGAGCACTAAATACATTCCGGCCAG
1485	TATAATGCTGTAGCTCCTCATCTTGGTCAATCATAAGG
1486	CGGCAACGCGGTCCCAGCCCGAGGAGTCCACTATTAAAG
1487	TGATTCCAAGAGGTCGTACCTTTTTTGAAAGA
1488	AACCTGTCCATCACGCAGTAATAACATTTTCACTTGAGT
1489	CTATTATACAAATATCCCAAATAGGACAGATGAACGGTGGGAACGAG
1490	GTCTGGTCCGGCCAACAGTGAGACGGGCAACAAAAGAAT
1491	CATCAACATCTGGCCTTCCTGTAGTTTTTAACCATTGCATCACACGACC
1492	GAACCCCTTACATTGGCAGATTCCAGAAGCATTTTGCGGAACAAAACAGCG
1493	TGGATTATAACAGGAAAATTTTTCGCTCATGGAA
1494	ATCGCCATTAAATCTCTGATAGCGGCTATTAGTCTTTAA
1495	AACGAAAGTGCCGACTTGTTAGAAGGCAGCCTTGAGGATTACTCGTAT

TABLE 9-continued

Staple sequences used for the T3 triangle 5 folding.	
SEQ ID NO:	
1496	GAGCGGAATTATCATTTCATATTCTCGTCTGAA
1497	CGTTATTAAATCCCGTGATCAAACGGAAGAGACGCAGCCAGCTTT
1498	TATTACTTTTGGCAGTTAGACTTGAAGGTTATGGTGCTGGGCAGCAC
1499	TCTCCGTGGGAACATAACACGTTAATAAACGAACT
1500	ATTCGCGTTAAATGTGAGCGAGTAACAACCAGGCATA
1501	TTCCAGTCGGAAACCTTTTTTCGTGCCAGCTGCAT
1502	AAAAATCTGCCAAAGGAATTACGCGTCGGATATTCATTG
1503	AGAGCGGCCAGCTTACGTCGGTGGTGCCATCAAATATCAATCTTTAG
1504	AAAAAAGCCGCGACAGCGGCTTGCAAAGCGGAATAGGAAGTTAAATC
1505	AGCGATTATACCAAGCATCTTAATTGCTGAAGCAACTAAA
1506	GTTAAACGATGCTGATCATAAAAAATTTTAAAGTTAACCACCACACCCG
1507	ATTTTTCGCGCAGACTGACCCCCATTAAACGGGTAAAAACGTAAT
1508	TGCGCGAAGCGAACGAGTTTAGCTAAAGACTTGTGAGAAGCAAAGA
1509	ATATAAAGGACATTTTTTGAATCCTAAACGTGGCACA
1510	AATTTGTAACGTGGACTTTTCCAACGTCAAATCGGCAAAATC
1511	CCACGCTGGTTTGCCGTTTTTCGTCTTTTCACCGCGCGGGG
1512	ATCAATATCTGGTCAGTTGGTTTTTAAATCCAGCCAGC
1513	AAGTACAAAGGACTAAAGACTTTTTCATGAGGAAGTTTC
1514	AACAGAGGTCTGACCTGAAAGCGTCAGAGATATAAATCCTTTGCAAG
1515	GATAAAACAGGACGTTGGGAAGGGCTCATTGTAATAGT
1516	CAGAGGGGATACCAAGTGCCTTTAAGGTCAGGATTAGAGA
1517	AGGGTAAAAGCACATCCTCAACTGACCAACTAATTGCTCGAAGCCCG
1518	GCGCGCTCGAGTAAAGAGTAGTTGATTTTAGGAATTGAG
1519	AATATACCGAACGAACCAACAGCACGCTGAACCTTGCTGAACCTC
1520	GCCACTACGAATACACTAAAAAATAATCTCTTTTGAT
1521	ACGAGTAGTAAATTGGGCTTGATCAAGAGTAATCTTAGGCTTTG
1522	CGATTAATTTTGGGATTTTAAACATTTTGGAGCTACAAACA
1523	GAGAGATAAGAATTTTAGCGAATGTTTACCAGTCCCG
1524	GACAATATCTGGCCAAAGAATACAATAGATACAACTAACGGAACG
1525	CAGCAGGCTTTTAAATCCTGTTTGATAAGCCGGCGAAGCTGGC
1526	CGGCTGGAGGTGTCCAGCATCAGCTAATGAATAGCAGCAAGCATCA
1527	GACTTCTCCGTGGTGAAGGATAGCTCTCACTTAAATTT
1528	TACAGGTAAGGCTTGCTGTGAATTAAATTGTGTCGAAATTTTTCGCGACCTGC
1529	TAATGTGTAGGTAAAATTTATCATCTTAAACAGCTTGA
1530	AGCCCCAAAAACACCATAAATCAACGTAAACAATAATAA
1531	TTTATTTCAACGCAAGGATAAAGACCCCTGTGAGATCTA

TABLE 9-continued

Staple sequences used for the T3 triangle 5 folding.	
SEQ ID NO:	
1532	TTAGCGACCCCTGAACAAAGTCAGCGCAACTGCACCACGG
1533	AAGACAAACATTAAAGCCCAAAGTTAAATAATTATACA
1534	TACGGTGTCTGGCCAAAGGCCGAGACAGTCGCATAAC
1535	AAAATGTTTTTTTTGACTGGACCTTATGCTTTTATTT
1536	CAGTACAACTCATAGTTAGCGTAACGATCAAGCAGCGCCA
1537	AGACTCCTTCTTTCCAAGCGCCATTTCGCCATTCAGGCTG
1538	TGGAGCAATCGGTTTAAAGGAACAATAAGCGACGACA
1539	TAATTGTAACAAGAGACAATCATAATTAGCAGTGGCAT
1540	CAATTCTATTAAATATTAAGCAAATATTTAAATATTTTC
1541	AATAGTGAGCAGGGAGTTAAGGCTCGGTCGCTGAGGCTT
1542	CAGAAAACGAGAAGCATCGTAACCGTTACGTTGGTGT
1543	CATTCCACGATTGAGGAGTAGCATTAAACACACCCTCAGAG
1544	ACGGAACAACATTTTTATGCAGATACAAACGGC
1545	GCCGGAACACCGGTTGTACCAAAACATTATATTTTGTAG
1546	CGCCTGATATCGCCCAAAATCACCATCTTTTATATGAT
1547	CATCTTCTATCGATAGCAGCAGACTCCTCAAGAG
1548	CTCCAAAAGGAGCCTTTACCGATAGGGTAGCTAGGGTGAG
1549	CAACAGTTAATTCAACCAGACAGCCACTACAAC
1550	TTAGCAAGATTGACGGCACTCCAATAAATCATAACAGGCAA
1551	AGATGGGCACTCCAGCCAGCTTTCGCCTCAGG
1552	GGTTATATTGACCGTAATCAGTAGGTAGCGCGTTTTCA
1553	ATCAAAATCGTCACCACATTACCAATAAGGCGAACTGGCA
1554	ATATTTTACAATAACAAAGCCTGTTTAGTATATAGGCGTT
1555	TACCGTCGTATCAGGTTGCCGTGAGAACCTCATATATTTGCCTTTAG
1556	TGCGGGAGAAGAAGTTCTTTTTTCCATATAACAGTTTAAATAT
1557	AGTAATTTTCATTTGGGGCGCTTTTAGCTGAAAAGTTTCGCAAATGGT
1558	ACATGATTTTGTAGTATTAAGAGGCAACTATAT
1559	GATTCAAAATTTTGAATACTTTAAATTAAGCAATAAAGGGCAAAGA
1560	TTTTCATACGTCAGACCGACAGAATCAAGTTTCTTTTAATTTAATGG
1561	GATTGACCGTAATGGGATAGGAATACCACATTCAACATTAT
1562	GAAAACTTTTTCAAATGTAATGCCTATTAATTAATTTCCCTTAGAA
1563	AATAACCTGTAGATTTAGTTTTTTTGACCATTACGGTAATGCCGGAGA
1564	AACTTTATTTTATCATCCTGACGACCCAAATCAACGTAACAGCAGCGA
1565	ACCTCCCGACTTGCGGTTTCATCGTAGGAAGAAAACATGTTCAAGCTAAT
1566	TCAGAAAGTAGATTTTGCTAAACAACCTT
1567	CATAATTATCTAAGAACGCGAATAAGCAAACGTAGAAAATACATACA

TABLE 9-continued

Staple sequences used for the T3 triangle 5 folding.	
SEQ ID NO:	
1568	CCTCAGAGGAGAGTCCAAAGGCAGGTGAATAATCATAGAAGAGTC
1569	CATAAAGCTAAATCGGAACCGAGCTCAGAGCAAATTATTAGGGCGAC
1570	GGATGGCTTAGAGCAACCGTTTTTCTAGCTGATAAATTAAT
1571	GACAAATGACAACAACCATCTCCAAGGAAGCTTTTAACTCCAAC
1572	TGTACCCCGGTTGGAGAGAATAGTCAGCTTGCTTACCGTGTGATAA
1573	AACGGCTACAGGACAAGAATTTTCGGATATTCATTAGAAACACCAG
1574	GCCAGTTTGAGGGGAGAATTGCGTAGCATGTATCGATGAAGATACA
1575	AATTCGCATTTTTTAAATTTTCGCCATCAAAATGCTTTAAACAGT
1576	GACGACGAGTTAATTACCGGAATAGCCAGCATCACCGTCACCG
1577	ACCAGACCAAAAAAGGTTTTTTCACGTTGAAAATAGCGTCCAAGCATCT
1578	CGGAGTTTTTTTTGTATCATAAGACAGCATCGGAACGAGGGTAG
1579	ATTCGATTTTCTTCAAAGCGAAAATCAGGTTTTTCTTTACCCCTGA
1580	AGGGTAATAGTATGTTACGGAATAGTGAATTAAATCACCAGAACCGCC
1581	CCTCCGGCAAACATAGCGATAGCTTAGATTAACCTAATTGTTTGAAA
1582	TCAGAACCGCCACCCACCCAGTAGCACATGAAACCGACCTAAA
1583	CAAAATCGAGAAACCAATCAAGGTTCCGCTCACAATCCACACTAAT
1584	TGAGAACTTCCTTATTCCTGAATTAACGTCATTCCAGT
1585	CCGTAATCGGCTGTCTATACGAGCCAGCCATACTTACCAACGCTAGTTT
1586	TAGCGTCTTTCAATGAAAAAGCTTTCAGTTGTAAA
1587	TGGGTAACCTCACATAGCTGTTTCTGTGTGAAAAGCCTGTCACAGTT
1588	TGTGGTGCTAATCATGGAGCCTCCGGGTGCCTAATGAGTGAGCGAGT
1589	TAGCTACAAATAAGAGAGGATCCTGAGCGGCGGGCGTTCTGAGAAG
1590	TTCAGGTTTTATTATACGTCAAACAGAGCCTAATTGCCAGTCCGT
1591	TCTGGCACTTAGCCTACAGACCTTAATTTCTAAGAACT
1592	GCGAAACATCCATGTTTGACCTTCAGATGGTAGGCGT
1593	TTTTACCTTCTTTAGTGATGAATTAAGAGGAGCCGTC
1594	GCGAGAGGCAGACGACAACACTATCATAAT
1595	AGGCAAAAGAAGGCACCAACCTAACCACGCAATTTGCGTAT
1596	TGCCAGGGTGGTTTTTCTCGTCGCTCGTCAGCG
1597	AGAAAGGAATTGGGCGGTTGTGTGGAACAAATAGGGTTGAGTGT
1598	TCAGTTTACATCGATGCCGTTTCATTTCGACA
1599	TTTTTGAGCAAGAAACAATTTTTAAGCCAGGCAAGACGTTAGTAAATGA
1600	TGTATGGTCGCGGCACCGCTTCTGGTGCCGAAAAAAGTAAGCATTTTT
1601	TTTTTCAGTTGAGATTTAGGTCACGCGGAATCGTTTTTTATAAAT
1602	TTTTTTTAACAACGCCAACAGTATAAAGCCAACGCTTTTTT
1603	ATCTTACCAACCGAGCACAAGAATTAGGCAGAGGCATTTTCGAGTTTTT

TABLE 9-continued

Staple sequences used for the T3 triangle 5 folding.	
SEQ ID NO:	
1604	AAGTAATAATATAAAGTACCGACAGAATCGCCATATTTTTT
1605	TTTTTCAACAGTAGGGGACGCTGAGGTCTGAGAGACTACAAATGCAA
1606	CAGAAGGAGAAGCCCTGAAATAGCAATAGCTTGAGCGCTAATATTTTTT
1607	TTTTTCAGAGAGATAACCGAAACGCACATATGCGGAATAAAC
1608	TTTTTTTCAGTGAATAGAAAGATTCATTTTTT
1609	AAAAGGTAAATTCTTACATGTAATTGAGTTAAGCCCAATAATAATTTTT
1610	TTTTTCCAGTAATAAGAGTCTGTCCATCCTTGATTAGGTTG
1611	TTTTTGATAGCCGAACAAAGTTACTGATTTTCAGCGGAGTAGGTAAATCATTCGCCT
1612	GTTGCGCCCGATATATCGCTTTTGCGGGATCGTCACCTCAAAGCTGCTCATTTTT
H1 design oligo sequences (5'-3')	
1613	GCAGTAGAGTAGGTAGAGATTAGGCAGGCAATTCGAACCCATCAGCATTGACGACCGC
1614	GCAGTAGAGTAGGTAGAGATTAGGCAACGACGGCATCTCATTTTCAGGGAACACTGAG
1615	GCAGTAGAGTAGGTAGAGATTAGGCAGGCGCTAGGGCGCATAGAGCTTGA
1616	GCAGTAGAGTAGGTAGAGATTAGGCAAGCTCATTAGAAACCACACCAGTCAAACCGAA
1617	GCAGTAGAGTAGGTAGAGATTAGGCACTGCTCATCGTTGAGTAACATTAT
1618	GCAGTAGAGTAGGTAGAGATTAGGCAATCCCCCTCAAATAAGTTTTGCGTAAGAGC
1619	GCAGTAGAGTAGGTAGAGATTAGGCAGCCTGTAGTGTAACGCTAATAGTGAGGATAA
1620	GCAGTAGAGTAGGTAGAGATTAGGCAAAGATCGCTGAGGAAGATTGTATTGTATA
1621	GCAGTAGAGTAGGTAGAGATTAGGCAAATTAAAGTTTTGTCGTATTACGC

TABLE 10

Staple sequences used for the T3 triangle 6 folding.	
SEQ ID NO:	
Core structure oligo sequences (5'-3')	
1622	CGCGCTTAATGCGCCGCTACGAATCAGAATCGGAACACCGGGG
1623	AGTACATATTAACCTCCGGTTTTTTTAGGTTGGTTATCGCGCA
1624	TCCGTGAGCATTAAATTGCGTTGCGGGTCACGCCGGTTACCTGCAG
1625	ATTAGGCAGGTCAGACGATTGGCCAAAGCCAGATAGCCCCGTCGA
1626	TATGAGCCGCTCACTGTCAGTGCAGCAGCAGCAACAGCTGA
1627	ATCATTACCGATAACTCATCGAGTTTTTACAAGCATATAACTA
1628	TAACGGATTTCAGATGATGATACGAACCGCCCAAAACGCGCCA
1629	GTCACCGAGCGCCACCATTTTTTAACCACCACAGAGCGGCC
1630	AGAACGCGCAATCAATAATCGGCTGAACCTCCTTCATCGTAGG
1631	CGACTTAGTACCGCAATAAAGAAATTGATTCGACTTGC
1632	CCAAGTTTTTAACGGGTATTAATTCTAAGTTTTTACGCGAGGC

TABLE 10-continued

Staple sequences used for the T3 triangle 6 folding.	
SEQ ID NO:	
1633	TGCGGCTGGTAATGGGAATTCGTAGGCCAGAATGCGTTTGAGGGTTG
1634	CCTCAGAGACTGGATTATACCTCTTTTGTATCCCAATACAAAATA
1635	CGTTCCTGGTAATATCATACCTACCCCTCGTTAAGGGCGCG
1636	ACATCGGGTACACTGGTCTTTGCTCGTCATAAAAAGAAGAAGATAAGT
1637	GCCACCACCTCATCAATAGAAAATTTTTTATATGGTTT
1638	GGAAGGGCGATCGGTGCGGGCCTCTTTTTTGCTATTACGCC
1639	AATGGAACTAATCTGTCCAGATTTTTGACGACA
1640	GAGAGAATAACATAAAAGCTGGCGATATAATTTTAAGAAACGCAAA
1641	TTACATTTAACAATTCATTGAATTTTTTCCTTTTTT
1642	GCGCAGTGCCCGCTTTTGAGACGGGTGCCTGTCCGGGTCCAC
1643	TGTGTTTCAGCAAATCTAACAGTGTGTTTTATTGCAGTAAGCGTCAT
1644	AGTTTATTTTTTTGTGACAGGCGATTAAAGAAC
1645	CAGAACAAACGGTACGGAGGCCACCACTCTGTCATCAGATTGTTGCCC
1646	TAGAAGAACTCAAACCTTTGATTAAATTAACCGTTGTAGGTTAACGG
1647	TCCTCATTTTGATATTACCCCTCAGAACCAGCACCATATCTAACAGTA
1648	AAGCCGTTTTTCGTAGATTTTCAGGTAGAAACCCTGTTTATCAACAAT
1649	TTATTCATCAAAACATCAAGAAAACAAATTAAGCTAATGCGTTACAAA
1650	ATAAAGGTGGCTTGGGTAACGCCACCTTTACAGGGAGGTTTGAAGCC
1651	CCCGTATAGGGTCAGTGCCTTGAGCAATACTTTATCGGCCTAATCAGT
1652	AACAGGGAAGCGCATTTTTTAGACGGGAGAATTAAC TGAA
1653	GCTGGTAGGGCGAAAAACCGTCGCGCCATGAAGGAGCG
1654	CTCCCTCAGACATAGCCCCCTTATTTTTTAGCGTTTGCCA
1655	GGAATAGGATATACAGAAAATTATTATTATTTTAACGTCA
1656	TACAATTTTATCCTGAATCTTACCTGCCAGTTCCAAATAA
1657	CGATGGCCTTTTGGGGTCGAGGTGCTTCGCGGTTTCTGC
1658	ACGAGCATGTTTAACGTGCGCTGATGCTCAGTTTCGGAAC
1659	TATCAGGGGGGGAAAAGCACTAAGCGGGAGCAGACAGGA
1660	GCAATTCATCAATATAGCGGAATTTGCTTTG
1661	ATCGGCATCTATTATTCTGAAACAGAGCGAAAATACCAA
1662	AACGGATAGAAACGATTGAATAATGGAACCTACTATGGTATCATCAAGAAACCA
1663	CCGTAACACAAGCCCAATAGGAAAATCCTGATTGTTGTTGTGGAGCCGCCACGGG
1664	GGATCCCCCTGCGGGCCGTTTTCTGAGAAGCGTGCTTT
1665	GTTTTAGCGTCTTTCCTTATCATTATAAACAACATGTTCA
1666	CCGATTTAAGATGATGGGAAACAATCGGCGAAACGTACA
1667	AAAACAGACACCTCAGTACTCAGGAGGTTTCGGGGTTTTTGCTTTG
1668	TGATGAAATTCATTACCTGAGCACTGCCTATACCAGGCGTACAGGAAAATAAA

TABLE 10-continued

Staple sequences used for the T3 triangle 6 folding.	
SEQ ID NO:	
1669	TTTCGGTGTGTACTGGTAATAAGTTTGATGAGATAAGTG
1670	GAAGCATAAAGTGTAATTGTTATCCGAGCTCGTAAAGGTT
1671	ACATGGCTTTTAAACGAACAGTTAATGCCCCACATCCCT
1672	TGACAGGAGGTTGTTGACGCTCAACGGAACAATATTCCTGATTGGCAA
1673	AAAGGGGATGTGCTGTGCCAAGCAATAGCAGGGGTTTC
1674	GAGGGGTTAGAACCTACGTATAAATATAAGTGAATTTAC
1675	ATTAGGATTAGGAGCCATTTTTTTGGGAATT
1676	GTAAGTTACATCCTCATATGACCAACTTTGATTTTTGCGCGTTTTA
1677	GAACCCCTTACATTGGCAGATTTTAAAGCCTTTGCCCCAACGTCAGCGGAT
1678	GTCATTGCAGGCGCTTCTTTTTCACTCAATCCGCCGGGCGCGTTGCGG
1679	CTTATAAAGTATTGCAAAAAATAACAATTCGACAAACCACCAACCCG
1680	CAACTAATCCATTGCCATTAAAAATCCTGA
1681	CCGAACACGATTGAGGACCTGCAACGTGAGGCGGTCAGTATAGTACGGT
1682	AAAAGCCGCACAGGCGGCCTTCAATAAGAGGATCGCGTCTAATAGGAA
1683	CAGCGTGGTGGTCAGTCAATGAAGAACGAA
1684	GAAAGAGGCGGACTTGTAGAACGTAGCCTCCGAGCACTAAACATTTGA
1685	CTTTACACCCGTAACAAACTTAAAAAGAGACGCAGAAAGCTTTCAT
1686	CTGGAGGTGTCCAGCATCAGCGGATGAATCGAGCAACCGTCAGAC
1687	CCACCAGCAGAAGATAAGCCCTAACACAGTT
1688	TGGTGAAGGGATAGCTTCCAGTTATCGACATCGTTCCGGTAGATAAT
1689	CAACATTAGGCCTTCTGTAGCCAAAAATAATAGCCGAAACACGACC
1690	AGAAAGGAAGGCGGTTGTGTACTGGAACAAGAAATCGGCAAAATCC
1691	GAATAAAGGGACATTCTGGCCAAAAGAATACTAAATAGAAAGGA
1692	TGAATGGCTATTAGACCATTAGATCGCGAGCTATTGAGCAAAAAGAT
1693	AGATTTAGTAGAGCTTAAGAGGTCAAAGAGGA
1694	CAAGCGCGTAAAGCTCAACATGTTGTTTCATTTC
1695	TATTACTTTTTGCCAGAGATTAGATCAATATCTGCTGGTCAGCACCGT
1696	TGCCCTTCACCGCTTTTTTCGTCTCGTCGCTGGC
1697	GATCCAGCCGGTGCTTTTCCCTGCACAAGAATG
1698	AATTGCTGCAGACGGTCCCCAGCTAAACGGGTAAAATACGTAATGCC
1699	TGGATTATAACAGGAAAAATTTTCGCTCATGGAA
1700	AACATCGCTCTGACCTGAAAGCGTCAGAGATAGGATTTAGAAGTCTT
1701	TTTGTCTTTAATGCGGAACTGATAAACAGAGAGTGCCACGCTGAGA
1702	AAGAAGTAAACAGTAATGTGAGCGAGTAACAACCCGTCATAACC
1703	TTTGAGTAACATTATTTTTCATTTTGTGCTCTGAA
1704	ACTACGAATACACTAAAACTCTAACACCGGGATGGCT

TABLE 10-continued

Staple sequences used for the T3 triangle 6 folding.	
SEQ ID NO:	
1705	GACACGAACTAACGGGTAAGAGCAACACTATCGGATTCTCATTGAAT
1706	GTGTAGCGGTTTTTCACGCTGCGCCGCCAGCAGTTGGGGTTTCTCCG
1707	CCGTGGGAACAAAGCATAAACAACATTATTACAGGT
1708	TTGTGAGAACGTGGACTTTTTCCAACGTCAATTGCCCCAGCAG
1709	AAATCTAAAGCATCACCTTGTTTTTTGAACCCAGCGGT
1710	AAACAAGATGTTACTCCCAATCATTTTAACGCATAGG
1711	AGAGGCTTTTGTAGTGATGAAGGCAAATATCAGGTTATC
1712	GCCCTGAGTTTTTGAGTGCAGCAAGCTAAGCCGGCGAACGTGGC
1713	CCAGTCGGGAAACCTGTTTTTTGTGCCAGCTGCATTA
1714	TTAAATATGCAACTAAATCTTTGACAATCATAAGGGAA
1715	CGATTAAATTTTGGGATTTTAAACATTTTGAGGCGCCGTCAA
1716	TAGCCGGAAGACCAGGGAACGGCCTACGTTAATAAAGAT
1717	CATTGTGAATTACCTTATGCGAACGTAACAAAGCTGCTTTGAGG
1718	AGGGGTAACAAAACAGAACAGAAATGCTCCTTTTGAT
1719	TACAACGGAATAAGACTTTTTCATGAGGAAATTCATTA
1720	CGAAAATCCTGCGCCAGGTTGGTTTTTTTTCTTTTACCAG
1721	CGCCTGTGCGAGTAAAGAGTAAATATTTTAAACCTCAA
1722	GAACGTGCCAAAAGAAGGCACCAACCTAAAACCGCAACCA
1723	GCTTACGGCGGTGGTGCCATCCAGCCAGCAGTGGCAAAT
1724	CAACGCGGTCCGCTGGTGGTTCCGAGTCCACTATTAAAG
1725	AGATAGACAAGAATTTTAGCGATTACCAGTCCCGGAA
1726	CTCCTGTCCATCACGCAGTAATAACATTTTCACTTGACC
1727	GATTGTATCGTTTGCTAAACAACCTTCAA
1728	GGGTGAGAAAGGCCGATAGCGATTAAACAGCTTGATAC
1729	TCCACAGACCGATTGAATACAGGCAAGGCACCACTCAG
1730	AGAGGGTAGCAGTATGTAACGGAAGGTGAATCAAATCACCGGAACCG
1731	TTTTTAGAACCTCATATATTTAAGCCTTTATTGCCTG
1732	AGGTCAGGAAAGGCTTTTCACGTTGAAAATCGCGTCCAATACTCTGCC
1733	TCATAGGTGACCACCGTAATCAGTACTGTAGCGCGTTTT
1734	TGAGATTTTGTAGATGTCAGGACGTTGTGTCGAAATCCGTTTTTGACCTGCTCC
1735	AGTAAAAGGTGGCATCAATTCTTTTACTAATAGTATTAGCTATATTTT
1736	TCTTTTCAAGCGTCAGAGCGACAGAATCAAGTGAGTCAATACTTTTTC
1737	TAAGCAAATATTTACCCTGACTATTGTACCCCAATAATTT
1738	CAAAAGGAGCCTTTAACGATAGTTCAAAGGCTCAAATCAC
1739	TCAGGTCTCGTAACCGTGCATGCGGTGTAGA
1740	TTTTAAGGGAGCAAAATTCAAAATGCCTGAGTAATGTGTTGCCTTT

TABLE 10-continued

Staple sequences used for the T3 triangle 6 folding.	
SEQ ID NO:	
1741	ATGTAAATGCTGATGCCTACCTTTAATCAATATATGTGAGTGAATAAC
1742	TCATTATTTTTTACCAGGTTTAATTCCTGACGAGAAACACAGCGAAAG
1743	TAATCAAAAACGTCACACCATTACTTCTGACCAGAAGCTGG
1744	AGAAGGCTTATCCGGTAACCAAGTACCGCAATCCGACAAAAGGTAAAG
1745	TGAAAGTTTTTATTAAGAGGCTGACTGAGAGA
1746	CAGATATCACCTGAACAAAGTCAACTGTTGGACACCAC
1747	AAACATTATGACACCGGAACAGCCCTCAGAGGAAATTTAAAGGGCG
1748	TGGGCGCACCAGCCAGCTTTCCGGTCAGGAAG
1749	CATTAGCAATATTGACGCCAAAGAATTAGCAAAATTAAGC
1750	AGGCCGGAATCCCTGTAATACTTTTGCGGGAGAAATGCAA
1751	ACGGTAATGTTTATCAGGAACAATAAGGAACGACAGTA
1752	CTGATAAAGCCACGCAACCGTTCTAGCTTTTGATAAAT
1753	CAGTTTGATACATTCAACAGCCCTCACAACGCC
1754	CAAAGACATTCAATTAATACCCAAATAAATTTACGGAATC
1755	GAGACAGTATCAGGTCATTTCAACCATAAAGCTAAATCGGAATAAAGC
1756	TAACATCCTAAATTTTTTAATATTTTGTTAAATTCGT
1757	TTGAAAACGGGAGTTAAGGCCGCGTCGCTGAGGCTTGCA
1758	TGTTAAATCAGCTCATTTTTTTTTTAACCTCAGAAA
1759	GAAAGATAAAGTTTTTGAATTACGAGCGGCGGA
1760	TACAACTATAGTTAGCGTAACGATCTAAATAACGAGCGC
1761	GGCTACAGAGGCTCATTCATTTTTTGAATAAGGCTTGTCAACTTTAA
1762	AAATCCAAATAAAGTAAAGGCGTTAAATAAGACAAGCAAAT
1763	TGACCGTAATGGGATAGGCAGATACATAACGCCTCATCAGT
1764	ATTTGGGGACATTTTCGCAATTTTTTGGTCAATCAATCATAGAGATCTA
1765	CAAAGAACCCTCGATAGCAGTCCTCAAGAGAAG
1766	GCAAGGATAAGATTCCCATTTTTTCTGCGAACGAGTGTCTGGAA
1767	AGTTTGAGGGACGATTGCGAATGGTTGATATAGCATGTAACCTGT
1768	ACCGTGTGCGCCCAATAGCAATAATTAGCAAACGTAGAAAAATACATA
1769	CCTCAGAACCGCCAAGCCACCAGTAGCCAATGAAAGCGAGAAA
1770	AATATAATGCTGTTGCCGATTTTTAGGGTAGCTATTTTTGA
1771	AGTGAATTCTGTAAATCGTCGCTATTAATTAATATACAAAAATATA
1772	ATATAACAGTTAAACATCAATATGATATTCATAACCGA
1773	TTGTACCATCGATGAAGAGTCTGAATTCAGCTTAGATAGAATCC
1774	CCGGAATTTTTCAAACCTCAACTATAGTCATTTTTAAGCAAAGCGG
1775	AATGACAACAACCATCTCCAAAAAATTAGATTTTTAGTACCTT
1776	ATAGTAAATGTTTAGATTTTTTGGATATTGGGAAGATTTTTAAAT

TABLE 10-continued

Staple sequences used for the T3 triangle 6 folding.	
SEQ ID NO:	
1777	TAAGACTCTTCCAGACGCCATTGCGCATTCAGGCTGCGC
1778	AAGAGAATTCGCAAGAAAATACCGAGAGCCAGTATCACCGTCAC
1779	AGCCCCAAAAACAGGAATAGAAAGCTTGCTTTCGGTTAATTTTCATC
1780	TTGTATCGCGTAAACATCAGAAACTCAGAGGTAGCAT
1781	AGATTTTTTTGTATCATCGCACAGCATCGGAACGAGGGTAGCAA
1782	AGAAACAACCTGAACAAGAAATACCGCTCACAATTCACACAAAAC
1783	TTTCCATCCTAATTTTAAATCATTGCACGTCCAGTCAC
1784	GGGAATAATATCCCATACGAGCCGAACAGCCAAGATTAGTTGCTAAGTT
1785	TAATTTTTCACCCAGCAAAATGAATTCAGAGGTAAACG
1786	TTTTACTCACATAGCTGTTTCTGTGTGAAAAGCCTGGGCAGTTGACACCCAA
1787	GGTGCTGCATCATGGTCCTCCTCAGTGCCTAATGAGTGAGCTATCAT
1788	TTTGTTTCCAGAGCCTAATTAACGCTAATGAACCATACCTCACC
1789	CCAGAATCCACGGTCACCTAAAGGATCAAGTTCACTACGCGAGCGTCTTCT
1790	TTCTTGACAAGAACCGGATGTTTCCATGATTATAC
1791	ATTGCATCTTCAAAGCAAATAGCGCAGATGAACCTTCATCAAT
1792	TTACCAGACCTGGCTGACGGTGACACGAGGCG
1793	ACGAGAATGAATGCTTTTTTGCAGCTCACGT
1794	CCAACGGCTGGTCAGCGCCAACGCGCT
1795	TAGGCGGTTTGCGTATTGGGTTTGATG
1796	TCTTTAGGGCCAGAGCAAACGATGTCAAAGAATT
1797	TAGATAGGGTTGAGTGTGCTCACGGAATTTCTG
1798	TTTTTATCAGAGAGATAAAGGAAACGATAAACACATGGTTTG
1799	TTTTTGTAAATTGGGCAGGAATACCACTTTTT
1800	ACCAGAAGCCGAAGCAATGAAATAGCAATAGATTGAGCGCTAATTTTTT
1801	TTTCGAGAACATGTAATTTAGGCTTCTTACCAGTATTTTTT
1802	ATGGGATGCCACCGCTTCTGGTGCCGAAACCAAGAAAAGTAAGTTTTT
1803	AGAGGCATATAATTACAACAGTAGAATTGAGTTAAGCCCAATAATTTTTT
1804	TTTTTATTCAACTAATGTCACGTTGGAATCGTCATTTTTTAATATT
1805	TTTTTATTTAACAACGCCCCAGTAATCTTGCTTTATCAAAA
1806	TTTTTAAAGCCAACGCTCTAGAAAAAGCCTGTTTAGTTTTTT
1807	TTTTTAAGAGCAAGAAACCCTTTTTAGGCAAAGCGTTAGTAATGAATT
1808	CTATCTTAGAAACCGCCCAAGGGCTTAATTGAGAATCGCCATTTTTT
1809	TTTTTCAGATAGCCGAACAAAGTTCATCAGCGGAGTGAGGAAGTAACAAGAGAA
1810	TTTTTATCATATGCGTTTTTCCCTTTAAGACGCTGAGAAAGGTAAAG
1811	GCGCCGACTATATTCGTTTTGCGGGATCGTCACCTCAGCCAGAACGAGTATTTTT

TABLE 10-continued

Staple sequences used for the T3 triangle 6 folding.	
SEQ ID NO:	
H1 design oligo sequences (5'-3')	
1812	GCAGTAGAGTAGGTAGAGATTAGGCACCGAAGGCCCATGTAGCCAGCATACGGCCAC
1813	GCAGTAGAGTAGGTAGAGATTAGGCAACGGCCAGCAATTTTCAGGGATAGCTGAGTTT
1814	GCAGTAGAGTAGGTAGAGATTAGGCAGGCGCTAGGGCGCATCGAGCTTGA
1815	GCAGTAGAGTAGGTAGAGATTAGGCACGCCATCATATTAATTCACCAAGTCAGAATTAGA
1816	GCAGTAGAGTAGGTAGAGATTAGGCATCTATTTGCGTACTCGTATTAAAT
1817	GCAGTAGAGTAGGTAGAGATTAGGCACCCCTCAACCATAATTTTTTCAAAAA
1818	GCAGTAGAGTAGGTAGAGATTAGGCATGTAGCATATTCGCATAATAAATCGGGAGGAA
1819	GCAGTAGAGTAGGTAGAGATTAGGCAATCGCACTTTTAAATTGTAAACG
1820	GCAGTAGAGTAGGTAGAGATTAGGCAGGAAGTTTGTGCTCTCTTATTAC
DNA sequence complementary to DNA handle of H1 to H3 design oligos	
1821	TGCCTAATCTCTACCTACTCTACTGC

[0139] It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination. All combinations of the embodiments pertaining to the invention are specifically embraced by the present invention and are disclosed herein just as if each and every combination was individually and explicitly disclosed. In addition, all sub-combinations of the various embodiments and elements thereof are also specifically embraced by the present invention and are disclosed herein just as if each and every such sub-combination was individually and explicitly disclosed herein.

[0140] The present invention is not to be limited in scope by the specific embodiments described herein. Indeed, various modifications of the invention in addition to those described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are intended to fall within the scope of the appended claims.

[0141] To the extent possible under the respective patent law, all patents, applications, publications, test methods, literature, and other materials cited herein are hereby incorporated by reference.

[0142] The following Examples illustrates the invention described above, but is not, however, intended to limit the scope of the invention in any way. Other test models known as such to the person skilled in the pertinent art can also determine the beneficial effects of the claimed invention.

EXAMPLES

Introduction

[0143] Effective broadband antiviral platforms that can act on existing viruses and viruses yet to emerge are not available, creating a need to explore treatment strategies beyond the trodden paths. Here, we report virus-encapsulating DNA origami shells that achieve broadband virus

trapping properties by exploiting a widespread background affinity of viruses to heparan sulfate proteoglycans (HSPG). With a calibrated density of heparan sulfate (HS) derivatives crafted to the interior of DNA origami shells, we could successfully encapsulate adeno-, adeno-associated-, chikungunya-, dengue-, human papilloma-, noro-, polio-, rubella-, and SARS-CoV-2 viruses or virus-like particles, in one and the same HS-functionalized shell system without the need for virus-type-specific binders. Our HS-functionalized shells amplify the individually weak and reversible interactions of HSPG to viral surfaces through strong avidity effects that emerge when curvature-matching HS-coated shells engulf the virus particles. Depending on the relative dimensions of shell to virus particles, multiple virus particles may also be trapped per shell, and multiple shells can also coordinate and enclose clusters containing dozens of virus particles. Since steric occlusion in virus-engulfing shells can prevent viruses from interactions with host cells, the heparan sulfate-coated virus-engulfing shells open an attractive path for establishing a broadband antiviral treatment strategy.

Example 1: Shell Design and Synthesis Principles

[0144] Here, we address the challenge of creating a broad-spectrum antiviral by exploiting the conserved background binding of HSPG to viruses to irreversibly trap viruses in HS-functionalized neutralizing shells (FIG. 1A, right panel).

[0145] To capture differently sized viruses, we fabricated three DNA origami shell variants and functionalized their interior with the same HS derivative. We used the previously described octahedral and T=1 icosahedral half shell designs (O and T1, respectively; Sigl et al., loc. cit.) featuring 40 nm and 85 nm wide cavities, respectively (FIG. 1D). We also developed a new T=3 icosahedral half shell design, termed T3, for the encapsulation of larger virus particles that do not fit into O and T1 shells (FIG. 1E). The T3 design is a finite-size higher-order assembly consisting of a total of 30 triangular subunits, partitioned as five copies of six different full-size DNA origami triangle designs with specific edge docking rules (FIG. 6). The resulting shell has a cavity

diameter of approximately 150 nm. Negative stain transmission electron microscopy (TEM) images validate the successful assembly of T3 shells (FIG. 1F and FIG. 7).

[0146] Staple strands for origami folding reactions were purchased from Integrated DNA Technologies (IDT) and used with standard desalting purification unless stated otherwise. DBCO-modified handle strands were purchased from Biomers at HPLC grade. Azide-modified heparan sulfate derivatives were purchased from Glycan Therapeutics (catalog references: **1a**: GT24-AZ-021; **1b**: GT24-AZ-005; **1c**: GT18-AZ-003; **1d**: customized product). VLPs were purchased from The Native Antigen Company, Creative Biostructure and Creative Biolabs (catalog references can be found in Table 12).

TABLE 12

VLP providers and catalog references.		
VLP	Provider	Reference
Poliovirus type 3	Creative Biolabs	VLP-003YF
Dengue type 1	The Native Antigen Company	DENV1-VLP-100
Norovirus G IL4	The Native Antigen Company	REC31620-100
HPV 16	Creative Biostructure	CBS-V641 HPV16
Chikungunya	The Native Antigen Company	CHIKV-VLP-10
SARS-CoV-2	Creative Biolabs	VLP-050YF
Rubella	The Native Antigen Company	REC31651-100

[0147] Folding of DNA origami triangular subunits

[0148] DNA origami structures were folded in one-pot reactions containing 50 nM of single-stranded scaffold DNA (M13, 8064 bases) and 250 nM of each staple strand in a standardized “folding buffer” (FoBx) containing x=20 mM MgCl₂, 5 mM Tris Base, 1 mM EDTA and 5 mM NaCl at pH 8.00. Scaffold M13 was produced as previously described based on M13 8064 as scaffold sequence (SEQ ID NO: 1; Engelhardt, F. A. S. et al., ACS Nano 13 (2019) 5015-5027). All folding reactions were subjected to optimized thermal annealing ramps (Table 13) in a Tetrad (Bio-Rad) thermal cycling device. It should be noted at this point that any variant of M13 8064 or in fact any other single-stranded DNA of sufficient length could have been used as scaffold sequence together with a correspondingly designed set of staple strands. Alternatively, DNA origami structures of the type used in the present application could be constructed by using different sets of overlapping single-stranded oligonucleotides and standard DNA origami techniques.

TABLE 13

Temperature ramps and scaffold used for each DNA origami triangle subunit. For scaffold sequence see SEQ ID NO: 1 in Table 1. For staple sequences see Tables 2-10.				
Structure	Denaturation step (15 min) (° C.)	Temperature ramp (1° C./1 h) (° C.)	Storage temperature (° C.)	Scaffold
T _{octa}	65	60-44	20	M13 8064
T1 (pentamer triangle)	65	58-54	20	M13 8064
T1 (ring triangle)	65	65-52	20	M13 8064
T3 (6 triangles)	65	56-52	20	M13 8064

Purification of Triangle Subunits and Shells Self-Assembly

[0149] All origami structures were purified using agarose gel extraction (1.5% agarose containing 0.5×TBE and 5.5 mM MgCl₂) and centrifuged for 30 min at maximum speed for residual agarose pelleting. If the origami needed a concentration step, ultrafiltration (Amicon Ultra 500 µl with 100 kDa molecular weight cutoff) was performed prior to shell assembly. For shell assembly, the purified triangles were mixed in 1:1 ratio. Typical triangle subunit concentrations ranged from 5 to 400 nM, while assembly times depended on the shell type. Table 14 summarizes and offers a comparison on the optimized salt concentrations, temperature, and self-assembly times required for all shells used in this study.

TABLE 14

Half-shells assembling conditions.			
Structure	[MgCl ₂] (mM)	Temperature (° C.)	Time
O shell	40	40	overnight
T1 shell	40	40	24 h
T3 shell	25	40	8 weeks

[0150] The assembled shells were UV cross-linked for 1 h at 310 nm using *Asahi Spectra Xenon Light source 300 W MAX-303* (Gerling, T. et al., Sci. Adv. 4 (2018) eaau1157). Buffer exchange to 1×PBS containing 10 mM MgCl₂ was performed prior to VLP encapsulation experiments using ultrafiltration (Amicon Ultra 500 µl with 100 kDa molecular weight cutoff) or dialysis (D-Tube™ Dialyzer Mini, MWCO 12-14 kDa, 2×500 ml exchanges over 8 h, r.t.).

Heparan Sulfate Attachment to DNA

[0151] We used a strain-promoted azide-alkyne 1,3-dipolar cycloaddition reaction (SPAAC) to covalently attach a heparan sulfate derivative to a DNA oligonucleotide (FIG. 1B, FIG. 5) which can hybridize to specific acceptor sites in the interior of the DNA origami shells: single-stranded DNA extensions termed “handles”. For the coupling, we used azide-group modified HS derivatives containing either 8 or 18 saccharide monomers (**1a** and **1c**, respectively), including monomers such as N-acetyl-glucosamine and glucuronic acid which characterize HS polymers. As controls, we used 8-mer and 18-mer polysaccharides lacking the sulfate and sulfonate groups (**1b** and **1d**, respectively). The DNA oligonucleotide to be clicked to the different HS polymers was modified with a dibenzocyclooctyne (DBCO) moiety (**2**), and the SPAAC reaction occurred rapidly upon mixture of both components. We analyzed the reaction products (**3a-d**) by polyacrylamide gel electrophoresis (PAGE), which revealed different electrophoretic mobilities for the different product versions consistent with expectation (FIG. 1C). Higher molecular weight reaction products had slower mobility, and the sulfate-containing products migrated faster in the gel compared to the products lacking the sulfate groups, which we attribute to the additional negative charges.

[0152] Excess of azide-modified heparan sulfate derivatives (**1a-d**) were mixed in a 4:1 ratio with DBCO-modified DNA to form the respective products (**3a-d**). MgCl₂ was added to a 0.5 M concentration and the mixture was left overnight at 37° C. to achieve >90% conversion. The products were run in a preparative 10% PAGE gel for 2 h at 35 W. Subsequently, the product bands were cut away and were crushed. 1×TEN buffer (10 mM Tris-HCl, 1 mM

EDTA, 100 mM NaCl, pH 8.00) was added to dissolve and recover the modified oligonucleotides, and EtOH precipitation was used for concentration and buffer exchange. The pure products were redissolved and kept in double distilled H₂O at either 4° C. or -20° C.

Attachment of Heparan Sulfate-Modified DNA Constructs to DNA Origami Shells

[0153] We then hybridized the HS-modified DNA oligonucleotides to sequence-complementary single-stranded DNA handles protruding from the target DNA origami shell's interior surface.

Testing of Different Heparan Sulfate-Coupled DNA Origami Shells

[0154] In initial experiments with adeno-associated virus serotype 2 (AAV2) we explored three different DNA handle designs: proximal (H1), distal (H2), and branched (H3) to determine the type and density of the HS modifications required for efficiently trapping viruses (FIG. 1G). These initial experiments with O half-shells showed that H1 was least efficient, and both H1 and H2 designs were not as efficient for virus trapping as the H3 branched handle design. Samples were analyzed via negative stain transmission electron microscopy (TEM), where images were collected using an automated montage setup to minimize bias. Blind TEM quantification of particles revealed about 96% of shells to be occupied with AAV2 when H3 was hybridized to the 18-mer HS derivative (3c), improving from the about 30 and 84% of occupied shells achieved with H1 and H2, respectively (FIG. 8). We therefore used the branched handle design H3, and the HS 18-mer variant (3c) henceforth, unless otherwise specified. We confirmed that the interaction with AAV2 is due to the sulfate and sulfonate groups present in the HS structure, as the 3d HS derivative used as negative control did not demonstrate any binding (FIG. 9). Importantly, all AAV2 particles were trapped with O shell excess (FIG. 10).

Example 2: Trapping of Different Viruses by DNA Origami Shells

[0155] With the HS handle design thus established, we tested the HS-modified DNA origami shells for their ability to trap a variety of exemplary viruses and virus-like particles (VLPs) (Zeltins, A., Mol. Biotechnol. 53 (2013) 92-107). Our target virus library sampled enveloped and non-enveloped particles, particles from different viral families, and particles with dimensions ranging from 25 to 90 nm (Table 11, see also FIG. 11 for TEM images).

Maturation of Dengue VLPs

[0156] Dengue VLP maturation was adapted by published methods (Yu, I.-M. et al., Science 319 (2008) 1834-1837;

Yu, I.-M. et al.; J. Virol. 83 (2009) 12101-12107). Briefly, dengue VLP sample (10 µl, 0.39 mg/ml, The Native Antigen Company, cat. no. DENV1-VLP) was added to MES buffer (10 µl, 50 mM, pH 6.00) and gently mixed. Next, CaCl₂ (aq) (0.75 µl 0.1 M) and furin (3.9 µl, 2000 U/ml, New England Biolabs, cat. no. P8077) were added and mixed, and the sample was incubated at 30° C. for 16 h. After incubation, Tris buffer (25 µl 100 mM Tris-HCl, 120 mM NaCl, pH 8.00) was added to the sample, and the sample was immediately dialyzed against 1×PBS (D-Tube™ Dialyzer Mini, MWCO 12-14 kDa, 2×50 ml exchanges over 24 h, 4° C.). Matured dengue VLP sample was used immediately and stored at 4° C.

Viruses and VLPs Encapsulation

[0157] We used HS-modified O shells to sequester AAV2, poliovirus, mature dengue, and norovirus (FIG. 2a); HS-modified T1 shells to trap human papilloma virus 16 (HPV 16), SARS-CoV-2, chikungunya and rubella particles (FIG. 2b); and the HS-modified T3 shell for enclosing adenovirus 5 (FIG. 2c and FIG. 12 for TEM tomography).

[0158] Pre-assembled and UV-welded shells in 1×PBS containing 10 mM MgCl₂ were mixed with a VLP sample in the appropriate ratio to achieve either shell or VLP excess. The MgCl₂ concentration was adjusted to 10 mM and the samples were incubated at RT for 2 h. Usual amounts of sample for TEM analysis range from 5-10 µl total solution at about 10 nM triangle origami concentration. Negative stain TEM grids were prepared immediately after the 2 h incubation.

Negative Staining TEM

[0159] Samples were incubated on glow discharged (45 s, 35 mA) formvar carbon-coated Cu400 TEM grids (Electron Microscopy Sciences) for 90 to 120 s depending on origami and MgCl₂ concentrations. Next, the grids were stained for 30 s with 2% aqueous uranyl formate containing 25 mM NaOH. Imaging was performed with magnifications in between 10,000× and 42,000× in a SerialEM at a FEI Tecnai T12 microscope operated at 120 kV with a Tietz TEMCAM-F416 camera. TEM micrographs were high-pass filtered to remove long-range staining gradients and the contrast was auto-leveled using Adobe Photoshop CS5. To obtain TEM statistics in an unbiased fashion, automatic grid montages were acquired. For detailed information on selected particles, negative stain EM tomography was used as a visualization technique. The tilt series were performed from -50° to +50° and micrographs were acquired in 2° increments.

[0160] Tilt series were processed with Etomo (IMOD) to acquire tomograms (Kremer, J. et al., J. Struct. Biol. 116 (1996) 71-76). The micrographs were aligned to each other by calculating a cross correlation of the consecutive tilt series images. The tomogram is then generated using a filtered back-projection. The Gaussian-Filter used a cutoff between 0.25 and 0.5, and a fall-off of 0.035.

TABLE 11

Target viruses and virus-like particles tested in this study.						
Virus/VLP	Family	Enveloped	Surface details	Genome type	Approx. size (nm)	References
AAV2*	Parvoviridae	No	3 capsid proteins	ssDNA	25	1
Poliovirus type 3	Picornaviridae	No	4 capsid proteins	ssRNA [†]	30	2
Dengue type 1	Flaviviridae	Yes	1 envelope protein	ssRNA [†]	30-40	3
Norovirus GII.4	Calciviridae	No	1 capsid protein	ssRNA [†]	30-45	4
HPV 16	Papillomaviridae	No	2 capsid proteins	dsDNA [†]	35-50	5

TABLE 11-continued

Target viruses and virus-like particles tested in this study.						
Virus/VLP	Family	Enveloped Surface details		Genome type	Approx. size (nm)	References
SARS-CoV-2	Coronaviridae	Yes	1 envelope, 1 spike protein	ssRNA [‡]	30-70	6
Chikungunya	Togaviridae	Yes	2 envelope proteins	ssRNA [‡]	65-70	7
Rubella	Matonaviridae	Yes	2 envelope proteins	ssRNA [‡]	65-80	8
Adenovirus 5*	Adenoviridae	No	3 capsid proteins	dsDNA	90	9

*Infectious virus;

[‡]Data referring to the infectious virus which is modelled by a VLP in this study.1. Liu, A. P. et al., *J. Pharm. Biomed. Anal.* 189 (2020) 113481;2. Hogle, J. M., *Annu. Rev. Microbiol.* 56 (2002) 677-702;3. Kuhn, R. J. et al., *Cell* 108 (2002) 717-725;4. Chan, M. C. W. et al., *P. K. S.* 51-63 (2017) doi: 10.1016/B978-0-12-804177-2.00004-X;5. Goetschius, D. J. et al., *Sci. Rep.* 11 (2021) 3498;6. Yao, H. et al., *Cell* 183 (2020), 730-738.e13;7. Yap, M. L. et al., *Proc. Natl. Acad. Sci.* 114 (2017) 13703-13707;8. Mangala Prasad, V. et al., *PLOS Pathog.* 13 (2017) e1006377;9. Russell, W. C., *J. Gen. Virol.* 90 (2009) 1-20

[0161] Interestingly, in many instances the multivalent interactions between the HS coating on the shell interior and the virus particles appeared sufficiently strong to support substantial elastic deformations of the surrounding shell. For example, the T3 shell material deformed from spherical to elliptical around adenovirus particles, presumably driven by maximization of the number of molecular interactions between the HS moieties on the shell interior surface and the viral surface, at the expense of elastically deforming the shell. The O shells deformed occasionally so that up to four AAV2 particles were accommodated in its cavity (FIG. 3a), even though by design the O shell has only room for one AAV2 particle if it were completely rigid. The T1 shell also flexed to fit up to three HPV 16 copies (FIG. 3b). Depending on the relative stoichiometry between shells and virus particles, we also observed sandwich-like structures where two shells coordinated one virus particle (e.g., with HPV 16 and O shells, FIG. 3c). If the shell diameter exceeded substantially the target virus dimensions, multiple target particles could be sequestered. For example, we observed up to six AAV2 per T1 shell (FIG. 3d), and up to three chikungunya in T3 shells (FIG. 3e and FIG. 13 for TEM tomography). Furthermore, multiple copies of HS-modified shells could also cooperatively encapsulate dozens of AAV2 particles in clusters surrounded and protected by DNA origami shell material (FIG. 3f). These results support the notion that the shells are flexible to adapt and capture also more pleomorphic virus particles.

[0162] In the negative staining TEM images, we saw that the chikungunya VLP particles appeared to completely fill the T1 cavity. Presumably due to the resulting high degree of shape-complementarity, we could efficiently trap chikungunya particles within T1 shells using any of the different handle designs described in FIG. 1G in high yields (H1, 3a, 90% full shells). In fact, we could trap chikungunya even with the 3b negative control, which are shells with a coating lacking the sulfate and sulfonate groups, albeit at lower yield (H1, 3b, 54% full shells, see also FIG. 14). We interpret this phenomenon as a manifestation of molecular recognition on the mesoscale. The effect is presumably due to cooperative amplification of weak electrostatic interactions between the negatively charged DNA shells and the chikungunya particles as they interact over extended surface areas. Incidentally, this observation also suggests another route for modification-free virus trapping which considers precisely tailoring shells to the dimensions of the target virus.

[0163] Dengue virus, as well as some other viruses, present two distinct “mature” and “immature” conformations. The viral surface proteins must undergo certain conformational changes to become infectious, allowing them to move between vector and host, and/or infected and healthy cells (Yu, I.-M. et al., *Science* 319 (2008) 1834-1837; Yu, I.-M. et al., *J. Virol.* 83 (2009) 12101-12107; Lim, X.-X. et al., *Nat. Commun.* 8 (2017) 14339; San Martín, C., *Virus Maturation*. In: *Physical Virology: Virus Structure and Mechanics* (ed. Greber, U. F.) 129-158 (Springer International Publishing, 2019), doi: 10.1007/978-3-030-14741-9_7). While the usage of VLPs is highly convenient for safety reasons, we do acknowledge some limitations. For instance, initially our dengue VLP samples contained a high percentage of immature particles which did not bind to our HS-functionalized shells. To overcome this, we induced enzymatic maturation of the dengue VLPs, as would occur in vivo, and observed binding of the matured particles (FIG. 2a dengue and FIG. 15).

[0164] We also performed cryogenic electron microscopy (cryo-EM) measurements of HPV 16 and chikungunya VLPs trapped inside O and T1 shells, respectively (FIG. 4).

Cryo-EM

[0165] DNA origami shells were prepared and functionalized, and viruses trapped as described above. Samples (O+HPV: 70 nM triangles; T1+chikungunya: 200 nM triangles) were incubated 60 s on glow-discharged lacey carbon 400-mesh copper grids with an ultrathin carbon film. Subsequently, the grids were plunge frozen in liquid ethane with a FEI Vitrobot Mark V (blot time: 2.5 s, blot force: -1, drain time: 0 s, 22° C., 100% humidity, 3 µl sample). Cryo-EM imaging was performed with a spherical-aberration (Cs)-corrected Titan Krios G2 electron microscope (Thermo Fisher) operated with 300 kV and equipped with a Falcon III 4k direct electron detector (Thermo Fisher). Automated image acquisition was performed in EPU 2.6 (dose: 42-45 e⁻/Å², exposure time: 3-5 s, 12 fractions, pixel size: 0.23 nm (O+HPV) and 0.29 nm (T1+chikungunya), defocus: -1.5 to -2 µm). Micrographs were processed in RELION-3 (Zivanov, J. et al., *eLife* 7 (2018) e42166) using MotionCor2 (Zheng, S. Q. et al., *Nat. Methods* 14 (2017) 331-332) and CTFFIND4.1 (Rohou, A. & Grigorieff, N., *J. Struct. Biol.* 192 (2015) 216-221). Particles were automatically picked with cryYOLO 1.7.6 (Wagner, T. et al., *Commun. Biol.* 2 (2019) 1-13). Extracted particle images were

classified and selected by visual inspection through multiple rounds of 2D and 3D classifications. Initial models were generated in silico in RELION-3. 3D reconstructions and multibody refinement yielded electron density maps with resolutions of 26 Å for O shells trapping HPV (EMD-13884, 1× O+HPV: 7834 particles, 2× O+HPV: 4634 particles) and 36 Å for T1 shells trapping chikungunya (EMD-13883, 1259 particles, C₅ symmetry).

[0166] Two-dimensional (2D) class average images and 3D cryo-EM reconstructions confirmed that the VLPs were successfully trapped within the respective shell's cavities (FIG. 4*b,e* and FIG. 16-17). While one O shell is not sufficiently large to encapsulate an entire HPV 16 particle, two O copies can coordinate and completely cover an entire VLP (FIG. 4*c*). 2D class averages of free HPV 16 showed a variation in particle sizes within the VLP sample (FIG. 18). Consistently, we also found that the gap distances in between O shells (indicated by the white arrows in FIG. 4*b*) varied depending on whether a smaller or larger HPV 16 particle was trapped. The cryo-EM map that we determined for the complex consisting of a chikungunya VLP in a HS-modified T1 shell reveals the near-perfect fit between the two particles (FIG. 4*e,f*). The cryo-EM maps provide compelling illustrations of the extent of relative dimensions of the artificial DNA origami shells relative to their viral targets and the extent of surface occlusion that can be achieved by sequestering viruses in shells.

Example 3: Stability of DNA Origami Shells Encapsulating Viruses/VLPs

[0167] Finally, to test the stability of virus trapping by the shells, we subjected exemplarily a sample consisting of AAV2 encapsulated in O shells to a dilution series. The fraction of occupied shells remained the same prior and after 100-fold dilution and incubation for 14 days in the diluted sample relative to the non-diluted sample (FIG. 19), suggesting that the spontaneous dissociation rate for the complexes formed between AAV2 particles and the surrounding HS-modified shell is at least on the scale of weeks under the conditions tested. The high stability is avidity-driven and can be understood by considering that a spontaneous dissociation of AAV2 from a surrounding shell requires simultaneously breaking dozens of bonds formed between HS chains on the engulfing shell and the virus surface. The likelihood for such event to happen decreases exponentially with the number of HS bonds formed.

Example 4: Efficiency of Encapsulation of Viruses/VLPs by DNA Origami Shells

1-Handle Design Development for Efficient Virus Trapping

[0168] To optimize and calibrate the density needed of our heparan sulfate derivatives, we exemplarily explored the trapping efficiencies of AAV2 with O half-shells using three different handle variants (FIG. 1*G* and FIG. 8*a*). The proximal handle (H1) was the shortest tested design and consisted of a DNA extension of 26 nucleotides, positioning the heparan sulfate modification in a proximal arrangement. The distal handle design (H2) included a single stranded extension of 20 thymidines (polyT extension), allowing for the heparan sulfate group to reach further from the origami surface and increase the chances of multivalent binding events. Finally, the branched design (H3) mimicked a branched polymer having two heparan sulfate modifications per handle unit, doubling the local heparan sulfate density.

[0169] The three handle designs were tested in parallel with O half-shells and excess of AAV2 particles. Samples were analyzed via negative stain TEM, where images were collected using an automated montage setup to minimize data collection bias. Particles were quantified blindly to estimate the number of full vs. empty shells for all three handle variants hybridized to heparan sulfate 3*c*. These experiments revealed that H1 was not as efficient for virus trapping as the longer and denser H2 and H3 handles. With H1, only 20% of O shells were occupied with AAV2, whereas with H₂ and H₃ the trapping increased to 84% and 96%, respectively (FIG. 8*b*). The branched handle design (H3) hybridized to the heparan sulfate 18-mer variant 3*c* was used henceforth, unless otherwise stated.

2-Multi-Virus Trapping and Trapping of Different Virus Types in the Exact Same Shell Unit

[0170] To test if our system could be used as a true broad-spectrum virus-trapping platform, heparan sulfate-modified T1 half-shells were challenged to trap a cocktail of viruses consisting of AAV2, HPV16 and Chikungunya particles. Negative stain TEM characterization of such sample revealed the trapping of all virus types present in the cocktail (FIG. 21). The field of view micrographs in FIGS. 21*A* and *B* exemplify the good performance of the system. Some shells were found to encapsulate individual viruses such as one Chikungunya particle (FIG. 21*C*), one HPV16 (FIG. 21*D*) and one AAV2 (FIG. 21*E*), but multiple particles were also trapped at once in the exact same shell unit as seen with several AAV2 (FIG. 21*F* top), HPV16-Chik (FIG. 21*F* bottom), AAV2-Chik (FIG. 21*G*), and AAV2-HPV16 (FIG. 21*H*).

3. Cooperative Shell Trapping of Virus Clusters

[0171] If the shell diameter was substantially larger than the target virus dimensions, we observed that multiple virus particles could be sequestered. Interestingly, multiple copies of HS-modified shells could partition over and cover the surface of AAV2 clusters (FIG. 22).

SUMMARY

[0172] In conclusion, here we presented a viral trapping system that targets features of viruses that are conserved across many families through the usage of HS derivatives. Overall, we achieved encapsulation of nine different virus and VLP test samples, each representing a different viral family, and different sizes and surface complexities. Our modular shell system creates a locally curved environment within the cavity that enables highly multivalent binding, and that can be optimized according to size and ligand density/type to realize an irreversibly binding broad-spectrum antiviral platform. Our shells can flex and adapt to a certain degree to the shape of trapped virus particles, suggesting that the shell system can also adapt to pleomorphic virus particles.

[0173] We envision that our HS-modified DNA origami shells can act as a cellular surface decoy, sequestering the viruses and preventing interactions with cell surfaces, and thus reduce the effective viral load in acute infections. Testing the therapeutic potential of this system to reduce viral load in vivo remains an important task for the future. Beyond virus neutralization, our system may also serve as a sink for trapping associated viral proteins (FIG. 20), and other side products such as subviral particles that could potentially overwhelm the immune system (Zelikin, loc. cit.; Chai, N. et al., J. Virol. 82 (2008) 7812-7817). Overall, our results strongly indicate that our heparan sulfate-modified shell library has potential to become a relevant therapeutic platform to combat viral infections.

SEQUENCE LISTING

The patent application contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US20250283055A1>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

1. A DNA-based nanostructure, wherein said DNA-based nanostructure is a shell comprises a cavity enclosed by said DNA-based nanostructure, wherein said DNA-based nanostructure is formed by self-assembling DNA-based building blocks, wherein each of said self-assembling DNA-based building blocks is formed by a single-stranded DNA template strand and a set of oligonucleotides complementary to said single-stranded DNA template, wherein each of said oligonucleotides is either complementary to one contiguous DNA sequence stretch or to at least two non-contiguous DNA sequence stretches on said single-stranded DNA template, wherein each of said self-assembling DNA-based building blocks is a triangular and/or a rectangular prismoid, particularly a triangular prismoid, and wherein a subset of one or more of said oligonucleotides in one or more of said self-assembling DNA-based building blocks is/are each linked to a construct comprising at least one sulfonated or sulfated polysaccharide group pointing to the interior of said cavity, particularly a construct comprising one or two sulfonated or sulfated polysaccharide groups, wherein each said construct comprises (i) a handle comprising at least one binding site for said sulfonated or sulfated polysaccharide group, and (ii) said sulfonated or sulfated polysaccharide group(s) bound to said handle; wherein said handle has a length corresponding to at least to the length of a single-stranded oligonucleotide comprising 30 nucleotides.
2. The DNA-based nanostructure according to claim 1, wherein each of said handles comprises two binding sites for said sulfonated or sulfated polysaccharide groups.
3. The DNA-based nanostructure according to claim 1, wherein each of said sulfonated or sulfated polysaccharide groups is independently selected from the group consisting of heparin, heparan sulfate, hybrid heparan sulfates, carrageenans, cellulose sulfate, and dextran 2-sulfate.
4. The DNA-based nanostructure according to claim 3, wherein each of said sulfonated or sulfated polysaccharides is independently selected from a heparan sulfate and a hybrid heparan sulfate, in particular a heparan sulfate.
5. The DNA-based nanostructure according to claim 1, wherein one or more of said self-assembling DNA-based building blocks in said subset comprise n single-stranded oligonucleotides as said handles, wherein each handle is independently linked to at least one of said sulfonated or sulfated polysaccharide groups, wherein n is an integer independently selected from 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12, particularly wherein n is 9.
6. The DNA-based nanostructure according to claim 1, wherein said handles are single-stranded oligonucleotides having a length of between 30 and 60 nucleotides, in particular between 40 and 55 nucleotides, more particularly between 45 and 50 nucleotides.
7. The DNA-based nanostructure according to claim 1, wherein each of said sulfonated or sulfated polysaccharide groups comprises an oligonucleotide having a sequence that is complementary to an oligonucleotide stretch comprised in said handles.
8. The DNA-based nanostructure of claim 1 comprising, a closed three-dimensional geometric shape, wherein the closed three-dimensional geometric shape is selected from the group consisting of: a sphere, a spherocylinder, a polyhedron, a tetrahedron, an octahedron and an icosahedron, wherein the DNA-based nanostructure is formed in situ from said self-assembling DNA-based building blocks in the presence of said viruses or viral particles to be encapsulated.
9. The DNA-based nanostructure of claim 1 comprising, a shell with an opening for accessing said cavity.
10. The DNA-based nanostructure of claim 1 comprising, a combination of a first and a second subshell, wherein each of said first and said second subshell comprises an opening to access a first and a second inner cavity, respectively, wherein said first and said second inner cavity together form said cavity, optionally wherein said first and said second subshell are connected by at least one linker.
11. The DNA-based nanostructure of claim 1 comprising, an icosahedral structure.
12. The DNA-based nanostructure of claim 11, wherein said DNA-based nanostructure is a DNA-based nanostructure formed by self-assembling DNA-based building blocks, wherein each of said self-assembling DNA-based building blocks is a triangular and/or a rectangular prismoid, optionally wherein each of said self-assembling DNA-based building blocks is a triangular prismoid.
13. The DNA-based nanostructure of claim 12, wherein each said triangular and/or a rectangular prismoid is formed by m triangular, or rectangular, respectively, planes, wherein m is an integer independently selected from 4, 5, 6, 7 and 8, in particular independently selected from 5, 6 and 7, more particularly wherein said integer is 6, the three, or four, respectively, edges of each of said m planes are formed by n parallel stretches of DNA double helices, wherein n is an integer independently selected from 1, 2, 3, 4, 5 and 6 in particular independently selected from 2, 3, 4 and 5, more particularly independently selected from 3 and 4, wherein each plane is connected to a plane above and/or a plane beyond said plane (i) by stacking interactions between the DNA double helices forming said planes, and (ii) partially by DNA stretches within said single-stranded DNA template and/or said oligonucleotides forming said DNA-based building block bridging at least two of said planes, and wherein at least two of the three, or four, respectively, side trapezoids comprise a specific pattern of recesses and/

or extrusions formed by missing or additional DNA double helical stretches for specific interaction with a complementary pattern on the side trapezoid of another one of said self-assembling DNA-based building blocks.

14. The DNA-based nanostructure of claim **1**, wherein said DNA-based nanostructure is a half shell selected from the group consisting of:

- (a) a half octahedron DNA-based nanostructure comprising T_{octa} self-assembling DNA-based building blocks, which consists of a set of four copies of a triangular frustum, wherein the base-pair stacking contacts on one of the triangular edges of the triangular frustum are inactivated by either strand shortening or by adding unpaired thymidines;
- (b) a T=1 half shell comprising T_{1_pentamer_triangle} self-assembling DNA-based building blocks, which consists of two sets of in each case five copies of two different triangular frusta, wherein the five copies of the first set form a closed pentamer, and the five copies of the second set dock onto the edges of said pentamer; and
- (c) a “trap” T=1 half shell with a missing pentagon vertex comprising a combination of T_{1_pentamer_triangle} and T_{1_ring_triangle} self-assembling DNA-based building blocks, which consists of three sets of in each case five copies of three different triangular frusta, wherein the five copies of the first set form a closed pentamer, the five copies of the second set dock onto the edges of said pentamer the five copies of the second set dock onto the edges of said pentamer, and the five copies of the third set dock into the gaps between the five copies of said second set; and
- (d) a T=3 icosahedral half shell comprising T_{3_6_triangle} based self-assembling DNA-based building blocks, which consists of a total of 30 triangular subunits partitioned as five copies of six different full-size DNA triangle designs with specific edge docking rules.

15. The DNA-based nanostructure of claim **12**, further comprising

- (a) one or more types of DNA brick constructs, each type of such DNA brick constructs being characterized by one or more interaction sites for specific interaction by edge-to-edge stacking contacts with one or more complementary interaction sites present on the plane of a triangular or rectangular frustum on the outer surface of said DNA-based nanostructure, wherein said DNA brick constructs cover the free space between the three, or four, respectively, edges of said plane;
- (b) one or more cross-linkages within one of said triangular or rectangular prismoids, and/or between two of said triangular and/or rectangular prismoids; and/or
- (c) at least one moiety specifically interacting with said viruses or viral particles.

16. A composition comprising a DNA-based nanostructure according to claim **1** encapsulating one or more viruses or viral particles.

17. A method for encapsulating one or more viruses or viral particles, comprising the steps of: providing a DNA-based nanostructure according to claim **1**, and contacting said DNA-based nanostructure with a medium comprising, or suspected to comprise, said viruses or viral particles.

18. The method of claim **17**, wherein (i) a DNA-based half shell nanostructure comprising T_{octa} self-assembling DNA-based building blocks is selected for a virus of a size up to 50*50*50 nm³; (ii) the DNA-based half shell nanostructure comprising T_{1_pentamer_triangle} self-assembling DNA-based building blocks is selected for a virus of a size between 15*15*15 and 100*100*100 nm³; (iii) the DNA-based half shell nanostructure comprising a combination of T_{1_pentamer_triangle} and T_{1_ring_triangle} self-assembling DNA-based building blocks is selected for a virus of a size between 15*15*15 and 100*100*100 nm³; and (iv) the DNA-based half shell nanostructure comprising T_{3_6_triangle} self-assembling DNA-based building blocks is selected for a virus of a size of 50*50*50 nm³ or larger.

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